2026 AMS Madison Summit

Complete schedule of oral sessions, panels, events, and poster presentations

Madison, Wisconsin and online
July 29 and August 2-7, 2026

All presentation authors are reproduced as listed in the official program. Presentation titles link to their Confex abstract pages.

190 sessions | 1014 scheduled items | 366 poster presentations

Wednesday, July 29, 2026

Joint Poster - Online Poster Showcase

1:00 PM - 2:30 PM | Description | Eighth Conference on Weather Warnings and Communication

Submitter: Lisa Harris / Boston, MA

1:00 PM - Poster V.1 ATRAD - Set 1 of CERES-Libera Data Products
Miklos A. Zagoni, Eotvos Lorand Univ., Budapest, Hungary

1:00 PM - Poster V.2 ATRAD - Balloon-Borne Observations of Aerosol Optical Properties During Long-Range Transport Using a Low-Cost AeroSonde System.
Michalina Anna Broda, Uniwersytet Warszawski, Warsaw, MZ, Poland; Uniwersytet Warszawski, Warszawa, 5250011266, Poland; and O. Zawadzka-Mańko, M. Chiliński, K. Nurowska, S. Kłapiński, K. Markowicz, and P. Makuch

1:00 PM WITHDRAWN
Poster # V.3 Withdrawn: ATRAD - Radiative Closure in Complex Terrain: Topographic Effects on Surface Radiative Fluxes in the East River Watershed of Colorado During SAIL Michael J. Iacono, AER, Lexington, MA; and E. Mlawer, I. Polonsky, D. Feldman, and W. Rudisill Cancelled

1:00 PM - Poster V.4 ATRAD - NASA CERES Science: Supplementary Notes on Surface Energy Budget and Atmosphere-Surface Coupling
Miklos A. Zagoni, Eotvos Lorand Univ., Budapest, Hungary

1:00 PM - Poster V.5 ATRAD - Integer Structures in Earth's Global Mean Radiant Energy Flow System
Miklos A. Zagoni, Eotvos Lorand Univ., Budapest, Hungary

1:00 PM WITHDRAWN
Poster # V.6 Withdrawn: CLOUDPHYS - Improving Forecasts of Persistent Contrails through Ice Deposition Adjustments Zane Dedekind, EC, Vancouver, BC, Canada Cancelled

1:00 PM - Poster V.7 CLOUDPHYS - In-Situ Measurements of Droplet Clustering and Turbulence in Shallow Cumulus Clouds
Gholamhossein Bagheri, Max Planck Institute for Dynamics and Self-organization, Göttingen, NI, Germany; and Y. Kim, B. Thiede, and E. Bodenschatz

1:00 PM - Poster V.8 CLOUDPHYS - Examining the Effects of Various Measured Aerosol Regimes on Fog Properties in Iqaluit, Canada
Joelle Dionne, Dalhousie University, Halifax, NS, Canada; and M. Leriche, P. Gauvin-Bourdon, F. Merrick, Z. Mariani, N. T. O'Neill, and R. Y. W. Chang

1:00 PM - Poster V.9 CLOUDPHYS - Relationships between ATMS Brightness Temperatures and Liquid Water Clouds using the VIIRS Cloud Product
Aditya Kumar, CIMSS, Madison, WI; and J. A. Jung, A. Heidinger, S. E. Nebuda, H. Sun, I. Genkova, S. Nadiga, L. Soulliard, and Q. Zhao

1:00 PM - Poster V.10 CLOUDPHYS - Cloud Physics in the CERES (Clouds and the Earth's Radiant Energy System) Project
Miklos A. Zagoni, Eotvos Lorand Univ., Budapest, Hungary

1:00 PM - Poster V.11 CLOUDPHYS - Cloud Condensation Nuclei (CCN) Production by Fires in the United States
Aditya Kumar, CIMSS, Madison, WI; and C. Schmidt

1:00 PM - Poster V.12 CLOUDPHYS - Global Distribution of Ice Clouds using the VIIRS Cloud Product
Aditya Kumar, CIMSS, Madison, WI

1:00 PM - Poster V.12A CLOUDPHYS - Quantifying Cloud Entrainment and Drizzle In the Marine Stratocumulus Boundary Layer Using Stable Isotopes In Water Vapor from the SCILLA Airborne Campaign
Nimya Sheena Sunil, Purdue University, West Lafayette, IN; and L. Welp, P. Chuang, A. Molino, M. D. Leandro, A. Bucholtz, and M. K. Witte

1:00 PM - Poster V.12B CLOUDPHYS - Climate Warming Could Weaken Aerosol-Cloud Interactions in Subtropical Marine Stratocumulus
Hongwei Sun, University of Hawaii at Manoa, Honolulu, HI; and P. Blossey, R. Wood, E. Erfani, S. Doherty, and J. Chun

1:00 PM - Poster V.12C MOUNTAIN - Idealized Simulations of Supercell Tornadoes Within the Southern and Central Appalachian Mountains
Logan Twohey, Univ. of North Carolina at Charlotte, Charlotte, NC

1:00 PM - Poster V.12D MOUNTAIN - Physics-Informed Deep Learning for Complex Terrain with Explainable AI
Arnau Toledano Rubí, University of Barcelona, Barcelona, Spain; and M. Udina, E. Busquets, Y. Sola Salvatierra, and J. Bech

1:00 PM - Poster V.13 SLS - Artificial Intelligence Outperforms Human Experts in Classifying Tornado, Hurricane, and Straight-Line Wind Damage from Photographic Evidence
C. Justin Van De Wiele, APEC Engineering & Laboratory, LLC, Shallowater, TX; AEOLUS Analytics, LLC, Shallowater, TX; and M. B. Phelps, C. B. Fedler, and K. Sherman-Morris

1:00 PM - Poster V.14 **SLS - Wind Jerk Can Cause Damage to Structures Below Design Wind Speeds**
C. Justin Van De Wiele, APEC Engineering & Laboratory, LLC, Shallowater, TX; AEOLUS Analytics, LLC, Shallowater, TX; and M. B. Phelps, D. V. Day, C. B. Fedler, and C. T. Williams

1:00 PM - Poster V.15 **SLS - The Implementation and Application of the Enhanced Fujita Scale in Canada: An Update**
David M. L. Sills, Canadian Severe Storms Laboratory, London, ON, Canada; and C. S. Miller, A. Jaffe, and G. A. Kopp

1:00 PM - Poster V.16 **SLS - Entropic Balance and Vorticity in Tornado Theory**
Mikhail M. Shvartsman, University of St.Thomas, St. Paul, MN; and P. Belik, D. P. Dokken, and K. Scholz

1:00 PM - Poster V.17 **SLS - Heavy Rainfall Simulations in the Piura Region, Peru: A Parametrization Sensitivity Study with the WRF and Eta Models**
Juan Carlos Tufino Bernuy, National Agrarian University La Molina, Lima, LIM, Peru; and A. Huerta, A. Llacza, A. Moya, J. Llamocca, and M. Bojorquez

1:00 PM - Poster V.18 **SLS - Meso-γ-scale Dynamic and Thermodynamic Mechanisms in an Extreme Rainfall Event in Zhengzhou, China**
Wenhua Gao, Chinese Academy of Meteorological Sciences, Beijing, 11, China; and C. Li

1:00 PM - Poster V.18A **SLS - Environmental Influences of the Chesapeake Bay on Convective Storm Morphology and Severe Weather Production**
Roger Ralph Riggin IV, University of North Carolina at Charlotte, Charlotte , NC; University of North Carolina at Charlotte, Charlotte, NC; and C. E. Davenport

1:00 PM - Poster V.19 **WAF - Liquid Water Path Thresholds for Precipitation using Scattering Index Tests and the VIIRS Cloud Product**
Aditya Kumar, CIMSS, Madison, WI; and J. A. Jung, A. Heidinger, S. E. Nebuda, H. Sun, I. Genkova, S. Nadiga, L. Soulliard, and Q. Zhao

1:00 PM - Poster V.20 **WXWARNING - Disaster! – A Serious Game about Atmospheric Hazards**
Nathan M. Hitchens, Ball State University, Muncie, IN

Sunday, August 2, 2026

Event - Weather-Ready Nation Scout Event

7:45 AM - 12:30 PM | Description | Events

Monday, August 3, 2026

Session 1 - Frontiers of AI/ML in Severe Local Storms I

8:15 AM - 10:00 AM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Aaron J. Hill / University of Oklahoma, School of Meteorology / Norman, OK / Julia Buhrman / Texas Tech University / Lubbock, TX

8:15 AM Welcoming Remarks

8:30 AM - 1.1 Explainable Deep Learning for Probabilistic Nowcasting of Radar Reflectivity in Tornadic Storms
Nathan Erickson, University of Oklahoma, Norman, OK; and A. McGovern and A. J. Hill

8:45 AM - 1.2 A Comparison of Image-based and Tree-based Tornado Nowcasting Models with a Nature Dataset
John L. Cintineo, NOAA / OAR / National Severe Storms Laboratory, Madison, WI; and T. Sandmael, R. B. Steeves, K. M. Calhoun, and P. A. Campbell

9:00 AM - 1.3 Hyperlocal Nowcasts of Hail across the CONUS with Ensemble Deep Learning
Ryan Lagerquist, MyRadar, Ward, CO; MyRadar, Ward, CO; and J. P. Gibbons and S. Garimella

9:15 AM - 1.4 Thunderquakes and Beyond: Machine Learning-Based Seismic Signal Discrimination to Detect Thunderstorms in Data-Poor Regions
Alexandra Anderson-Frey, Univ. of Washington, Seattle, WA

9:30 AM - 1.5 Extending Predictability Limits of Tornadic Supercell Thunderstorms Using Self-Organizing Maps
Luke Justin LeBel, The Pennsylvania State University, State College, PA; and B. A. Garcia and P. M. Markowski

9:45 AM - 1.6 Predicting Biases in Day 1 Probabilistic Convective Outlooks from Time-Evolving Environmental Conditions with Convolutional Neural Networks
Miles Epstein, University of Washington, Seattle, WA; and A. Anderson-Frey and V. Staneva

Joint 1 - From Metrics to Minutes: Tornado Risk and Response

8:30 AM - 10:00 AM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Chair: Kodi L. Berry / NOAA/OAR/NSSL / Norman, OK

8:30 AM - 1.1 Professional Meteorologist Risk Communication Prior to and during the Rolling Fork and Amory Supercell
Kodi L. Berry, NOAA/OAR/NSSL, Norman, OK; NOAA/OAR/NSSL, Norman, OK; and M. Krocak, Z. Rosen, H. Obermeier, D. Hogg, T. Maciag, T. Sempier, J. E. Sharpe, J. E. Trujillo-Falcón, A. R. Gaviria Pabon, and J. Tessitore

8:45 AM - 1.2 The Influence of Tornado Warning Exposure on Reception, Understanding, and Response Measures
Makenzie Krocak, NOAA National Severe Storms Laboratory, Norman, OK; and J. T. Ripberger and M. D. Flournoy

9:00 AM - 1.3 How Possible is it? A Geometric and Operational Analysis of the Severe Thunderstorm Warning With the 'Tornado Possible' tag
Jacob Ryan Reed, Mississippi State University, Mississippi State, MS; and G. Slade

9:15 AM - 1.4 The Comprehension Gap: Public Understanding of Tornado Emergencies vs. Tornado Warnings in the Southeastern U.S.
Kelsey Othling, Mississippi State University, Baton Rouge, LA

9:30 AM - 1.5 Long-Term Tornado Warning Metrics Revisited
Harold E Brooks, NSSL, Norman, OK; and M. E. Baldwin and J. Correia Jr.

9:45 AM - 1.6 Actions Taken During the April 28, 2025 Severe Weather Outlook
Jacob Beitlich, NWS, Chanhassen, MN

Session 1 - Atmospheric Radiative Transfer and Light Scattering Theory

8:30 AM - 10:00 AM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / Jet Propulsion Laboratory, California Institute of Technology / Pasadena, CA

Chair: Xiquan Dong / The Univ. of Arizona / Tucson, AZ

Cochairs: Ryan Kramer / GFDL / Alexandria, MD

8:30 AM - 1.1 Theoretical and Operational Advancements in Two-Stream Shortwave Radiative Transfer Modeling

Dion Jia Xu Ho, LDEO, New York, NY

8:45 AM - 1.2 An Accurate, Efficient, and Flexible Atmospheric Radiation Scheme for Satellite Earth Energy Budget Monitoring: From Heritage to Innovation
Jian Wei, Texas A&M Univ, College Station, TX; and P. Yang

9:00 AM - 1.3 Impact of Mineral Dust Aerosol Inhomogeneity on Optical Properties, Microphysical Properties, and Satellite Retrievals
Dongchen Li, Texas A&M University, College station, TX

9:15 AM - 1.4 Advancements in PCRTM: Efficient Forward Modeling and Inversion Across MW, IR, and Solar Spectra
Xu Liu, LRC, Hampton, VA

9:30 AM - 1.5 RTSOS: A Polarized Radiative Transfer Package for the Earth System: Theory and Applications
Peng-Wang Zhai, University of Maryland Baltimore County, Baltimore, MD; and Y. Hu and M. Gao

9:45 AM - 1.6 Examining the Feasibility of Passive Satellite Remote Sensing of Ocean Microplastics With New High-Resolution Multiple Scattering Simulations
Gili Kurtser, The Hebrew University of Jerusalem, Jerusalem, Israel; and C. Haspel

Session 1 - High-Latitude Cloud, Precipitation and Radiation Processes I

8:30 AM - 10:00 AM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Andrew Michael Dzambo / Cooperative Institute for Severe and High Impact Weather Research and Operations / Norman, OK / Sabrina Schnitt / University of Cologne / Cologne Germany / Yi Huang / The University of Melbourne / Melbourne, VIC / Australia / Michael Tjernstrom / Stockholm Univ. / Stockholm, Uppland and Södermanland / Sweden

8:30 AM - 1.1 Surface Albedo Parameterization in the Arctic: Effects of Melt Ponds and Clouds
Patrizia Schoch, Leipzig University, Leipzig, Germany; and E. Jäkel, T. Sperzel, W. Dorn, and M. Wendisch

8:45 AM - 1.2 The Maintenance of Phase-partitioning in “Warm” Arctic Mixed-phase Stratocumulus due to Microphysical-Dynamical Interactions
Amy B. Solomon, CIRES and NOAA/PSL, Boulder, CO

9:00 AM - 1.3 Liquid Water Path is Largely Insensitive to Meteorology in Arctic Winter
Kara Hartig, CIRES, Boulder, CO; and J. J. Cassano, M. D. Shupe, and A. B. Solomon

9:15 AM - 1.4 Stochastic Cloud Fluctuations Drive Arctic Winter Radiative Bistability
Jung-Sub Lim, CIRES, Boulder, CO; and G. Feingold

9:30 AM - 1.5 Examining the Thermodynamics and Dynamics of Arctic Goldilocks Clouds During the ARCSIX Field Campaign
Bradley Fletcher Lamkin, University of Oklahoma, Norman, OK; and J. Redemann, C. J. Flynn, L. Gao, E. Crosbie, A. R. Nehrir, B. Collister, K. L. Thornhill, and K. S. Schmidt

9:45 AM - 1.6 Arctic Cloud Regimes and Their Radiative Effect Based on MOSAIC Observations
Hartwig M Deneke, Leibniz Institute for Tropospheric Research, Leipzig, Germany; and H. J. Griesche, J. Happich, A. Macke, M. D. Shupe, H. Sundermann, and J. Witthuhn

Session 1 - Meso to Subseasonal Scale Meteorological Phenomena Part I

8:30 AM - 10:00 AM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Tomer Burg / WindBorne Systems / Palo Alto, CA / Austin Coleman / NOAA

8:30 AM - 1.1 A Summary of Fire Behavior and Outlook Forecasting with Machine Learning Methods over CONUS
Christina E. Kumler, CIRES, Boulder, CO; CIRES, Boulder, CO; and M. L. Breeden, R. Worsnop, and R. Lagerquist

8:45 AM - 1.2 Subseasonal Predictability of Fire Weather Indicators in the Western United States via Linear Inverse Modeling
Alejandro Nino, University of California Santa Barbara, Goleta, CA; and C. Jones

9:00 AM - 1.3 Investigating Surface and Boundary Layer Thermodynamic and Kinematic Changes Across Fronts in a Semi-arid Region
Danielle E. Harr, Texas Tech University, Lubbock, TX; and S. Pal, T. R. Lee, and M. Hamel

9:15 AM - 1.4 Vertically Resolved Atmospheric Cloud Distribution Through Supervised Classification
Iain Lothian McConnell, CIMSS, Madison, WI; and A. J. Wimmers, S. T. Wanzong, R. Lagerquist, J. P. Gibbons, and S. Garimella

9:30 AM - 1.5 Spatially Heterogeneous Arctic Warming Impacts on Polar Low Genesis Conditions: A 42-Year ERA5 Period Comparison
Aarav Mishra, Metea Valley High School, Naperville, IL

9:45 AM - 1.6 A Climatology of Tropopause Polar Vortex Induced Jet Superposition Events
Eleanor Salm, University of Colorado Boulder, Boulder, CO; and A. C. Winters and C. A. Reiher

Session 1 - Model Center Overviews

8:30 AM - 10:00 AM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Jack Lind / NWS / Lenexa, KS / Jun Du / NOAA

8:30 AM - 1.1 The NOAA Modeling Moonshot Initiative to Rapidly Advance Operational Numerical Weather Prediction Capabilities
Daryl T. Kleist, NWS, College Park, MD; and J. E. Ten Hoeve III, K. Garrett, R. Bandy, V. S. Tallapragada, and C. R. Alexander

8:45 AM - 1.2 NWS Transformation: Environmental Modeling Center and Meteorological Development Laboratory Unification
Richard Bandy, NWS, Silver Spring, MD; and D. T. Kleist

9:00 AM - 1.3 An Overview of NOAA's Unified Forecast System (UFS) Applications for Operational Forecasting Across Scales
Fanglin Yang, NOAA, ELLICOTT CITY, MD

9:15 AM - 1.4 Transitioning the NWS Production Suite to the Joint Effort for Data assimilation Integration (JEDI)
Cory R Martin, PhD, NOAA/NWS, College Park, MD; and A. Collard, C. A. Thomas, C. S. Draper, D. Holdaway, G. Vernieres, R. Treadon, and S. Liu

9:30 AM - 1.5 The U. S. Navy Earth System Prediction Capability: Overview and Future Plans
Carolyn A. Reynolds, NRL, Monterey, CA; and C. Barron, C. Barton, H. Christophersen, J. Christophersen, W. Crawford, D. Hebert, D. Heinzeller, M. A. Janiga, T. Jensen, D. D. Kuhl, F. Liu, S. Paturi, P. A. Reinecke, E. Rogers, S. S. Rushley, J. Shriver, C. Snyder, G. J. Theurich, P. Thoppil, M. Ulate, J. Zappala, and L. Zamudio

9:45 AM - 1.6 WRF Ensemble Updates at the US Air Force's 16th Weather Squadron
Burkely Twiest Gallo, 557th Weather Wing, Offutt AFB, NE; and A. J. Elliott, N. F. Wright, E. L. Kuchera, S. Rentschler, G. Creighton, S. Childs, G. R. Brooks, J. Keane, M. T. Vaughan, C. J. Melick, PhD, and W. T. Sedlacek

Session 1 - Land-Atmosphere Interactions in Mountains and Valleys

9:00 AM - 10:00 AM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 22nd Conference on Mountain Meteorology

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Cochairs: Natalie Wagenbrenner / USDA / Missoula, MT / Yicheng Li / Davis, CA

9:00 AM - 1.1 Seasonal Characteristics of Surface Energy Fluxes By Dominant Cloud Types In the High-Altitude Rocky Mountains
Joseph Sedlar, CIRES, Boulder, CO; and T. P. Meyers

9:15 AM - 1.2 Aerosol-Cloud-Rainfall Interactions and Land-Atmosphere Coupling from Micron to Hectometer Scales in Complex Terrain
Lihui Ji, University of Illinois Urbana Champaign, Urbana, IL; and A. P. Barros

9:30 AM - 1.3 Multiscale Processes Affecting Wintertime Cold-Fog Formation and Evolution in Complex Terrain: Insights from the 2022 CFACT Field Campaign
Zhaoxia Pu, Univ. of Utah, Salt Lake City, UT; and C. Feng, G. Carrillo-Cardenas, A. G. Hallar, I. Gultepe, and E. R. Pardyjak

9:45 AM - 1.4 Macroclimates, Microclimates, and Climate Change in the Cederberg Poetry of Stephen Watson, 1954-2011.
Gavin Edward Craig Heath, PhD, University of KwaZulu-Natal, Ashwood, ZN, South africa

Coffee Break

10:00 AM - 10:45 AM | Grand Terrace (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

Session 2 - Advances in Severe Local Storm Diagnosis and Prediction

10:45 AM - 12:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Thea Sandmael / Cooperative Institute for Severe and High-Impact Weather Research and Operations, Univ. of Oklahoma / Norman, OK / Stephen M. Strader / Villanova Univ. / Villanova, PA

10:45 AM - 2.1 QTor: A New Parameter for Forecasting QLCS Tornado Potential
Kelsey C. Britt, CIWRO, Norman, OK; and P. Skinner, M. C. Brown, and A. J. Clark

11:00 AM - 2.2 An Analytical Model for Predicting Tornado Duration and Pathlength
Jonathan M. Garner, Private Meteorologist, Leadville, CO; and V. A. Gensini, J. M. Straka, and K. M. Kanak

11:15 AM - 2.3 How Sensitive are Convective Parameters to Vertical Resolution?

Robert A Warren, Bureau of Meteorology (Australia), Melbourne, VIC, Australia; and H. Richter, D. Sgarbossa, D. G. Udy, C. H. Su, and M. Thatcher

11:30 AM - 2.4 Sounding, Mesoanalysis, and Numerical Weather Model Postprocessing of Convective Fields at the Storm Prediction Center
Kelton T. Halbert, SPC, Norman, OK; SPC, Norman, OK; and I. L. Jirak and P. T. Marsh

11:45 AM - 2.5 KDP Cores! What Are They Good For?
Charles M. Kuster, NOAA/OAR/NSSL, Norman, OK; and K. Tuftedal, R. Bowers, K. Sherburn, and T. J. Schuur

Session 2 - Climate Forcing, Feedbacks and Natural Variability

10:45 AM - 12:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / Jet Propulsion Laboratory, California Institute of Technology / Pasadena, CA

Cochairs: Ryan Kramer / GFDL / Alexandria, MD / David Jeffrey Paynter / NOAA/GFDL / Princeton, NJ

10:45 AM - 2.1 A Strong Constraint on Radiative Forcing of Well-mixed Greenhouse Gases
Jing Feng, SLRSS - State Key Laboratory of Remote Sensing and Digital Earth, Beijing, Beijing, China

11:00 AM - 2.2 Developing Radiative Forcing and Feedback Climate Data Records for Understanding Observed Radiation Trends and Evaluating Models
Ryan Kramer, GFDL, Princeton, NJ; and A. Matus

11:15 AM - 2.3 Model Dependence on Cloud Radiative Kernels: Can We Use One Cloud Kernel for All Models?
Xiuhong Chen, Univ. of Michigan, Ann Arbor, MI; and X. Huang, S. Zhao, Q. Yue, K. Suzuki, and M. Zhao

11:30 AM - 2.4 Forcing-Dependent Arctic Longwave Cloud Feedback in E3SMv3
Suhui Zhao, University of Michigan, Ann Arbor, MI; and X. Chen, M. Amma, H. Wang, W. Lin, and X. Huang

11:45 AM - 2.5 The Crucial Importance of Solar Irradiance Observations in Developing and Validating Solar Irradiance Models During a Period of Rising Solar Activity
Odele M. Coddington, Laboratory for Atmospheric and Space Physics, Boulder, CO; and J. Lean, E. C. Richard, and P. Pilewskie

Session 2 - Distribution of Precipitation in Mountains and Valleys I

10:45 AM - 11:45 AM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 22nd Conference on Mountain Meteorology

10:45 AM - 2.1 Climatology of Cold-Season Emergent Convection in Frontal Systems and Its Impact on Orographic Precipitation
Francis Osei Tutu Afrifa, University of Wyoming, Laramie, WY

11:00 AM - 2.2 A Cyclone-Centered Examination of Low-Level Jet–Terrain Interactions on Observed Precipitation Patterns in Northeastern United States Winter Storms
Kaitlyn R. Jesmonth, Univ. of Illinois Urbana-Champaign, Urbana, IL; and S. W. Nesbitt and K. H. Lundstrom

11:15 AM - 2.3 Dominant Synoptic Systems for Summer Precipitation over the Complex Terrain of Southwestern China
Yin Zhao, Chinese Academy of Meteorological Sciences, Beijing, 11, China

11:30 AM - 2.4 Large-Scale Atmospheric Circulation and Land-Sea Breeze Controls on Precipitation Formation across Elevation Gradients across Western Peruvian Andes Using Stable Isotopes In Rain
Carol Kristi Romero Roldan, Purdue University, Lafayette, IN; Purdue University, West Lafayette, IN; and L. Welp

Session 2 - High-Latitude Cloud, Precipitation and Radiation Processes II

10:45 AM - 12:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Andrew Michael Dzambo / Cooperative Institute for Severe and High Impact Weather Research and Operations / Norman, OK / Sabrina Schnitt / University of Cologne / Cologne Germany / Yi Huang / The University of Melbourne / Melbourne, VIC / Australia / Michael Tjernstrom / Stockholm Univ. / Stockholm, Uppland and Södermanland / Sweden

10:45 AM - 2.1 Geographic Variability in Boundary Layer Cloud Properties: Insights from CAPE-k, COMBLE, and MARCUS
Zeqian(Hazel) Xia, Cooperative Institute for Severe and High Impact Weather Research and Operations (CIWRO), Norman, OK; School of Meteorology, University of Oklahoma, Norman, OK; and Y. Huang and G. M. McFarquhar

11:00 AM - 2.2 Investigation of Marine Boundary Layer Cloud and Drizzle Microphysics using Multi-Platform Observations over the Southern Ocean
Anik Das, The University of Arizona, Tucson, AZ; and X. Dong and B. Xi

11:15 AM - 2.3 Increasing Stratiform Rainfall Frequency and Duration at a Long-term Arctic Observatory
Yan Xie, University of Michigan, Ann Arbor, MI; University of Oklahoma, Norman, OK; and C. Pettersen and F. King

11:30 AM - 2.4 Cloud-Radiative Effects in the Arctic Based on NABOS Shipborne Observations

Alexandra Igorevna Narizhnaya, IAP, Moscow, Moscow, Russia; IAP, Moscow, Moscow, Russia; and A. Chernokulsky, D. Chechin, and I. Repina

11:45 AM - 2.5 Tracking the Floes, Following the Clouds: The Science of ARCSIX
Konrad Sebastian Schmidt, University of Colorado Boulder, Boulder, CO; and A. B. Solomon, V. Nataraja, K. Hirata, and Y. W. Chen

Session 2 - Modernizing the Message: Advances in AI, Technology, and the Human Connection

10:45 AM - 12:00 PM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

Submitter: Robin Tanamachi / Purdue University / West Lafayette, IN

Chair: Gina M. Eosco / NOAA Federal / Silver Spring, MD

10:45 AM - 2.1 Bridging Technical Public Alerts and Public Action with AI: The United National Weather Team alert system.
Alicia Kay Hardimon, UNITED NATIONAL WEATHER TEAM, Eagle Bend, MN

11:00 AM - 2.2 AI In the Forecast: Identifying Public Support and Opposition to the Use of AI In Weather Forecasting Activities
Abigail Lee Bitterman, Institute for Public Policy Research and Analysis, Univ. of Oklahoma, Norman, OK; Institute for Public Policy Research and Analysis, Univ. of Oklahoma, Norman, OK; and J. T. Ripberger and M. Krocak

11:15 AM - 2.3 Integrating Social Science into the Future of Weather Forecasting
Gina M. Eosco, Ph.D, NOAA Federal, Silver Spring, MD; and A. Krepp

11:30 AM - 2.4 Preparing the Future of TV Warnings and Alerts: Examining User Preferences
Michael S. Michaud, University at Albany, Albany, NY; and H. Obermeier, K. L. Berry, N. Johnson, and L. L. Sun

11:45 AM - 2.5 Hazard Services Version 4: A Modernization in Issuing Short-Fuse Convective Warnings for the National Weather Service
Darrel M. Kingfield, NOAA, Boulder, CO; NOAA/Global Systems Laboratory, Boulder, CO; and E. E. J. Schlie, C. Golden, N. R. Hardin, K. L. Manross, J. Ramer, G. J. Stumpf, D. Tomalak, S. Zhuo, and M. Armstrong

Session 2 - Testing, Metrics, and Validation of Artificial Intelligence for Numerical Weather Prediction (AI4NWP) Part I

10:45 AM - 12:00 PM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitters: Austin A. Coleman / CIRES / WPC / Boulder, CO / Christina E. Kumler / CIRES / Boulder, CO / Aaron J. Hill / University of Oklahoma, School of Meteorology / Norman, OK

Cochairs: Kara D. Lamb / Columbia University / New York, NY / Corey K. Potvin / NSSL / Norman, CO

10:45 AM - 2.1 Parameterization Development and Machine Learning Accelerated Calibration and Uncertainty Quantification for the NASA Giss Modele Earth System Model
Marcus van Lier-Walqui, Columbia University, New York, NY; and K. Loftus, G. S. Elsaesser, A. Z. Hu, and H. Morrison

11:00 AM - 2.2 Probabilistic quantitative precipitation forecasts from deterministic model guidance
Thomas M. Hamill, The Weather Company, Boulder, CO

11:15 AM WITHDRAWN
2.3 Withdrawn: TianJi: An Artificial Intelligence Scientist for Autonomous Discovery in Earth System Science kaikai Zhang, Nanjing University of Information Science and Technology, Nanjing, China; and F. Meng, PhD Hall of Ideas - EFG (Monona Terrace Community and Convention Center) Cancelled

11:30 AM - 2.4 Validation of Cool-Season Precipitation Forecasts By Both Conventional and AI Numerical Weather Prediction Models over the Contiguous United States
Peter Veals, Univ. of Utah, Salt Lake City, UT; and A. J. Schwartz and S. Flampouris

11:45 AM - 2.5 Use of a Convolutional Neural Network to Evaluate the Hourly Extreme Wind Climatology In a 42-Year WRF PGW Climate Simulation
Benjamin M. Kiel, Iowa State University, Ames, IA; and W. A. Gallus Jr.

Session 2 - Tropical Cyclones: Processes and Prediction

10:45 AM - 12:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Diana Rose Jesberg / ESRL / Boulder, CO / Leanne Blind-Doskocil / Valparaiso University / Valparaiso, IN

10:45 AM - 2.1 Characterizing Overpredicted Tropical Cyclone Intensity in NHC's Official and Numerical Model Forecasts
Yunji Zhang, PhD, Pennsylvania State Univ., University Park, PA

11:00 AM - 2.2 Investigating Sources of Forecast Error for non-Intensifying Tropical Storm Rene
Jordan Rendon, DAO, State College, PA

11:15 AM - 2.3 Does Our Approach to Calculating Shear Matter? An Assessment of Three Shear Methods for Tropical Cyclone Rapid Intensity Change
Autumn Toms, Mississippi State University, Mississippi State, MS; and K. Wood and J. E. Rudzin

11:30 AM - 2.4 Tropical Cyclone Downshear Reformation in the Hurricane Analysis and Forecast System
Brian H. Tang, PhD, University at Albany, SUNY, Albany, NY; and G. R. Alvey III

11:45 AM - 2.5 North Pacific Tropical Cyclone Lionrock (2016) Revisited: The Many “Flavors” of a Long-Lived North Pacific ("Rock Star") Tropical Cyclone
Lance F. Bosart, University at Albany/SUNY, Albany, NY

Lunch Break

12:00 PM - 1:45 PM | View Related | Eighth Conference on Weather Warnings and Communication

Joint 3 - The Weight of the Warning: Ethics, Trust, and Public Impact

1:45 PM - 3:00 PM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Chair: Michael S. Michaud / Newark, DE

1:45 PM - 3.1 Can a Tornado Warning Cause More Damage Than the Tornado Itself?
Matt E. Jeglum, NWS, Salt Lake City, UT; NOAA/NWS, Salt Lake City, UT

2:00 PM - 3.2 "Weather is Your Mood; Climate is Your Personality": A Consideration of Metaphor and Simile As Tools for Messaging Science
Susan Jasko, Retired, TUSCALOOSA, AL

2:15 PM - 3.3 SHOW THE SNOW MAP and Other Uncomfortable Truths about Weather Communication
Christopher A. Vagasky, National Lightning Safety Council, Fort Atkinson, WI; University of Wisconsin-Madison, Madison, WI

2:30 PM - 3.4 Excellent Forecasts and Major Weather Disasters: Why Don't We Do Better?
Clifford F. Mass, University of Washington, Seattle, WA

2:45 PM - 3.5 When Communication Fails: Impacts on Weather Risk, Public Trust, and Decision-Making
Kala Golden, Weather Modification International, Fargo, ND

Session 3 - Advances in Cloud Parameterizations

1:45 PM - 3:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Julia A Shates / SIO / La Jolla, CA

Cochairs: Adel L. Igel / University of California, Davis / Davis, CA / Fan Yang / Brookhaven National Laboratory / Mastic, NY

1:45 PM - 3.1 How Entrainment Shapes Droplet Size Distributions and Controls Precipitation Onset in Warm Clouds
Steven K. Krueger, Univ.of Utah, SALT LAKE CITY, UT; and C. Bois and G. Daniels

2:00 PM - 3.2 Supersaturation-Droplet Interactions In Idealized Turbulent Cloud Convection and Evaluation of a New Subgrid Model
Kamal Kant Chandrakar, PhD, NCAR, Boulder, CO; and H. Morrison and R. A. Shaw

2:15 PM - 3.3 Better Rainfall Prediction without Autoconversion: Advances in Single Liquid Category Bulk Scheme
Arthur Zhuoqun Hu, Columbia University, New York, NY; and M. van Lier-Walqui, S. P. Santos, K. Loftus, and H. Morrison

2:30 PM - 3.4 Toward an Operational Data-Driven Parameterization for Cloud Microphysics
Emily Katherine de Jong, Lawrence Livermore National Lab, Livermore, CA; and N. Gunawardena, H. Morrison, H. Beydoun, and P. M. Caldwell

2:45 PM - 3.5 Influence of Aggregation Efficiency Parameterization for Ice Particles at Very Low Temperatures on Global Cloud Distributions
Tatsuya Seiki, JAMSTEC, Yokohama, 14, Japan; and T. Nagao

12:00 AM WITHDRAWN
Withdrawn: Global In Situ Evidence of Narrow Droplet Size Distributions In Non-Precipitating Marine Boundary Layer Clouds Yi Huang, The University of Melbourne, Melbourne, VIC, Australia; University of Melbourne, Melbourne, ACT, Australia; and L. Ger B. Aragon, J. Crosier, P. T. May, P. J. Connolly, and S. Abel Madison Ballroom D (Monona Terrace Community and Convention Center) Cancelled

Session 3 - Distribution of Precipitation in Mountains and Valleys II

1:45 PM - 2:45 PM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 22nd Conference on Mountain Meteorology

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Chair: Gavin Edward Craig Heath / University of KwaZulu-Natal / Ashwood, ZN / South africa

Cochairs: Holly J. Oldroyd / University of California, Davis / Davis, CA

1:45 PM - 3.1 Orographic Influences on Precipitation in the Wasatch Mountains as Observed by Mountain and Valley Profiling Radars
Michael L. Wasserstein, University of Utah, Salt Lake City, UT; and A. N. Evans, P. Veals, D. E. Kingsmill, and J. Steenburgh

2:00 PM - 3.2 Multi-Scale Characteristics of Extreme Cool-Season Precipitation in the Park Range of Colorado.
Annegret Lang, University of Utah, Salt Lake City, UT; and J. Steenburgh, A. G. Hallar, and G. G. Mace

2:15 PM - 3.3 Introducing the Snow Sensitivity to Clouds in a Mountain Environment (S2noCliME) Field Campaign
Claire Pettersen, Univ. of Michigan, MADISON, WI; and G. G. Mace, L. A. McMurdie, A. Rowe, A. G. Hallar, B. Dolan, and M. Oue

2:30 PM - 3.4 Evaluation of ERA5-Land Precipitation Phase Classification in the Tropical High Andes
Ludving A Cano Fernandez, University of Nevada Reno, RENO, NV; and L. B. Perry, M. Andrade, A. Seimon, N. Pokhrel, T. Matthews, B. Graves, N. Montoya, M. Rado, and S. Arias

Session 3 - Forecast Verification Methods and Performance: CAM Verification

1:45 PM - 3:00 PM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Clark Evans / Michelle A. Harrold / NCAR / Boulder, CO

1:45 PM - 3.1 Addressing the Precipitation Prediction Grand Challenge with Rapidly Updating Numerical Weather Prediction: How Are We Doing?
Eric P. James, GSD, Boulder, CO; and S. G. Benjamin

2:00 PM - 3.2 Evaluating MPAS-based RRFS Prototype Forecasts using METplus
Michael J. Kavulich Jr., NSF-NCAR, DTC, Boulder, CO; and K. M. Newman, M. A. Harrold, C. Evans, and D. Dowell

2:15 PM - 3.3 Evaluating the Impact of Non-Gaussian Radar Data Assimilation in a Warn-on-Forecast System
Patrick Skinner, CIWRO, Norman, OK; and T. A. Jones and L. J. Wicker

2:30 PM - 3.4 Evaluating Model Performance with Convective Storms with a Cloud Tracking Algorithm
Sean W. Freeman, University of Alabama in Huntsville, Huntsville, AL; and J. D. Goode

2:45 PM - 3.5 Using Machine Learning to Improve Ensemble Quantitative Precipitation Forecast Displacement and Intensity Errors
Tyreek JaQuan Frazier, Iowa State University, Ames, IA; and A. Pathak, W. A. Gallus Jr., K. J. Franz, and S. Dutta

Session 3 - Forecasting Tools and Techniques across Weather Scales: Atmospheric Rivers

1:45 PM - 3:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Tomer Burg / CIRES @ NOAA/NWS/WPC/HMT / College Park, MD / Diana Rose Jesberg / ESRL / Boulder, CO

1:45 PM Developing the prototype MPAS-JEDI basedAtmospheric River Prediction System (AR-PS)
Guoqing Ge, CIRES/NOAA GSL, LONGMONT, CO; CIRES and NOAA/OAR/GSL, Boulder, CO; and S. Pan, H. Lin, K. C. Eure, S. Trahan, J. Beck, and T. T. Ladwig

2:00 PM An Evaluation of TEMPO Microphysics In MPAS for Atmospheric-River Prediction
Anders A. Jensen, NOAA, Boulder, CO; NOAA, Boulder, CO; and G. Ge, M. J. Barlage, C. Evans, and J. Olson

2:15 PM - 3.3 The Influence of Atmospheric Rivers on Northeast U.S. Extreme Precipitation
Kristen L. Corbosiero, Univ. at Albany, Albany, NY; and E. Belkin, R. D. Torn, and J. Cordeira

2:30 PM - 3.4 Using Self-Organizing Maps to Identify Synoptic-Scale Patterns Associated with Northeast U.S. Extreme Precipitation Events
Evan Andrew Belkin, State University of New York at Albany, Albany, NY; and K. L. Corbosiero, R. D. Torn, and J. Cordeira

2:45 PM Drivers of Extreme Streamflow During a Snowmelt-Enhanced Heavy Rainfall Event in the Catskill Mountains of New York
Christopher Gilberti, University at Albany, Cortlandt Manor, NY

Session 3 - Parameterizations of Radiative Processes and the Role of Artificial Intelligence

1:45 PM - 3:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / Jet Propulsion Laboratory, California Institute of Technology / Pasadena, CA

Cochairs: Daniel Feldman / LBNL / Berkeley, CA / Aaron J. Hill / University of Oklahoma, School of Meteorology / Norman, OK

1:45 PM - 3.1 Machine Learning Emulation of 3D Shortwave Radiative Transfer for Shallow Cumulus Cloud Fields

Jui-Yuan Christine Chiu, Colorado State University, Fort Collins, CO; and S. Pfreundschuh, O. Pierpaoli, J. J. Gristey, T. Yamaguchi, G. Feingold, and W. I. Gustafson Jr., PhD

2:00 PM - 3.2 Machine-Learned Emulator for 3D Radiative Transfer Effects from 1D Solutions
Jonas Albert Actor, Sandia National Laboratories, Albuquerque, NM; and J. L. Torchinsky and B. R. Hillman

2:15 PM - 3.3 Prediction of Solar Variability by Cloud Type and Cloud Cover
Kelly A. Balmes, CIRES, Boulder, CO; NOAA GML, Boulder, CO; and L. D. Riihimaki, J. Sedlar, D. D. Turner, and K. O. Lantz

2:30 PM - 3.4 Using Artificial Neural Networks to Improve Near-Real-Time CERES Surface Solar Radiative Fluxes
Kuanman Xu, LRC, Hampton, VA; and P. Sawaengphokhai, H. Winecoff, J. Garg, and P. W. Stackhouse Jr.

2:45 PM - 3.5 A Latent Twin Architecture for Clear-Sky State Retrieval from Infrared Satellite Acquisitions
Michele Martinazzo, Univ. of Bologna, Bologna, BO, Italy; and C. Sgattoni, M. Menarini, C. Zugarini, L. Sgheri, and T. Maestri

Session 3A - Frontiers of AI/ML in Severe Local Storms II

1:45 PM - 3:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Alec Prosser / Norman, OK / Ryan A. Sobash / National Center for Atmospheric Research / BOULDER, CO

1:45 PM - 3A.1 Environment-Based Prediction of Severe Hazard Intensities in SPC Mesoscale Convective Discussions Using Random Forests
Grant Christian Talkington, Cooperative Institute for Severe and Hazardous Weather Research and Operations, University of Oklahoma, Norman, OK; and C. Karstens, D. R. Harrison, and I. L. Jirak

2:00 PM - 3A.2 Integrating MRMS Radar Products and Random Forest Method for Hail Detection and Nowcasting in North America
Chris Jing, EC, Coquitlam, BC, Canada; and D. Brunet and S. Boodoo

2:15 PM - 3A.3 Automated 3D Identification and Classification of Mesoscale Polarimetric Signatures: A Neuro-Fuzzy Approach for Phased-Array Radar
Edwin Lee Dunnavan III, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and A. A. Alford, T. J. Schuur, and A. E. Reinhart

2:30 PM - 3A.4 Low-Latency Multi-Spectral Geostationary Satellite Convection Nowcasting with CREDIT WXFormer
Kevin Yang, NCAR, Boulder, CO; and D. J. Gagne II, C. Becker, C. Spallone, and S. J. Greybush

2:45 PM - 3A.5 Leveraging Machine Learning to Study Storms for Scientific Discovery
Kristen Lani Rasmussen, Ph.D., Colorado State University, Fort Collins, CO; and H. Yu, I. Ebert-Uphoff, L. Ver Hoef, and E. A. Sherman

Session 3B - From Lines to Complexes: Advances in Organized Convective System Research I

1:45 PM - 3:00 PM | Hall of Ideas - HI (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Edward C. Wolff / Univ. of Illinois Urbana-Champaign / Urbana, IL / Andrew Wade / Univ. of Oklahoma / Norman, OK

1:45 PM - 3B.1 Recipe for Confusion: Challenges and Limitations of the Ingredients-Based QLCS Tornado Forecast Paradigm
Matthew C. Brown, NSSL, Norman, OK; and M. C. Coniglio

2:00 PM - 3B.2 When Lines Meet Cells: Processes Driving Tornado Development in HSLC QLCS-Supercell Interactions
Joshua S. Ostaszewski, Texas Tech University, Lubbock, TX; and C. C. Weiss

2:15 PM - 3B.3 Analyzing the Relationship Between Updraft Tilt and Shear/ColdPool Intensity in Quasi-Linear Convective Systems
Kate Olivia Stapleton, University of Nebraska-Lincoln, Austin, TX; and M. Van Den Broeke

2:30 PM - 3B.4 An Assessment of Mesovortices in Quasi-Linear Convective Systems From 2013-2023 Using GridRad-Severe
Hanna J McDaniel, University of Oklahoma, Norman, OK; and A. J. Hill and C. R. Homeyer

2:45 PM - 3B.5 Variability in High-Shear, Low-CAPE QLCS Environments in the Southeastern U.S.
Zachary Andrew Chalmers, SPC, Norman, OK; and M. D. Parker

Poster - 22nd Conference on Mountain Meteorology Posters I

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 22nd Conference on Mountain Meteorology

Submitter: Lisa Harris / Boston, MA

3:00 PM - Poster 44 Variability of Drivers of Wildfires in Mountainous Terrain during the 2019 and 2020 United States Wildfire Seasons
Aditya Kumar, CIMSS, Madison, WI; and C. Schmidt

3:00 PM - Poster 45 Recent Developments in the NSF NCAR Integrated Sounding System and LOTOS in support of Mountain Meteorology
William O.J. Brown, NCAR, BOULDER, CO

3:00 PM - Poster 46 Overview of Observations from the Jeff Dozier Snow Study Site on Mammoth Mountain, CA
Manda B. Chasteen, Leidos, Reston, VA; and E. Bair, N. Griessbaum, B. C. Wallace, L. Zeller, S. Carter, and R. Davis

3:00 PM - Poster 47 Evaluating Orographic Precipitation Biases and Observational Uncertainty Using S2noCliME Observations and WRF Simulations over the Park Range of Colorado
Troy Justin Zaremba, University of Washington, Seattle, WA

3:00 PM - Poster 48 A S2NOCliME Case Study: Investigating Moisture Transport Impacts on Orographic Precipitation Processes in the Park Range of Colorado Using Radar and WRF Simulations
Sofia Vakhutinsky, University of Washington, Seattle, WA

3:00 PM - Poster 49 Multi-ridge Effects on Cool-season Orographic Precipitation in the Wasatch Range, Utah, USA
Michael Wasserstein, Department of Atmospheric Sciences, University of Utah, Salt Lake City, UT; and J. Steenburgh and D. E. Kingsmill

3:00 PM - Poster 50 Dynamical Moisture Transport Patterns Affecting Storm Peak Laboratory, Colorado, during S2noCliME
Crista Aurora Castillo, University of Michigan, Ann Arbor, MI; and A. Raudzens Bailey, C. Pettersen, A. G. Hallar, M. A. Garcia, I. B. McCubbin, G. G. Mace, L. A. McMurdie, and A. Rowe

3:00 PM - Poster 51 Multiscale Radar Analysis of Precipitation Systems over the Complex Topography of Northern Utah during the SNOWSCAPE2026 Field Campaign
Zackery Zounes, University of Utah, Salt Lake City, UT; United States Air Force, Wright-Patterson Air Force Base, OH

3:00 PM - Poster 52 Machine Learning-based Synoptic Patterns for Mountain and Coastal Types of Orographic Precipitation in Korea
Junho Kang, Seoul National University, Seoul, 41, South korea; and J. H. Kim, Y. J. Park, J. H. Kim, Y. Shin, and B. Lim

3:00 PM - Poster 53 Momentum in QUIC: Generating High-Fidelity Training Sets for Learning Winds in Complex Terrain
David Robinson, LANL, Santa Fe, NM; LANL, Los Alamos, NM; and J. Slaten, S. Coffing, S. Brambilla, and A. Mohan

3:00 PM - Poster 54 Optimal Placement of Snow Survey Stations for Basin-Scale Snow Estimation
Brendan C. Wallace, Leidos, Reston, VA; and N. Griessbaum, M. Chasteen, E. Bair, L. Zeller, S. Carter, R. Davis, and E. Deeb

Poster - 30th Conference on Numerical Weather Prediction Monday Posters

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Lisa Harris / Boston, MA

3:00 PM WITHDRAWN
Withdrawn: Toward ML-Based Emulation of Micrometeorological Dispersion and Eddy Covariance Flux Footprints In Solar Arrays Atharva Tyagi, Ridge High School, Basking Ridge, NJ; and E. Mather and A. R. Desai Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM - Poster 56 Evaluation of Multi-Month Forecasts By Two NOAA Weather Forecast Models over the USA: AI Vs. Traditional NWP
Qihang Liu, Huron High School, Ann Arbor, MI; and X. Huang

3:00 PM - Poster 57 Using a Fully Coupled Atmosphere–Lake Model and Machine Learning to Simulate Lake-Effect Snowstorms over the Great Lakes Region and Understand Their Physical Drivers
Yun Qian, Pacific Northwest National Laboratory, Richland, WA; PNNL, Richland, WA

3:00 PM - Poster 58 Impact of Assimilating New York State Mesonet Observations on a 1-km WoFS for the 18-19 August 2024 Long Island Flash Flood Event
Derek R. Stratman, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and J. L. Tweedie, M.S., N. Yussouf, B. C. Matilla, C. A. Kerr, Y. Wang, and C. L. Yu

3:00 PM - Poster 59 Assessing the Impact of AR Recon Dropsonde and Flight-level Data on Forecasts of the December 2025 Pacific Northwest Precipitation Event
Minghua Zheng, Center for Western Weather and Water Extremes, Scripps Institution of Oceanography, Univ. of California San Diego, La Jolla, CA; and H. Jing, M. C. Johncox, J. Wang, A. Wilson, S. Yoon, PhD, Z. Zhang, M. Warner, L. Delle Monache, Z. Pu, and F. M. Ralph

Poster - 34th Conference on Weather Analysis and Forecasting Monday Posters

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Lisa Harris

3:00 PM - Poster 74 Use of Aerosol Wind Profiler Doppler Wind Lidar Profiles to Analyze GOES Atmospheric Motion Vectors and Their Height Assignment
Kristopher M. Bedka, LRC, HAMPTON, VA; and J. Cooney, K. Khlopenkov, J. M. Daniels, J. Apke, Y. J. Noh, W. Bresky, J. Carr, and H. Madani

3:00 PM - Poster 75 Analysis of Extreme Rainfall and Streamflow Events over the Catskill Mountains
Justin R Minder, Univ. at Albany, Albany, NY; and A. Ellingworth and J. Cordeira

3:00 PM - Poster 76 Predictability of Extreme Rainfall Events in the Catskill Mountains of New York
Chenhong chen Rao, University at Albany, Albany, NY; and B. H. Tang, PhD, J. R. Minder, J. Cordeira, and C. Gilberti

3:00 PM - Poster 77 Spatial Characteristics of Rainfall Associated with Tropical Cyclones Weakening over Open Ocean
Ander Sarai Ortiz, The University of Arizona, Tucson, AZ; and K. Wood and A. Toms

3:00 PM - Poster 78 A Deep Learning ML Framework for Nowcasting Flash Flood Warning Issuance
Nathan A. Snook, CAPS, Norman, OK; and E. Chladny, A. Berrington, and M. Xue

Poster - Cloud Physics Poster I

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

3:00 PM - Poster 9 Multi-Frequency Radar Estimates of Rainfall Rate and Drop Size at CAPE-k
Matthew D. Lebsock, JPL, Pasadena, CA

3:00 PM - Poster 10 Southern Ocean Shallow Cloud Precipitation Properties: Diurnal Cycle and Seasonal Variability
Dao Wang, University of Utah, Salt Lake City, UT; and G. G. Mace

3:00 PM - Poster 11 Understanding Southern Ocean Low-Cloud Optical Depth Feedbacks Using CAPE-k Observations
Kyndra Buglione, University of Washington, Seattle, WA; and R. Marchand

3:00 PM - Poster 12 Satellite-Derived Arctic Precipitation Climatology for Arctic Sea Ice Melt Onset
Ash Gilbert, University of Colorado at Boulder, Boulder, CO; CIRES, Boulder, CO; and J. Kay and H. Chepfer

3:00 PM - Poster 13 Microphysical Characteristics of Ice Particles in the Eyewall of Tropical Cyclones simulated by a Habit Prediction Scheme
Tempei Hashino, Kochi University of Technology, Kami, Kochi, Japan; and Y. Hirata, X. Zhang, and G. J. Tripoli

3:00 PM - Poster 14 Cloud Radiative Feedback as a Driver of Arctic Amplification: Insights from Cloud-Locking Experiments
Hailong Wang, PNNL, Richland, WA; and Y. Huo and B. Harrop

3:00 PM - Poster 15 Simulating Microphysical Processes in Hurricane Analysis and Forecast System
Xuejin Zhang, AOML, Miami, FL; and X. Yang and J. A. Zhang

3:00 PM - Poster 16 The Dependence of Cloud Phase Transitions and Precipitation over the Southern Ocean on Environmental Factors: Results from CAPE-k
Erika Elizabeth Pruitt, University of Oklahoma, Norman, OK; CIWRO, Norman, OK; and G. M. McFarquhar and Y. Huang

3:00 PM - Poster 17 The Impact of Lower Tropospheric Clouds on Seasonal Anomalies and Trends of Wintertime Radiative Heating Rates in the Arctic
Sohom Gupta, McGill University, Montréal, QC, Canada; and Y. Huang

3:00 PM WITHDRAWN
Poster # 18 Withdrawn: Characterization of Entrainment-Induced Droplet Clustering in Laboratory Generated Clouds Michael L Larsen, College of Charleston, Charleston, SC; Michigan Technological University, Houghton, MI; and S. P. Singh, J. C. Anderson, H. Fahandezh sadi, J. Yeom, W. H. Cantrell, and R. A. Shaw Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM - Poster 19 Hemispheric Asymmetry in Extratropical Marine Cyclones as Constraint on ERFaci
Grant Schlaff, Univ. of Wyoming, Laramie, WY; Univ. of Wyoming, Laramie, WY; and D. McCoy

3:00 PM - Poster 20 Eddy Fluxes and Atmospheric Stability in LES-LCM Simulations of Fog-Induced Turbulence
Rodrigo Rodakoviski, University of Notre Dame, Notre Dame, IN; and D. H. Richter

3:00 PM - Poster 21 Droplet Clustering in Shallow Convective Clouds in the Trade Wind Region of the Atlantic
Eberhard Bodenschatz, Max Planck Society, Goettingen, NI, Germany

3:00 PM - Poster 23 **Impacts of Atmospheric Rivers in Central Greenland: Snowfall, Clouds, and Atmospheric State**
Alanna Emma Wedum, University of Michigan-Ann Arbor, Ann Arbor, MI; and C. Pettersen, K. S. Mattingly, M. D. Shupe, H. Guy, and M. Gallagher

3:00 PM - Poster 24 **Turbulence Effects on Hydrometeor Fall Speeds and Collisions in Warm Clouds: WRF and Idealized KiD-A simulations**
Ming Zhu, University of Oklahoma, Norman, OK

3:00 PM - Poster 25 **Broadening of Adiabatic Droplet Spectra through Stochastic Condensation: 3D Cloud Model and 1D Adiabatic Updraft Model Simulations**
Wojciech W. Grabowski, NCAR, Boulder, CO; and K. K. Chandrakar, PhD and H. Morrison

3:00 PM - Poster 26 **Interacting Aerosol, Microphysical, and Dynamical Controls on Cloud Evolution During an Arctic Marine Cold-Air Outbreak**
Mikhail Ovchinnikov, PNNL, Richland, WA; and P. Wu, P. Blossey, T. W. Juliano, J. Tian, H. Xiao, A. Varble, PhD, and J. Fast

3:00 PM - Poster 27 **Super Droplet Cloud Microphysics in the Energy Research and Forecasting Model**
Hassan Beydoun, LLNL, Livermore, CA; and D. Ghosh, E. K. de Jong, D. McGuffin, C. Kendrick, J. lee, D. Gardner, and K. Lundquist

3:00 PM - Poster 28 **Constraints on Marine Boundary Layer Moisture Sources Using Stable Isotope Measurements in Water Vapor During the SCILLA Stratocumulus Airborne Experiment**
Lisa Welp, Purdue University, West Lafayette, IN; and P. Chuang, N. Sheena Sunil, M. K. Witte, A. Molino, and A. Bucholtz

3:00 PM - Poster 29 **Geometry-Aware Particle Identification from Non-Collocated Multi-Frequency Radar Observations during WINTRE-MIX**
Jairo Michael Valdivia Prado Sr., University of Colorado Boulder, Boulder, CO

3:00 PM - Poster 30 **Immersion Freezing of Dilute Solution Drops with Biological and Mineral Ice Nuclei: Implications for Ice Crystal Growth**
Hailey Zangara, Penn State University, Basking Ridge, NJ; and D. Rui, A. M. Moyle, and J. Y. Harrington

3:00 PM - Poster 31 **Marine Boundary Layer Cloud Evolution along an Arctic Cold Air Outbreak: Results from CAESAR Airborne Observations and Simulations**
Zeqian(Hazel) Xia, Cooperative Institute for Severe and High Impact Weather Research and Operations (CIWRO), Norman, OK; School of Meteorology, University of Oklahoma, Norman, OK; and G. M. McFarquhar, Y. Huang, B. Geerts, J. Haggerty, H. Vömel, S. Woods, and P. Zuidema

3:00 PM - Poster 32 **Examining the Weak Temperature Gradient Approximation in Reanalysis Data**
Vaidehi Ramachandrula, Brookfield Central High School, Brookfield, WI; and S. N. Stechmann

3:00 PM - Poster 33 **Sacrificing Complexity for Versatility: Building a Simplified Processing Script for Global Field Campaign Data.**
Jonathan Connor Douglas, The University of Oklahoma/CIWRO, Norman, OK; and G. M. McFarquhar, Y. Huang, P. A. Brechner, MS, A. M. Dzambo, J. Moreno, C. Knowles, and C. Kotlarz

3:00 PM - Poster 34 **Temperature Profiling in the Korea Cloud Physics Experimental Chamber (K-CPEC): Evaluation of Spatial Temperature Uniformity**
Sungmin Park, Pusan National University, Busan, Korea, Republic of (South); and J. Kim, M. Park, S. Jang, J. W. Cha, Y. S. Oh, M. Belorid, B. Y. Kim, S. Kim, S. S. Yum, J. Lee, and J. Um

3:00 PM - Poster 35 **A Closed to Open-Celled Mixed-Phase Cloud Transition Observed Over the Nordic Seas Under High Aerosol Loading**
Samuel Ephraim, Rosenstiel School of Marine, Atmospheric, and Earth Science, Miami, FL; and P. Zuidema, A. Bansemer, L. Cai, O. Cruikshank, S. Eckhardt, N. Evangeliou, R. Foskinis, J. R. French, B. Geerts, C. Grasmick, V. Kumar, S. Henning, A. Massling, G. M. McFarquhar, G. Noer, M. D. Petters, E. M. Rosky, L. L. Sorensen, A. Nenes, H. Skov, J. R. Snider, T. Tatro, Z. Wang, S. Woods, and L. Zhang

3:00 PM - Poster 36 **Cryo-SEM Imaging of Lab-Grown Cirrus Analog Ice Crystals: TCNJ ‘Cloud-In-A-Box’**
Nathan B. Magee, The College of New Jersey, Ewing, NJ; and E. J. Jansen, G. A. Lampmann, E. K. Ragbir, and M. Dhiman

3:00 PM - Poster 37 **An Analysis of the Relationship Between Ice Microphysics and Radar Reflectivity in Various Wintertime Cloud Structures During IMPACTS**
Kaylee Heimes Lundstrom, Univ. of Illinois Urbana-Champaign, Urbana, IL; and S. W. Nesbitt, R. M. Rauber, and J. Finlon

3:00 PM - Poster 38 **Isolating the Effect of Chosen Ice Nucleation Framework on Ice Crystal Budgets in Cold-Air Outbreak Clouds**
Yijia Sun, Stony Brook University, Stony Brook, NY; Stony Brook University, Stony Brook, NY; and F. Tornow, A. M. Fridlind, N. Riemer, and D. A. Knopf

3:00 PM - Poster 39 **Design of a Polarimetric Imaging System for Laboratory Cloud Droplet Size Distribution Characterization**
Zaid Bakri, Michigan Technological University, HOUGHTON, MI; and C. Mazzoleni, R. A. Shaw, and D. J. Miller

3:00 PM - Poster 40 **Multi-Instrument Observations of Aerosol-to-Cloud Particle Transition in the Korea Cloud Physics Experimental Chamber (K-CPEC)**
Jeonggyu Kim, Pusan National University, Busan, Busan, Korea, Republic of (South); and S. Park, M. Park, S. Jang, J. W. Cha, Y. S. Oh, M. Belorid, B. Y. Kim, S. Kim, S. S. Yum, J. Lee, and J. Um

3:00 PM - Poster 41 **New Pathways for Secondary Ice Production**
Andrew J. Heymsfield, NCAR, Boulder, CO; and A. Bansemer, A. DeLaFrance, and V. T. J. Phillips

3:00 PM - Poster 42 **Retrievals of Giant Cloud Condensation Nuclei (GCCN) in Marine Boundary Layers**
Virendra P. Ghate, ANL, Lemont, IL; and A. G. Carlton, D. B. Mechem, D. D. Turner, Y. Feng, and T. Subba

3:00 PM - Poster 43 **Connecting cloud dynamics to lightning observations with a community Lightning Modeling Grand Challenge**
Eric C. Bruning, Texas Tech University, Lubbock, TX; Texas Tech University, Lubbock, TX; and A. Back, S. Behnke, T. Edwards, S. Goodman, T. J. Lang, and J. Tilles

Poster - Eighth Conference on Weather Warnings and Communication Posters: Monday

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

Submitter: Lisa Harris

3:00 PM - Poster 79 **An Update on the Storm Surge Watch/Warning Product and Hazard Services**
Josh Thompson, CIRA, Boulder, CO

3:00 PM - Poster 80 **Toward an Operational Decision Arc for Severe Thunderstorm Hazards: Extending a User-Centered Framework for Warning Communication and Preparedness**
Kevin Ash, University of Florida, Gainesville, FL; and C. Williams

3:00 PM - Poster 81 **How Can We Create Meteorological Tools for Both the Scientist and the Non-expert?**
Isabelle Bogot, University of Wisconsin-Madison, Madison, WI

3:00 PM - Poster 82 **Authentic Meteorology Learning Experiences: Student Use of Analogies to Describe Convective Storms**
Courtney Sheckler, Temple University, Philadelphia, PA; and T. Shipley, P. M. Schneider, T. M. Bals-Elsholz, K. H. Goebbert, and P. M. McNeal

3:00 PM - Poster 83 **Authentic Meteorology Learning Experiences: How do Emotions Affect Memory and Learning During a Convective Field Study?**
Patrick Michael Schneider, Towson University, Towson, MD; and T. F. Shipley, C. Sheckler, T. M. Bals-Elsholz, K. H. Goebbert, and P. M. McNeal

3:00 PM - Poster 85 **How Does the Atmospheric Science Community Use Case Studies to Teach Mesoscale Meteorology?**
Casey Davenport, PhD, University of North Carolina at Charlotte, Charlotte, NC; and Z. J. Handlos, A. M. Klees, and D. M. Kopacz

Poster - POSTER Beyond Warnings: Exploring Effective Lead Time and Mesoscale Convective Messaging

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 60 **Communication and Validation of Real-Time Probabilistic Guidance for Severe Weather Hazards**
Patrick Adrian Campbell, OU/CIWRO, Norman, OK; and R. B. Steeves, C. N. Satrio, K. M. Calhoun, T. Sandmael, K. L. Manross, and K. L. Berry

Poster - POSTER Frontiers of AI/ML in Severe Local Storms

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 61 **Training Wofscast to Predict Derived and Observed Variables**
Corey K. Potvin, NSSL, Norman, OK; and J. A. Duarte and Z. Hua, PhD

3:00 PM - Poster 62 **What Features Matter When Generating Medium-Range Mpas-Based Convective Hazard Forecasts with Neural Networks?**
Ryan A. Sobash, National Center for Atmospheric Research, BOULDER, CO; and D. A. Ahijevych and C. S. Schwartz

3:00 PM - Poster 63 **Mechanistic Machine Learning for Scientific Discovery in Tornadogenesis**
Ryan Martz, University of Oklahoma, Norman, OK; NSF AI Institute for Research on Trustworthy AI in Weather, Climate, and Coastal Oceanography Univ. of Oklahoma (AI2ES), Norman, OK; and A. H. Fagg, A. Berrington, A. McGovern, C. K. Potvin, M. Xue, and N. A. Snook

3:00 PM - Poster 64 **Probabilistic Convection Initiation Nowcasting from Geostationary Satellite Observations using Deep Learning**
Christian Spallone, Penn State University, State College, PA; and S. J. Greybush, D. J. Gagne II, C. Becker, and K. Yang

3:00 PM - Poster 65 Automatic Detection of Mesoscale Convective Systems in Remote Sensing Data Based on Deep Learning Models:an Example for European Russia
Mikhail Krinitskiy, Shirshov Institute of Oceanology, Russian Academy of Sciences, Moscow, Moscow, Russian Federation; and A. I. Narizhnaya and A. Chernokulsky

3:00 PM WITHDRAWN
Withdrawn: Using a Physics-Informed Neural Network for Single Doppler Wind Retrievals Dylan Reif, Cotality, Irvine, CA Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM - Poster 66 Using ML to Identify Surface Features Involved in Convection Initiation over the Houston Area
Michael Self, Texas A&M Univ., College Station, TX; and C. J. Nowotarski, B. J. Mortazavi, S. Huang, M. Sharma, A. D. Rapp, and S. D. Brooks

Poster - POSTER High-Temporal-Resolution Radar Perspectives on Severe Convection

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 67 Tornado Debris Signature Characteristics Observed by a Phased-Array Radar
A. Addison Alford, NSSL, Norman, OK; and J. T. Carlin, A. W. Lyza, and A. E. Reinhart

3:00 PM - Poster 68 Using NSSL's Advanced Technology Demonstrator to Observe Precursors to Localized Heavy Rainfall
Emily Blumenauer, OU/CIWRO, Norman, OK; and A. A. Alford, D. J. Bodine, P. E. Kirstetter, R. A. Clark III, and A. E. Reinhart

3:00 PM - Poster 69 Close-Range RaXPol Observations of Numerous Tornadoes on 05 June 2025 Near Morton, TX
Robby M. Frost, Univ. of Oklahoma, Norman, OK; and D. J. Bodine, H. B. Bluestein, A. Schueth, A. Naik, J. Gebauer, and M. C. Coniglio

Poster - POSTER Societal, Economic, and Industrial Impacts of Severe Local Storms

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 70 Improving Understanding of Tornado Impact on Buildings
Shanshan Shi, Missouri University of Science and Technology, Rolla, MO; and G. Yan

3:00 PM - Poster 71 From Hail Swaths to Residential Roof Impacts: Footprint, Event Concentration, and Recent Change in a 15-Year Climatology
Dan M. Stechman, HailTrace, Edmond, OK; and B. Baranowski, MS and J. Choquette

3:00 PM - Poster 72 Spatial Characteristics of Tornado Exposure associated with Multi-Unit Dwellings in the Eastern U.S.
Cooper Corey, University of Tennessee, Knoxville, TN; and K. N. Ellis

3:00 PM - Poster 73 Evaluating Image-Based and Profilometric Methods for Measuring Hail Damage on Asphalt Shingles
Christopher Sanders, IBHS, Richburg, SC; Insurance Institute for Business and Home Safety, Richburg, SC; and I. M. Giammanco

Poster - Posters: Atmospheric Radiative Transfer and Light Scattering Theory

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / JPL / Pasadena, CA

Chair: Xiquan Dong / The Univ. of Arizona / Tucson, AZ

3:00 PM - Poster 1 Temperature-Dependent Refractive Index Compilations for Ice and Liquid Water from Ultraviolet to Microwave
Siheng Wang, Texas A&M University, College Station, TX; and P. Yang

3:00 PM - Poster 2 Spheroid Optical Properties under the Discrete-Dipole Approximation (DDA) and Separation of Variables Method (SVM).
Neil Harris Cutting, University of Texas A&M, College Station, TX; and P. Yang and S. Wang

Poster - Posters: Observations of the Far Infrared: from PREFIRE to FORUM

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Cochairs: Kyle S Mattingly / University of Wisconsin–Madison / Madison, WI / Xianglei Huang / University of Michigan / Ann Arbor, MI

3:00 PM - Poster 3 Using Orbital Far-Infrared Observations to Inform a Cloud Mask Algorithm

Tim Michaels, University of Wisconsin - Madison, Madison, WI; and B. Kahn, K. S. Mattingly, N. Miller, Q. Yue, and C. Bean

3:00 PM - Poster 4 Applications of PREFIRE Data to Far Infrared Radiation In High Altitude Environments at Low Latitudes
Aronne Merrelli, University of Michigan, Ann Arbor, MI; and D. Feldman, N. Nguyen, K. S. Mattingly, X. Huang, X. Chen, B. Drouin, and T. S. L'Ecuyer

3:00 PM - Poster 5 PREFIRE Radiance Comparisons to CrIS and IASI: Implications for the Level 2 ATM Retrieval of Column Water Vapor
Nathaniel Miller, CIMSS, Madison, WI; and A. Merrelli, L. Borg, and Q. Yue

3:00 PM - Poster 6 The PREFIRE Anc-Sat Product: Development and Applications of CubeSat Collocations with Other Satellite Data
Kyle S Mattingly, University of Wisconsin–Madison, Madison, WI; and C. Bean, X. Chen, B. Drouin, X. Huang, B. Kahn, T. S. L'Ecuyer, A. Merrelli, T. I. Michaels, N. Miller, and Q. Yue

Poster - Posters: Past and Future Observations of Earth's Radiation Budget: on the Road from ERBE to Libera and beyond

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / JPL / Pasadena, CA

Cochairs: Yolanda L. Shea / Hampton, VA / Odele M. Coddington / Laboratory for Atmospheric and Space Physics / Boulder, CO

3:00 PM - Poster 7 A Seasonal and Geospatial Comparison of CERES-Derived Downward Shortwave Surface Radiation Fluxes from 2018-2022
Bradley Fletcher Lamkin, University of Oklahoma, Norman, OK; and J. Redemann, C. J. Flynn, I. Chang, L. Gao, S. Kato, B. G. Illston, P. W. Stackhouse Jr., and F. Xu

3:00 PM - Poster 8 Preparing for Libera: Spatio-temporal Variability of Visible and Near-infrared Irradiance
Maria Z Hakuba, ; and P. Pilewskie and G. L. Stephens

Poster Viewing and Coffee Break

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

"Session 4 - In the Hot Seat: Power Shutoffs, Peak Heat, and Public Safety"

4:30 PM - 6:00 PM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

4:30 PM - 4.1 Public Safety Power Shutoff Notifications and Resident Decision Making: Investigating Communication Prior to and during Fire Weather Induced Power Outages
Julie Brooks, University of Oregon, Eugene, OR; University of Oregon, Eugene, OR; and A. M. Stasiewicz and A. Masciopinto

4:45 PM - 4.2 Revealing Hidden Dynamics: Exploring Unique Patterns in Fire Weather Information Use across Four "Extreme" Wildfires
Amanda M Stasiewicz, University of Oregon, Eugene, OR; and C. Tensuan and K. J. Bieber

5:00 PM - 4.3 When Official Alerts Fall Short: WatchDuty and the Future of Wildfire Public Warning On Wildfires
Christian Tensuan, University of Oregon, 1251 University of Oregon, OR; University of Oregon, Eugene, OR; and A. M. Stasiewicz

5:15 PM - 4.4 Wildfire on the Eastern Front: Fire Weather Communication in a Multi-Hazard Coastal Environment
Kaitlyn J. Bieber, University of Oregon, Eugene, OR; and A. M. Stasiewicz and C. Tensuan

5:30 PM - 4.5 NOAA NIHHIS Heat Services and User Engagement: Bridging Heat Products and Stakeholder Needs
Barbara Mayes Boustead, NOAA, Gretna, NE

5:45 PM - 4.6 Challenges in Communicating Increasingly Extreme Temperatures to the Hispanic Audience in Las Vegas.
Darielis Aguiar, Univision Las Vegas, Las Vegas, NV

Joint 4A - Beyond Warnings: Exploring Effective Lead Time and Mesoscale Convective Messaging

4:30 PM - 6:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

Submitter: Keith Sherburn / National Weather Service / Rapid City, SD

Cochairs: Abigail Lee Bitterman / Institute for Public Policy Research and Analysis, Univ. of Oklahoma / Norman, OK / Evelynn Mantia / Northern Illinois University / DeKalb, IL

4:30 PM - 4A.1 Remote Mesoanalysis Support: Measuring the Impact of Collaborative Mutual Aid on Warning Performance
Jennifer M. Laflin, NWS, Kansas City, MO; and R. A. Graham, NWS and Z. Uttech

4:45 PM - 4A.2 NWS Demonstration of a Cloud-based Warn-on-Forecast System for Operations
Patrick C. Burke, NSSL, Norman, OK

5:00 PM - 4A.3 Balancing Automation and Expertise: A Hazardous Weather Testbed Evaluation of PHI-Derived Warning Recommenders
Kristin M. Calhoun, NOAA, NORMAN, OK; and K. L. Manross, R. B. Steeves, T. Sandmael, J. Cintineo, M. Krocak, J. W. Monroe, J. G. Madden, N. R. Hardin, K. L. Berry, and P. A. Campbell

5:15 PM - 4A.4 Curating Effective Severe Weather Communication for Designated Decision Makers Using the Warn-On-Forecast System
Jackson Augustus Shumaker, University of Illinois Urbana-Champaign, Urbana, IL; and J. E. Trujillo-Falcón and S. R. Ernst

5:30 PM - 4A.5 A Real-World Warn-on-Forecast Story: Messaging to Emergency Managers during the 14 March 2025 Tornado Outbreak
Miranda K. Silcott, Cooperative Institute for Severe and High Impact Weather Research and Operations, Norman, OK; and M. Krocak, D. Hogg, P. C. Burke, and H. Obermeier

5:45 PM - 4A.6 Exploring NWS Graphical Risk Messaging within the Watch-to-Warning Gap
Sean Robert Ernst, IPPRA, Pepperell, MA; and B. J. Fellman, L. A. Harvey, M. Krocak, and J. T. Ripberger

Joint Panel Discussion 4 - Best Practices for AI Integration into Weather Forecast Operations

4:30 PM - 6:00 PM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitters: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO / Christina E. Kumler / CIRES / Boulder, CO / Aaron J. Hill / Univ. of Oklahoma / Norman, OK / Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Jack Lind / NWS / Lenexa, KS / Aaron J. Hill / University of Oklahoma, School of Meteorology / Norman, OK

Panelists: James A. Nelson / NWS / College Park, MD / Patrick C. Burke / NSSL / Norman, OK / Thea Sandmael / Cooperative Institute for Severe and High-Impact Weather Research and Operations, Univ. of Oklahoma / Norman, OK / David R. Harrison / CIWRO / Norman, OK

4:30 PM Welcoming Remarks

Session 4 - Downslope Windstorms

4:30 PM - 5:30 PM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 22nd Conference on Mountain Meteorology

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Chair: James D. Doyle / NRL / Monterey, CA

Cochairs: Sonia Wharton / Oakland, CA

4:30 PM - 4.1 Future Changes to Extreme Downslope Windstorms Across the Northern Colorado Front Range
McKenzie L. Larson, University of Colorado Boulder, Boulder, CO; and A. C. Winters, G. A. Meehl, C. A. Shields, and E. M. Dougherty

4:45 PM - 4.2 Gap Flow and Downslope Winds: The Two Flavors of Santa Ana Winds
Clifford F. Mass, University of Washington, Seattle, WA; and R. G. Fovell and D. Ovens

5:00 PM - 4.3 The Potential Role of a Downslope Windstorm and Associated Hydraulic Jump In the Formation of a Tornado-Scale Fire Whirl during the 2018 Carr Fire
Natalie Wagenbrenner, USDA, Missoula, MT; and J. Forthofer

5:15 PM - 4.4 Generation Mechanisms of Extreme Santa Ana Winds during the Catastrophic Los Angeles Fires in January 2025
Yewon Shin, Seoul National Univ., Seoul, South korea; and J. H. Kim, S. H. Kim, and J. D. Doyle

Session 4 - Meso to Subseasonal Scale Meteorological Phenomena Part II

4:30 PM - 6:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Andrew C. Winters / Michael L Brown

4:30 PM Comparing Forecast Techniques for Overlapping Tornado and Flash Flood Events
Christian McGinty, University of Oklahoma, Norman, OK; and A. J. Hill and E. R. Nielsen

4:45 PM - 4.2 Unraveling Left-Moving Supercell Environments with Self-Organizing Maps
Aaron W. Zeeb, Central Michigan Univ., Mount Pleasant, MI; and J. T. Allen

5:00 PM - 4.3 A Climatology of Coupled Tracks of Extratropical Cyclones and Mesoscale Convective Systems
Gabriel Lach, McGill University, Montreal, QC, Canada

5:15 PM **Testing an Approach to Forecasting Windstorms produced by Mesoscale Gravity Waves**
Anton Seimon, Bard College, Annandale-on-Hudson, NY; and L. F. Bosart

5:30 PM - 4.5 **The Influence of Subseasonal-to-Seasonal Modes of Variability on Medium-Range North Pacific Jet Stream Forecasts**
Stephanie A Henderson, Univ. of Wisconsin-Madison, Madison, WI; and A. C. Winters

5:45 PM - 4.6 **Subseasonal-to-Seasonal Predictability of North American Weather Regimes Using Global Wind Oscillation Phase-Space Trajectories**
Caitlin M Roufa, Northern Illinois University, Dekalb, IL; and V. A. Gensini

Session 4 - Past and Future Observations of Earth's Radiation Budget: on the Road from ERBE to Libera and beyond

4:30 PM - 6:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / Jet Propulsion Laboratory, California Institute of Technology / Pasadena, CA

Cochairs: Odele M. Coddington / Laboratory for Atmospheric and Space Physics / Boulder, CO / Maria Z Hakuba / JPL / Pasadena, CA / Yolanda L. Shea / Hampton, VA

4:30 PM - 4.1 **Earth Radiation Budget First Science Results and Lessons Learned from the First Three Decades of Satellite Observations**
Thomas H. VonderHaar, Colorado State Univ., Fort Collins, CO

4:45 PM - 4.2 **Exploring the angular dimension of Earth’s Radiation Budget with Libera**
Jake J. Gristey, CIRES, Boulder, CO; and K. S. Schmidt, D. Harber, M. Z. Hakuba, and P. Pilewskie

5:00 PM - 4.3 **Radiance-to-Irradiance Conversion for the Libera Split-Shortwave Channel**
McKenzie Hawkins, University of Colorado Boulder, Boulder, CO; Laboratory for Atmospheric and Space Physics, Boulder, CO; and J. J. Gristey, D. Feldman, and P. Pilewskie

5:15 PM - 4.4 **Correcting AIRS shortwave channel drift through SRF retrieval and comparisons to CERES and MODIS**
Daniel Zetterberg, University of Michigan, Ann Arbor, Ann Arbor, MI; and X. Huang, X. Chen, Z. Wei, D. Feldman, and Q. Yue

5:30 PM - 4.5 **Earth's Radiation Budget Observations from Space: Progress on and Risks to Continuity**
Peter Pilewskie, Laboratory for Atmospheric and Space Physics, Boulder, CO; and M. Z. Hakuba, O. M. Coddington, E. C. Richard, and G. L. Stephens

5:45 PM **WITHDRAWN**
4.6 Withdrawn: Bridging the Gap: Addressing Earth Radiation Budget Continuity via a Novel SmallSat Approach Alexander William Jarnot, NASA Langley Research Center, Hampton, VA; and P. Coleman, M. Shankar, and N. G. Loeb Madison Ballroom C (Monona Terrace Community and Convention Center) Cancelled

Session 4 - Turbulence and Entrainment

4:30 PM - 6:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Julia A Shates / SIO / La Jolla, CA

Cochairs: Kamal Kant Chandrakar / NCAR / Boulder, CO / Fan Yang / Brookhaven National Laboratory / Mastic, NY

4:30 PM - 4.1 **Turbulence and Entrainment Effects on Droplet Clustering in Shallow Cumulus Clouds**
Yewon Kim, MPI, Göttingen, Germany; and B. Thiede, E. Bodenschatz, and G. Bagheri

4:45 PM - 4.2 **Characterization of Subsiding Shells by Dual-Wavelength Radar Observations**
Bernd Mom, University of Helsinki, Helsinki, 18, Finland; and D. Moisseev

5:00 PM - 4.3 **Fewer Drops, but They Look the Same: Why Inhomogeneous Mixing Appears to Dominate in Cumulus Clouds**
Nithin Allwayin, Michigan Technological University, Houghton, MI; Stony Brook University, Stony Brook, NY; and G. Roberts, E. M. Rosky, K. Bala, A. Bansemer, K. Ranjbar, and R. A. Shaw

5:15 PM - 4.4 **Dynamics of Thermals Impinging on a Sharply Stratified Interface**
Atif Khan, IIT Bombay, Powai, Maharashtra, India; and A. Aiyer, PhD and R. Sivaramakrishnan

5:30 PM - 4.5 **How Droplet Size Distributions Age in Stratocumulus: Lagrangian Pathways in the Phase Space of Mean Radius and Relative Dispersion**
Jung-Sub Lim, NOAA, Boulder, CO; and F. Hoffmann

5:45 PM - 4.6 **What Fraction of Giant Sea Salt Particles Reach Clouds? Insights from Lagrangian Modeling**
Katherine L. Ackerman, Univ. of Hawaii at Manoa, Honolulu, HI; University of Notre Dame, Notre Dame, IN; and D. H. Richter and E. Melvin

Session 4B - High-Temporal-Resolution Radar Perspectives on Severe Convection

4:30 PM - 6:00 PM | Hall of Ideas - HI (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Morgan E. Schneider / Canadian Severe Storms Laboratory / London, ON / Canada / Sarah M. Stough / Cooperative Institute for Severe and High-Impact Weather Research and Operations, The University of Oklahoma / Norman, OK

4:30 PM - 4B.1 The RAPTOR Mobile Radar Suite: New Observing Tools at NSSL
Jeffrey C. Snyder, NSSL, Norman, OK; and P. Heinselman, D. Wasielewski, J. J. Gourley, G. Kunkle, Z. Barney, A. A. Alford, V. C. Chmielewski, M. Coniglio, M. D. Flournoy, K. Howard, A. W. Lyza, A. E. Reinhart, and J. Zhang

4:45 PM - 4B.2 On the Evolution and Structure of an Anticyclonic Tornado and an Anticyclonic Satellite Vortex in the Selden, Kansas, Supercell of 24 May 2021
Howard B. Bluestein, Univ. of Oklahoma, Norman, OK; and J. A. Margraf, T. Greenwood, S. Emmerson, J. C. Snyder, L. J. Wicker, and D. J. Bodine

5:00 PM - 4B.3 Observations and Insights from a Tornadic Supercell-QLCS Merger with the Horus Fully Digital S-band Polarimetric Phased Array Radar
Brandon K. Cohen, Univ. of Oklahoma, Norman, OK; and D. J. Bodine, J. C. Pehl, J. C. Snyder, D. Schvartzman, C. M. Kuster, A. A. Alford, T. J. Schuur, J. B. Boettcher, T. Y. Yu, and M. Yeary

5:15 PM - 4B.4 Examining Rapidly-Evolving Processes Associated with Several Hail-Producing Storms Observed by the Horus Fully- Digital Polarimetric Phased Array Radar
Laura Shedd, University of Oklahoma, Norman, OK; and D. J. Bodine, D. Schvartzman, A. E. Reinhart, and J. C. Snyder

5:30 PM - 4B.5 NSSL PAR Rapid Volumetric Update Data As a Tool to Better Understand Storm Processes
Anthony E. Reinhart, NOAA/OAR/NSSL, Norman, OK; and R. Mendoza, A. A. Alford, D. Wasielewski, K. Tuftedal, C. M. Kuster, T. J. Schuur, and S. M. Torres

5:45 PM - 4B.6 Rethinking Radar Scanning: Leveraging Phased-Array Radar to Sample Storms Differently
Kristofer S. Tuftedal, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and A. A. Alford, C. M. Kuster, T. J. Schuur, and A. E. Reinhart

Welcome Reception (Sponsored by Rain Enhancement Technologies)

6:00 PM - 7:30 PM | Rooftop Terrace (Monona Terrace Community and Convention Center) | Events

Tuesday, August 4, 2026

"Session 5 - Memorial Session: Honoring the Legacy of Prof. Yuk Ling Yung in Atmospheric Radiation"

8:30 AM - 10:00 AM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

8:30 AM Introductory Remarks

8:45 AM - 5.1 Bringing the Field of Atmospheric Radiative Transfer into the Modern Era: How Yuk Yung Organized an Entire Discipline and Continues to Impact Science and Society
Daniel Feldman, LBNL, Berkeley, CA; and X. Huang

9:00 AM - 5.2 Recent Progress Toward Exact Solutions to the Optical Properties of Spheroidal Particles
Ping Yang, Texas A&M Univ., College Station, TX; and S. Wang

9:15 AM - 5.3 New Products and Tools from Hyperspectral Infrared Sounders for Characterizing Earth's Radiation Budget
Qing Yue, California Institute of Technology, Pasadena, CA; and X. Huang, J. P. Teixeira, X. Chen, T. Wang, T. Do, T. S. Pagano, B. Lambrigtsen, H. Nguyen, and G. Chen

9:30 AM - 5.4 Aerosol Effective Radiative Forcing Increases Earth’s Energy Imbalance In Recent Decades
Tianle Yuan, UMBC JCET/NASA GSFC, Greenbelt, MD; and H. Song, R. Kramer, L. Oreopoulos, S. P. Raghuraman, R. Wood, and M. Chin

9:45 AM - 5.5 Light Scattering by Super-spheroids with Applications to Aerosols and Ice Clouds
Lei Bi, Zhejiang University, Hangzhou, Zhejiang Province, China

Session 5 - AI for Weather Research and Operations

8:30 AM - 10:00 AM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Corey K. Potvin / NSSL / Norman, CO / Kara D. Lamb / Columbia University / New York, NY

8:30 AM - 5.1 What Does the Modeling Community Need for ML/AI Training? Findings from a Nationwide ML Training Needs Assessment
Bryan Guarente, UCAR, Boulder, CO; and M. G. Cains, D. J. Gagne II, C. Becker, T. Martin, A. M. Smith, M. Barlow, N. Corbin, and J. Russell, PhD

8:45 AM - 5.2 Systematic Evaluation of NOAA’s Artificial Intelligence Modeling Suite at the National Centers for Environmental Prediction
L. Gwen Chen, EMC, College Park, MD; and J. J. Levit, J. Du, Q. Shi, A. M. Bentley, J. Wang, J. R. Carley, L. Cui, B. Cui, J. Liu, B. Fu, and H. Guan

9:00 AM - 5.3 Development of a Deep Learning Model for Hourly NBM Precipitation Forecasts
Sidney Lower, Arch Systems, Baltimore, MD; NWS, Silver Spring, MD; and E. F. Engle, D. E. Rudack, and G. S. Manikin

9:15 AM - 5.4 An AI/ML regional downscaling model based on CONUS404 dataset
Yongxin Liu, Colorado State University, Fort Collins, CO; Colorado State University, Fort Collins, CO; and H. Chen and L. Xue, PhD

9:30 AM - 5.5 Impact of Surface and Vertical Loss Weighting on Tropical Cyclone Track Prediction In AI Weather Model
Hua Chen Leighton, University of Miami, Miami, FL; NOAA, Miami, FL; and X. Zhang, Z. Zhang, and L. Zhu

9:45 AM - 5.6 Rocket Launch Probabilistic Wind Forecasts in Complex Terrain with Machine Learning
Zackery Zounes, MS, University of Utah, Salt Lake City, UT; United States Air Force, Wright-Patterson Air Force Base, OH

Session 5 - Extreme Rainfall: Processes and Prediction Part I

8:30 AM - 10:00 AM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitters: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO / Sarah Marie Trojniak / CIRES / Sterling Heights, MI / James Correia / CIRES @ NOAA/NWS/WPC/HMT / College Park, MD / Russ S. Schumacher / Colorado State Univ. / Fort Collins, CO

Cochairs: Sarah M. Trojniak / CIRES / College Park, MD / James Correia / CIRES @ NOAA/NWS/WPC/HMT / College Park, MD

8:30 AM - 5.1 Climatological and Dynamical Links Between Cut-Off Lows and Extreme Precipitation over the United States
Benjamin J. Moore, Center for Western Weather and Water Extremes, Scripps Institution of Oceanography, Univ. of California, San Diego, La Jolla, CA; and J. Cordeira and K. M. Lupo

8:45 AM - 5.2 Extreme Short-Term Precipitation in Gridded Precipitation Analyses and the CONUS404 Regional Climate Simulation
Russ S. Schumacher, Colorado State Univ., Fort Collins, CO; and A. J. Hill and A. J. Tomanek

9:00 AM - 5.3 Cold-Season Subtropical Air Mass Intrusions into the Saint-Lawrence River Valley in a Warming World: Dynamic-Thermodynamic impacts on Extreme Precipitation and Regional Climate
Florence Dion-Ladouceur, McGill University, Laval, QC, Canada

9:15 AM - 5.4 Synoptic Influences on Nocturnal Summertime Extreme Rainfall Events in the United States Northern Great Plains
Kelly M. Geiger, University of Oklahoma, Norman, OK; and A. J. Hill and R. S. Schumacher

9:30 AM - 5.5 A Database for Evaluating North American Extreme Precipitation during Atmospheric Blocking Events
Abby Lynn Hutson, University of Michigan, Ann Arbor, MI; and C. Pettersen, S. A. Henderson, and G. J. Gilcrease

9:45 AM - 5.6 Dynamically Linking Two High-Impact Weather Events
James D. Doyle, NRL, Monterey, CA; and K. Biernat, M. G. Fearon, and C. A. Reynolds

Session 5 - The In-Situ Collaborative Experiment for the Collection of Hail in the Plains (ICECHIP)

8:30 AM - 10:00 AM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Jacob Widanski / Cooperative Institute for Severe and High-Impact Weather Research and Operations / Cincinnati, OH / Lydia Kate Spychalla / The Pennsylvania State Univ. / University Park, PA

8:30 AM - 5.1 Overview of the In situ Collaborative Experiment for the Collection of Hail in the Plains (ICECHIP)
Rebecca D. Adams-Selin, AER, Papillion, NE; and J. T. Allen, V. A. Gensini, A. J. Heymsfield, B. M. Argrow, I. M. Giammanco, K. A. Kosiba, M. Kumjian, J. Wurman, A. C. Bernal Ayala, J. Brimelow, V. Chandrasekar, X. Chen, D. T. Dawson II, K. Friedrich, Y. Gao, C. R. Homeyer, J. M. Keeler, M. Kunz, Z. J. Lebo, K. Lombardo, E. R. Mansell, O. Martius, A. J. F. Protat, R. S. Schumacher, J. Soderholm, H. Vagasky, L. D. White, Y. Zhang, PhD, and C. L. Ziegler

8:45 AM - 5.2 ICECHIP: Hail...and Tornadoes
Karen A. Kosiba, University of Alabama Huntsville, Boulder, CO; University of Alabama Huntsville, BOULDER, CO; and J. Wurman and P. Robinson

9:00 AM - 5.3 Characterizing Hail Swaths Using In Situ and Post-storm Sampling During ICECHIP
John T. Allen, Central Michigan University, Mt Pleasant, MI; and I. M. Giammanco, S. N. Servey, J. Sorber, B. Meisenzahl, R. Adams-Selin, and V. A. Gensini

9:15 AM - 5.4 Comparison of Simulated and Observed Surface Rain and Hail Size Distributions in the 6 June 2025 Supercell during ICECHIP
Phillip Marmorino, Purdue University, Lafayette, IN; and D. T. Dawson II, A. James, Z. J. Lebo, and E. R. Mansell

9:30 AM - 5.5 Constraining Hail Melting and Shed Drop Size Distributions Using ICECHIP Observations and WRF Simulations
Anna James, University of Oklahoma, Norman, OK; and Z. J. Lebo, E. Mansell, D. T. Dawson II, and P. Marmorino

9:45 AM - 5.6 Advancing Hailstone 3D Modeling: A Dual-Method Evaluation of Photogrammetry and Laser Scanning Techniques
Anthony Crespo Bernal Ayala, AER, Lexington, MA; NCAR, Boulder, CO; and J. S. Soderholm, I. M. Giammanco, PhD, A. Bansemer, R. Adams-Selin, and A. J. Heymsfield

Session 5 - Turbulence and Drag over Complex Terrain

8:30 AM - 10:00 AM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 22nd Conference on Mountain Meteorology

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Cochairs: Bianca Adler / Brian J. Butterworth

8:30 AM - 5.1 The Influence of Horizontal Shear on Mountain Wave Propagation
James D. Doyle, NRL, Monterey, CA; and Q. Jiang

8:45 AM - 5.2 Modeling the Turbulence Dissipation Rate In Complex Terrain and Its Effect on Wind Turbine Wake Persistence
Adam S. Wise, LLNL, Livermore, CA; University of California, Berkeley, Berkeley, CA; and N. Wildmann, J. K. Lundquist, and F. K. Chow

9:00 AM - 5.3 Turbulence Transport and Pressure Redistribution in Katabatic Flows from Large-Eddy Simulations
Yicheng Li, University of California, Davis, Davis, CA; and H. J. Oldroyd

9:15 AM - 5.4 Energy Balance and Near-Surface Turbulence Under Contrasting Synoptic Wind Regimes in a High-Altitude Rocky Mountain Valley
Brian J. Butterworth, Cooperative Institute for Research in Environmental Sciences, CU Boulder, Boulder, CO; and B. Adler, C. J. Cox, J. D. Lundquist, T. P. Meyers, and E. Schwat

9:30 AM - 5.5 Turbulent Coherent Structures and Flux Dissimilarity Over a Forested Mountain Slope

Ting Diane Wang, University of California, Davis, Davis, CA; and H. J. Oldroyd and K. T. Paw U

9:45 AM - 5.6 Mean Momentum Balance over Non-Uniform Canopies Covering a Valley and Their Implications for Drag Representation at Local, Bulk, and Domain-Averaged Scales
Alec Petersen, University of California Irvine, Irvine, CA; and M. Placidi, M. Carpentieri, G. G. Katul, and T. Banerjee

Session 5 - Warm Clouds and Climate I

8:30 AM - 10:00 AM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Anita D. Rapp / College Station, TX / Mikael Kurt Witte / NPS / Monterey, CA / Christina S. McCluskey / NCAR / Boulder, CO

8:30 AM - 5.1 Atmospheric Control of Deforestation-Induced Cloud Regimes
Tom Dror, NOAA, Boulder, CO; and G. Feingold

8:45 AM - 5.2 Observational Constraints on Structural and Parametric Uncertainty in Simulated Aerosol-Cloud Interactions
Jacqueline M. Nugent, Univ. of Wyoming, Laramie, WY; and D. McCoy and D. Caulton

9:00 AM - 5.3 Warm-Phase Convective Invigoration from Aerosol-Cloud Interactions Inferred from Globally Distributed Aircraft Observations
Casey James Wall, Stockholm University, Stockholm, Sweden

9:15 AM - 5.4 Cloud Base Height Changes with Vertical Velocity Due to Hysteresis: Evidence from High-Resolution Lidar Observations
Fan Yang, ; and K. Lamer, P. Hou, E. Luke, Z. Zhu, Y. M. Sua, P. Kollias, and G. Feingold

9:30 AM - 5.5 Decadal Changes in Atmospheric Circulation Detected in Cloud Motion Vectors
Larry DiGirolamo, Univ. of Illinois at Urbana-Champaign, Urbana, IL; and G. Zhao, G. Zhang, Z. Wang, J. R. Loveridge, and A. Mitra

9:45 AM - 5.6 A Stereo Wind Approach to Investigating the Role of Warm Air Advection on Marine Stratocumulus Cloud Lifetimes
Stephen Windle, University of Maryland, College Park, USA, Lanham, MD; Univ. of Maryland, College Park, College Park, MD

Session 5 - Water on the Move: Advances in Flood and Surge Communication

8:30 AM - 10:00 AM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

Chair: Marinna Stopa / Oviedo, FL

8:30 AM Hurricane Evacuation Order Awareness and Efficiency: Insights from Hurricane Ian (2022), Henri (2021), and Laura/Marco (2020).
Marinna Stopa, NSF SOARS, Boulder, CO; University of Oklahoma, Norman, OK; and A. B. Schumacher and D. E. Purcell

8:45 AM - 5.2 Resident Perspectives on Flood Warning in Jakarta, Indonesia and the Potential for Forecast-Based Flood Watch using WRF-Hydro
ALIA RAHMI NASUTION, University of Illinois Urbana Champaign, Urbana, IL

9:00 AM - 5.3 When the Grid Goes Down, Neighbors Show Up: Understanding Social Capital in Western North Carolina during Hurricane Helene
Anthony Luis Corrales, University of Illinois Urbana-Champaign, Urbana, IL; and J. E. Trujillo-Falcón, Z. N. Caryl, S. T. Halvorson, J. A. Shumaker, E. De Jesus Otero, R. D. O. M. Gomes, J. Carstens, and Z. Wang

9:15 AM - 5.4 Communicating Prolonged Flood Threats: Flash Flood Warning and Duration in West Virginia's Complex Terrain
Ash Lazarus Orr, Independent Researcher, Pikesville, MD

9:30 AM - 5.5 Understandable, Localized, Accessible, Actionable – Fundamental Best Practices for Risk Communication from Lessons Learned over a Decade of Social Science Research about Flood Forecast Products
Kathryn Semmens, Nurture Nature Center, Easton, PA; and R. A. H. Carr, K. Maxfield, B. Montz, and P. Painter

9:45 AM - 5.6 Call it What it is: How AccuWeather's Tropical Rainstorm Classification Better Communicates Impacts from Tropical Systems
Daniel DePodwin, AccuWeather, Inc., State College, PA

Coffee Break

10:00 AM - 10:45 AM | Grand Terrace (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

"Session 6 - Extreme Rainfall: Processes and Prediction Part II"

10:45 AM - 12:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

10:45 AM - 6.1 Tyranny of the Tiny: Comparing CAMs to observations for tiny and mighty storms
James Correia Jr., NWS/Weather Prediction Center and CU Boulder CIRES, College Park, MD; and S. M. Trojniak

11:00 AM - 6.2 Understanding Rainfall-Temperature Relationship at Varying Spatial Scales Using Thermodynamic Variables
Mohammad Ashar Hussain, University Of Wisconsin-Madison, Madison, WI; University of Wisconsin, Madison, WI; and D. B. Wright and A. Alexander

11:15 AM - 6.3 Variability in U.S. Watershed Precipitation Forecast Skill Influencing the Potential National Expansion of FIRO
Emily A. Slinskey, Center for Western Weather and Water Extremes, UCSD/SIO, La Jolla, CA

11:30 AM - 6.4 Development toward a Proposed High-Resolution Reforecast Dataset at Convection-Allowing Scales Suitable for Extreme Precipitation Estimation: Background and Configuration
Amanda Back, NOAA, Fort Collins, CO; NOAA/OAR/GSL, Boulder, CO; and D. R. Jesberg, K. M. Mahoney, J. Beck, T. T. Ladwig, and P. N. Schumacher

11:45 AM - 6.5 Development toward a Proposed High-Resolution Reforecast Dataset at Convection-Allowing Scales Suitable for Extreme Precipitation Estimation: Evaluation
Diana Rose Jesberg, NOAA, Lakewood, CO; and A. Back and K. M. Mahoney

Session 6 - AI for Data Assimilation (Joint with Committee on Artificial Intelligence Applications to Environmental Science)

10:45 AM - 12:00 PM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Christina E. Kumler / CIRES / Boulder, CO / Fraser King / University of Wisconsin

10:45 AM - 6.1 Machine Learning Forecast Sensitivity
James D. Doyle, NRL, Monterey, CA; and D. J. Lloveras

11:00 AM - 6.2 An Adaptive U-net Architecture for Model Bias Correction on a Lorenz-96 Model
Georgios Britzolakis, NOAA/NWS/NCEP/EMC, College Park, MD

11:15 AM - 6.3 Machine Learning for Adaptive Sampling and Uncertainty Estimation of Satellite Observations
Hui Christophersen, NRL, Monterey, CA; and D. Sidoti and W. F. Campbell

11:30 AM - 6.4 Assessing Observation Type Impacts on AI Forecast Skill in the Arctic Using WeatherMesh
Christopher Riedel, PhD, Windborne Systems, Norman, OK; and S. Cavallo, J. Michaels, and J. Creus-Costa

11:45 AM - 6.5 Diffusion-Based Generative Modeling for Accelerating NASA GMAO's Atmospheric Data Assimilation System
Melissa Adrian, Purdue University, West Lafayette, IN; and T. Arcomano, A. Wikner, D. Chakraborty, S. J. Greybush, R. Todling, and R. Maulik

Session 6 - Advances in 3D Radiative Transfer in Atmospheric Radiation and Remote Sensing

10:45 AM - 12:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / Jet Propulsion Laboratory, California Institute of Technology / Pasadena, CA

Cochairs: Konrad Sebastian Schmidt / University of Colorado Boulder / Boulder, CO / Jake J Gristey / CIRES / Boulder, CO

10:45 AM - 6.1 What is the Optical Thickness of Tropical Cyclones? The Answer is in the Wings of Oxygen Absorption Lines.
Anthony B. Davis, NASA Jet Propulsion Laboratory / California Institute of Technology, Pasadena, CA; and M. J. Kurowski, M. T. Richardson, and L. Forster

11:00 AM - 6.2 Spatial and Angular Representation in a 3D Radiative Transfer Scheme for Numerical Weather Models
Ken Hirata, Laboratory for Atmospheric and Space Physics, University of Colorado Boulder, Boulder, CO; Department of Atmospheric and Oceanic Sciences, University of Colorado Boulder, Boulder, CO; and K. S. Schmidt

11:15 AM - 6.3 Cloud Geometry and Solar Zenith Angle Control 3D Radiative Effects
Margaret Powell, Columbia University, New York, NY; Columbia University, New York, NY; and C. van Heerwaarden, P. Gentine, and R. Pincus

11:30 AM - 6.4 Impact of Horizontal Resolution on Three-Dimensional Radiative Effects in the Shortwave
Jason Louis Torchinsky, Sandia National Laboratories, Albuquerque, NM

11:45 AM - 6.5 Characterization of Cloud Shadow Transition Signatures and 3D Radiatve Effects Using a Dense Pyranometer Network
Jonas Witthuhn, Leibniz Institute for Tropospheric Research, Leipzig, Germany; and H. M. Deneke, A. Macke, O. Ritter, J. Redemann, C. J. Flynn, A. A. Fakoya, B. F. Lamkin, E. D. Lenhardt, L. T. Mitchell, E. West, D. M. Romps, R. Oktem, and H. Kalesse-Los

Session 6 - Observations, Modeling, and Impacts of Ice Phase Processes

10:45 AM - 12:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Timothy J. Wagner / Univ. of Wisconsin-Madison / Madison, WI / Jeffrey R. French / Univ. of Wyoming / Laramie, WY / Samantha Turbeville / Sandia National Laboratory / Albuquerque, NM

10:45 AM - 6.1 Physics-Informed Machine Learning for Global Contrail Risk Prediction:Fusing OpenSky Flight Trajectories for Sustainable Aviation
Fathurrahman Yahyasatrio, University of Portsmouth, Guildford, SRY, United kingdom; and T. Elboghdadly

11:00 AM - 6.2 Ground-based Observations of Surface Radiative Forcing by Contrails
Timothy J. Wagner, Univ. of Wisconsin-Madison, Madison, WI; and T. Dean, T. Abbott, T. Robarge, and M. Shapiro

11:15 AM - 6.3 Sensitivity of Bulk Ice Microphysics Approaches for Nucleation and Depositional Growth on Cirrus Clouds
Julian Christopher Schima, Pennsylvania State University, University Park, PA; and H. Morrison and J. Y. Harrington

11:30 AM - 6.4 Where Did All the Ice Come From?
Paul Lawson, SPEC Inc., Boulder, CO; and A. V. Korolev, I. Heckman, P. Morris, R. J. Perkins, Z. Wang, K. S. Schmidt, and H. Maring

11:45 AM - 6.5 Secondary Ice Production Increases Condensate but Suppresses Precipitation in Simulated Subtropical Convection
Benjamin David Ascher, Free University of Berlin, Berlin, Berlin, Germany; and F. Hoffmann

Session 6 - Orographic Clouds, Precipitation, and Microphysics

10:45 AM - 12:00 PM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 22nd Conference on Mountain Meteorology

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Cochairs: Zhaoxia Pu / Univ. of Utah / Salt Lake City, UT / Ludving A Cano Fernandez / University of Nevada Reno / RENO, NV

10:45 AM - 6.1 Observed Sensitivity of Snow Microphysics in Complex Terrain to the Presence of Near-Surface Supercooled Liquid Water
Pierce Gardner, University of Utah, Salt Lake City, UT; and G. G. Mace

11:00 AM - 6.2 Multiple Mechanisms of Graupel Formation in Snow Clouds over Inland Hill Terrain from Balloon-Borne Particle-Imaging Sonde Observations
Kazuya Takami, Railway Technical Research Institute, Kokubunji-city, Tokyo, Japan; and K. Suzuki, Y. Hara, and Y. Fujimori

11:15 AM - 6.3 Contrasting Dynamical and Microphysical Controls on Mountain Snowfall
Sierra Liotta, University of Colorado Boulder, Boulder, CO; and K. Friedrich, B. Adler, and S. Matrosov

11:30 AM - 6.4 Environmental Ingredients and Microphysical Processes Associated with Heavy Rainfall In Mountainous Taiwan
Angela Rowe, Univ. of Wisconsin-Madison, Madison, WI; and I. C. Cornejo, K. S. Chung, S. K. Yang, M. M. Bell, J. C. DeHart, and W. Y. Chang

11:45 AM - 6.5 Terrain Impacts on Microphysical Processes of Deep Convective Storms in Central Argentina
Carly Night Zawlocki, University of Wisconsin-Madison, Windsor, WI; and A. Rowe and I. C. Cornejo

Session 6 - Severe Convective Storm Climatologies: Past, Present, and Future I

10:45 AM - 12:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Rachel L. Miller / Norman, OK / William Louis Faletti / CIWRO / Norman, OK

10:45 AM - 6.1 Quantifying the Role of the Dryline in the Severe Thunderstorm Climatology of the US Southern Great Plains
Michael J. Hosek, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and K. A. Hoogewind, A. J. Clark, and A. D. Justin

11:00 AM - 6.2 Impact of Climate Change on Intense Derecho Events in Convective-Permitting Pseudo-Global Warming Simulations
Josh Nathanael Schwarz, Iowa State University, Ames, IA; and W. A. Gallus Jr. and C. M. Patricola-DiRosario

11:15 AM - 6.3 Right- and Left-Moving Supercells in Europe: Climatology, Storm Structure, and Climate Change Response
Monika Feldmann, Institute for Atmospheric and Climate Science, ETH Zürich, Zürich, ZH, Switzerland; and S. Beer, M. Blanc, A. Zeeb, K. Brennan, L. Wilhelm, I. Thurnherr, O. Martius, and C. Schär

11:30 AM - 6.4 Climate-change Projections of Tornado Activity Using an Ensemble of Convection-permitting Dynamically Downscaled Simulations
Songning Wang, Univ. of Illinois, Urbana, IL; and R. J. Trapp, J. T. Allen, D. Gopalakrishnan, and E. Robinson

11:45 AM - 6.5 The Future of Cool-Season Severe Weather Outbreaks in the United States
Kyle D Pittman, Northern Illinois University, DeKalb, IL; and W. S. Ashley, V. A. Gensini, A. C. Michaelis, and A. Haberlie

Session 6 - The Unified Front: Advances in Weather Tools and Emergency Operations

10:45 AM - 12:00 PM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

Chair: Kaylie Marie Bonifer / University of Illinois Urbana-Champaign / French Lick, IN

10:45 AM - 6.1 Comparing Risk Perceptions of Weather Hazards Between Emergency Managers and the Public: Evidence from a Nationwide Survey
Kaylie Marie Bonifer, University of Illinois Urbana-Champaign, Urbana, IL; and J. E. Trujillo-Falcón, A. C. Wanless, A. L. Bitterman, and S. W. Nesbitt

11:00 AM - 6.2 New York State Weather Risk Communication Center Tools to Assess and Communicate Weather Risks
Nick P. Bassill, University at Albany, Albany, NY

11:15 AM - 6.3 Partnering with the New York State Weather Risk Communication Center: How Emergency Managers Leverage Winter Weather Products from a Statewide Operations Center
Kaitlyn R. Jesmonth, Univ. of Illinois Urbana-Champaign, Urbana, IL; and J. E. Trujillo-Falcón, M. S. Michaud, K. M. Bonifer, N. P. Bassill, and S. W. Nesbitt

11:30 AM - 6.4 Threats In Motion Is In Motion!
Gregory J. Stumpf, CIRA, Norman, OK; and D. M. Kingfield, K. L. Manross, N. R. Hardin, A. V. Bates, K. L. Berry, K. laws, J. J. Brost, K. Sherburn, R. Bowers, and K. Klockow McClain

11:45 AM - 6.5 An Update on Threats-in-Motion Iterative Research & Development: Survey Findings from Broadcast Meteorologists and Emergency Managers
Holly Obermeier, Cooperative Institute for Severe and High-Impact Weather Research and Operations, Boulder, CO; and K. L. Berry, K. L. Manross, G. J. Stumpf, M. Krocak, A. C. Wanless, J. T. Ripberger, D. Hogg, and T. Maciag

Lunch Break

12:00 PM - 1:45 PM | View Related | Eighth Conference on Weather Warnings and Communication

Joint 7 - Joint Session Honoring the Legacy of Roy Rasmussen I

1:45 PM - 3:00 PM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 25th Conference on Planned and Inadvertent Weather Modification

Submitter: Bart Geerts / Univ. of Wyoming / Laramie, WY / Chairs / Bart Geerts / Univ. of Wyoming / Laramie, WY / Lulin Xue / NCAR / Boulder, CO

Cochairs: Adele Igel / Nicholas Dawson / NCAR / Boulder, CO / Kristen Lani Rasmussen / Colorado State University / Fort Collins, CO

1:45 PM Joint 7.1
Roy Rasmussen: A Legacy of Accomplishments In Cloud Physics, Weather Modification, and Mountain Meteorology Robert M. Rauber, Univ. of Illinois Urbana-Champaign, Urbana, IL

2:15 PM Joint 7.2
Identifying and Addressing Causes for Dry Bias in NWP Model Initial Conditions with Impacts on Clouds and Precipitation Forecasts Stanley G. Benjamin, NOAA Global Systems Laboratory, Boulder, CO; and E. P. James, D. D. Turner, A. A. Jensen, T. T. Ladwig, and C. Evans

2:45 PM Joint 7.3
Ice Formation and Growth In Mixed-Phased Clouds Sarah A. Tessendorf, NSF NCAR, Boulder, CO; and L. Xue, PhD, S. Chen, and C. W. Karkkainen

Joint Panel Discussion 7 - Testbed Experiments - Past, Present, and Future

1:45 PM - 3:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Austin Coleman / NOAA / Kent H. Knopfmeier / NOAA

Panelists: Adam J. Clark / NOAA/OAR/NSSL / Norman, OK / Sarah M. Trojniak / CIRES / College Park, MD / Emily Nicole Tinney / NCAR / Boulder, CO / Jack Lind / NWS / Lenexa, KS / Thomas Galarneau / Norman, OK

1:45 PM Welcoming Remarks

Session 7 - Surface Energy Budget, Albedo, and Atmosphere-Surface Coupling

1:45 PM - 3:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / Jet Propulsion Laboratory, California Institute of Technology / Pasadena, CA / CoChair / Laura D. Riihimaki / Boulder, CO

Cochairs: Seiji Kato / NASA / Hampton, VA

1:45 PM - 7.1 Characterizing surface-cloud coupling states in the convective mixed layer using surface energy fluxes and boundary layer observations
Joseph Sedlar, CIRES, Boulder, CO; and V. Caicedo, B. Adler, and L. Bianco

2:00 PM - 7.2 An Update on Decadal Changes in Surface Solar Radiation
Martin Wild, ETH Zurich, Zürich, Switzerland

2:15 PM - 7.3 The All-Sky Radiative Response to Amazon Deforestation: The Role of Low-Level Clouds in Local Biophysical TOA Cooling
Tom Dror, NOAA, Boulder, CO; and G. Feingold

2:30 PM - 7.4 Disentangling Cloud Radiative Effects on the Albedo of Upper Colorado Snowpacks
William Rudisill, LBNL, Berkeley, CA; and D. Feldman

2:45 PM - 7.5 Trojan Horse Snowfall and Its Impact on Snowpack Albedo and Snowmelt
Daniel Feldman, LBNL, Berkeley, CA; and L. Gibson, F. Yu, M. Cowherd, A. C. Aiken, PhD, L. Riihimaki, and W. Rudisill

Session 7 - Warm Clouds and Climate II

1:45 PM - 3:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Anita D. Rapp / College Station, TX / Mikael Kurt Witte / NPS / Monterey, CA / Christina S. McCluskey / NCAR / Boulder, CO

1:45 PM - 7.1 Large-eddy simulation of aerosol competition in marine fog
David H. Richter, University of Notre Dame, Notre Dame, IN; University of Notre Dame, Notre Dame, IN; and K. H. Seh, R. Rodakoviski, G. Giacosa, R. Chang, L. Salehpoor, T. C. VandenBoer, K. Y. Huang, and S. Bardoel

2:00 PM - 7.2 How Does Measurement Scale Affect Derived Cloud Microphysical Parameters
Birte Thiede, Max Planck Institute for Dynamics and Self-Organization, Göttingen, NI, Germany; and M. L. Larsen, F. Nordsiek, O. Schlenczek, E. Bodenschatz, and G. Bagheri

2:15 PM - 7.3 Can Cloud Drop Recycling Explain Drizzle in Marine Stratocumulus?
Alessia Molino, Univ. of California Santa Cruz, Santa Cruz, CA; and M. D. Leandro, P. Chuang, L. Welp, N. Sheena Sunil, A. Bucholtz, and M. K. Witte

2:30 PM - 7.4 Lucky Collisions In Warm Stratocumulus
Kaitlyn Loftus, Columbia University, New York, NY; and M. D. Leandro, C. R. Williams, M. K. Witte, P. Chuang, A. Molino, M. van Lier-Walqui, K. K. Chandrakar, PhD, and H. Morrison

2:45 PM - 7.5 Dissipation of Shallow Cumulus Clouds in High-Resolution Large-Eddy Simulations
Yael Arieli, Weizmann Institute of Science, Rehovot, Israel; and A. Khain, O. Altaratz, E. Gavze, and I. Koren

Session 7A - Societal, Economic, and Industrial Impacts of Severe Local Storms

1:45 PM - 3:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Benjamin A. Schenkel / Norman, OK / Kyle D Pittman / Northern Illinois University / DeKalb, IL

1:45 PM - 7A.1 Estimation of Return Periods of Maximum Tornadic Windspeeds and Their Potential Future Changes for Building Codes and Standards
Robert J. Trapp, Univ. of Illinois, URBANA, IL; and J. T. Allen, M. Levitan, and S. Wang

2:00 PM - 7A.2 Characteristics of Near Surface Wind Fields During Hail-Producing Thunderstorms
Dennis Weaver, VDE Americas, San Jose, CA; and J. O. Allen and J. T. Allen

2:15 PM - 7A.3 The Center for Interdisciplinary Research on Convective Storms: A university-industry partnership to accelerate science and decision making
Daniel B. Wright, University of Wisconsin-Madison, Madison, WI; and V. A. Gensini, W. S. Ashley, A. Haberlie, A. A. L. Lang, and K. Cranmer

2:30 PM - 7A.4 Wintertime Severe Convective Storms in the United States: Climatology, Trends, and Compound Hazards
August Kaapcke LePique, University of Wisconsin-Madison, Madison, WI; and A. A. L. Lang

2:45 PM - 7A.5 Constructing Severe Convective Storm Event Sets using GCM and Reanalysis-Derived Environments Using Machine Learning
Victor A. Gensini, ; and A. Haberlie and W. S. Ashley

Session 7B - From Lines to Complexes: Advances in Organized Convective System Research II

1:45 PM - 3:00 PM | Hall of Ideas - HI (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Matthew C. Brown / Texas A&M University / College Station, TX / Anastasia Joy Tomanek / Colorado State University / Fort Collins, CO

1:45 PM - 7B.1 Analysis of QLCS Vorticity Organization Along the Cold Pool Edge Using CM1
Kyle Evan Watson, Texas Tech University, Lubbock, TX; and J. M. L. Dahl

2:00 PM - 7B.2 Impacts of Upper-Level Shear on the Formation of Deep Updraft Cores in an Ensemble of Simulated QLCSs
Edward C. Wolff IV, University of Illinois Urbana-Champaign, Urbana, IL; and R. J. Trapp and S. W. Nesbitt

2:15 PM - 7B.3 Influence of QLCS Merger Types on the Frequency and Timing of Tornado Production
Maliyah Smith, Purdue University, Carmel, IN

2:30 PM - 7B.4 Mesoscale Convective System Rear Inflow Contributions in Numerical Simulations
Dillon V Blount, Ohio University, Athens, OH; and C. Evans, R. Adams-Selin, and H. C. Vagasky

2:45 PM - 7B.5 Severe Convective Wind Events Associated with Bore-Driven Qlcss
Kevin Knupp, Univ. of Alabama in Huntsville, Huntsville, AL; and P. T. Pangle

Session/74756

1:45 PM - 3:00 PM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

1:45 PM - 7.1 Designing the Inclusive Severe Weather 101s -- Because Weather is For Everyone
Jennifer Walton, Girls Who Chase, Broomfield, CO

2:00 PM - 7.2 An Analysis of Student Approaches to Learning Severe Weather Concepts Using Gesture
Peggy M. McNeal, Towson University, Towson, MD; Towson University, Towson, MD; and T. F. Shipley, C. Sheckler, P. M. Schneider, T. M. Bals-Elsholz, and K. H. Goebbert

2:15 PM - 7.3 Who Gains the Most from Severe Storms Field Work Experiences?
Robin Tanamachi, Purdue University, West Lafayette, IN; and L. Carleton Parker, D. T. Dawson II, and M. Couillard

2:30 PM - 7.4 Generation Z and Weather Warning: Trust, Framing, and Platform Preferences for a Digital-First Generation
Rob Haswell, Digital Media, Mequon, WI

2:45 PM - 7.5 Tornado Tales: Comparing Results From Four Regions
Taylor Maciag, Cooperative Institute for Severe and High Impact Weather Research and Operations, Norman, OK; Univ. of Oklahoma, Norman, OK; Cooperative Institute for Severe and High Impact Weather Research and Operations, Norman, OK; and M. Krocak, H. Obermeier, D. Hogg, and M. K. Silcott

Poster - 22nd Conference on Mountain Meteorology Posters II

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 22nd Conference on Mountain Meteorology

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

3:00 PM - Poster 110 Modeling Orographic Cirrus Clouds Using a Suite of Regionally Refined Meshes in SCREAM
Samantha Turbeville, Sandia National Laboratory, Albuquerque, NM

3:00 PM - Poster 111 Impact of Valley Wind and Atmospheric Boundary Layer Evolution on Sublimation In a High-Altitude Mountain Valley
Bianca Adler, NOAA, Boulder, CO; and B. J. Butterworth, C. J. Cox, J. Sedlar, and V. caicedo

3:00 PM - Poster 112 NOAA's New Fire-Weather Observatories
David D. Turner, NOAA, Boulder, CO; and B. Adler, J. Sedlar, T. P. Meyers, J. Bednar, D. Costa, L. Bianco, L. D. Riihimaki, J. Kochendorfer, M. Brewer, and M. C. Mahalik

3:00 PM - Poster 113 Assessing Parametric Error in Orographic Drag Schemes With Idealized Ensemble Simulations over 2D Mountains
Georgios Thalassinos, University of Vienna, Vienna, Austria; and M. Weissmann and S. Serafin

3:00 PM - Poster 114 Advances in Physics-Informed Machine Learning in Mountainous Terrain in QUIC-Fire
Jesse Slaten, LANL, Los Alamos, NM; and D. Robinson, S. Coffing, A. Mohan, and S. Brambilla

3:00 PM - Poster 115 End-to-End Uncertainty Propagation in QPF: Understanding the Impact of QPF Variability on Flood Warning Effectiveness in Mountainous Terrain
Soyoung Sohn, University of Illinois, Urbana-Champaign, Urbana, IL; Univ. of Illinois Urbana Champaign, Urbana, IL; and A. P. Barros

3:00 PM - Poster 116 StormScape - Disentangling Topographic and Climate Change Influences in the Alpine Region
Monika Feldmann, Institute for Atmospheric and Climate Science, ETH Zürich, Zürich, ZH, Switzerland

3:00 PM - Poster 118 Convective Storm Processes Promoting Heavy Precipitation In the Sierras De Córdoba, Argentina
Stephen William Nesbitt, Univ. of Illinois Urbana-Champaign, Urbana, IL; and W. O'Brien

3:00 PM - Poster 119 Numerically Simulated Vertical Profiles of Wind Gusts Induced By Wildfires: Effects of Inflow Wind, Slope, and Fire Intensity
DongHyun Ko, Kongju National University, Gongju, Chungcheongnam-do, South korea; and H. LEE and G. Lee

Poster - 30th Conference on Numerical Weather Prediction Tuesday Posters

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Lisa Harris / Boston, MA

3:00 PM - Poster 120 Implementation of a Convective-Scale MPAS-JEDI Prototype in CWA: System Setup and Preliminary Results
Ying-Jhen Chen, Central Weather Administration, Taipei, TPE, Taiwan; and C. C. Tsai, D. S. Chen, T. C. Wu, C. S. Schwartz, J. Ban, and Z. Liu

3:00 PM - Poster 121 Initial Development of a Regional 3-to-1-km Variable-Resolution MPAS Framework for WoFS
Derek R. Stratman, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and C. A. Kerr, N. Yussouf, C. S. Schwartz, C. Liu, Y. Wang, and T. A. Jones

3:00 PM - Poster 122 The Impact of ESA-based Targeted WxUAS Observations on NWP Evaluated with OSSEs.
Millicent Tsey, Univ. of Nebraska-Lincoln, Lincoln, NE; and A. L. Houston, S. Murdzek, K. A. Swan, T. T. Ladwig, and E. P. James

3:00 PM - Poster 123 Using NWS Warnings for Verification of Short-Term Convection-Allowing Model Forecasts
Christopher A. Kerr, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and P. Skinner and D. R. Stratman

Poster - 34th Conference on Weather Analysis and Forecasting Tuesday Posters

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

3:00 PM - Poster # 166 An Observational Assessment of Local and Remote Influences on Deep Convection over the Tropical North Atlantic
Mark Delgado, The University of Arizona, Tucson, AZ; and K. Wood

3:00 PM - Poster # 167 Assessment of the Relationship between Theta-E Gradients and Upscale Convective Growth in the Great Plains of the United States.
Sophie Margaret Miller, Iowa State University, West Chester, OH; and W. A. Gallus Jr. and S. Ritter

3:00 PM - Poster # 168 Simulations of Bow Echoes with Varying Formation Mechanisms in 3-km and 1-km Grid Spacing - A Preliminary Analysis
James Surya Zinnbauer, Iowa State University, Ames, IA; and W. A. Gallus Jr.

3:00 PM - Poster # 169 The Application of RKW Theory and the Role of Misovotices in Lake Breeze Front Convection Initiation
Cassandra J. Isenberger, Texas Tech University, Lubbock, TX

3:00 PM - Poster # 170 Investigating the Relationship Between NGFS Fire Properties and WoFS Forecast Environmental Conditions
Sarah King, Cooperative Institute for Severe and High Impact Weather Research and Operations, Norman, OK; and T. A. Jones, P. S. Skinner, and J. A. Otkin

Poster - Cloud Physics Poster II

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

3:00 PM - Poster # 92 Transport of Aerosols by Cold Pool Trains
Christine A. Neumaier, Colorado State University, Fort Collins, CO; Colorado State Univ., Fort Collins, CO; and L. D. Grant, T. K. Feldman, S. M. Kreidenweis, R. J. Perkins, E. A. Stone, and S. C. van den Heever

3:00 PM - Poster # 93 New Insights into Hail Growth using Cryogenic Electron BackScattered Diffraction (EBSD)
Noah Le Brun, University of Wisconsin Madison, Madison, WI; and A. Rowe and C. Bonamici

3:00 PM - Poster # 94 Evidence for Enhanced Dynamical Efficiency of Organized Maritime Shallow Convection in Large-Domain LES

Carman Arnold, NPS, Monterey, CA; and M. K. Witte and M. Chinita

3:00 PM - Poster # 95 Particle-Based Simulations of Chemical and Physical Aerosol Processing in Marine Boundary Layers
Levin Rug, FU Berlin, Berlin, BE, Germany; and F. Hoffmann

3:00 PM - Poster # 96 Characterization of Three-Dimensional Structure of an Eastern North Atlantic MBLCC Using FLEXPART-WRF
Stephanie Braddock, Rutgers University, New Brunswick, NJ; and M. A. Miller and Q. Zheng

3:00 PM - Poster # 97 Unifying Stochastic Condensation and Entropy-Based Systems Theories for Understanding Droplet Size Distributions
Yangang Liu, Brookhaven National Laboratory, Upton, NY; and Y. Shu and R. McGraw

3:00 PM - Poster # 98 Vertical Profile Analysis of Cloud Feedbacks
Hideaki kawai, MRI, Tsukuba, Japan; and T. Koshiro and S. Yukimoto

3:00 PM - Poster # 99 Multi-Regime LASSO-ENA Shallow Maritime Cloud LES Library for the US DOE Atmospheric Radiation Measurement Eastern North Atlantic Site
William I. Gustafson Jr., PhD, PNNL, Richland, WA; and S. Giangrande, S. Endo, J. Rausch, H. Xiao, and D. Zhang

3:00 PM - Poster # 100 Cloud Coalescence Processing of CCN in Marine Low Cloud Regimes over the Eastern North Atlantic
David B. Mechem, University of Kansas, Lawrence, KS; and Z. J. Horning and V. P. Ghate

3:00 PM - Poster # 101 Simulating Marine Drizzling Stratocumulus Microphysics in the Cloud-Resolving E3SM Atmosphere Model: Comparison with ARM Observations and Sensitivity to Machine-Learning based Warm Rain Process Emulation
Lin Lin, LLNL, Livermore, CA; LLNL, Livermore, CA; and Y. Zhang, P. A. Bogenschutz, and X. Zheng

3:00 PM - Poster # 102 Droplet-Resolved Radiative Effects in Stratocumulus Clouds using the Superdroplet Method
Emma Ware, University of California, Davis, Davis, CA; Univ. of California Davis, Davis, CA; and A. L. Igel

3:00 PM - Poster # 103 Impacts of Longwave Radiative Cooling Structure on Stratocumulus Clouds
Adele L. Igel, ; and L. Nash, M. R. Igel, and A. Grimmett

3:00 PM - Poster # 104 Airborne Characterization and Mass Flux Estimations of Secondary Ice Production Mechanisms in Two Wintertime Storms from WINTRE-MIX
Eden Koval, University of Wyoming, Laramie, WY; and J. R. French, D. E. Kingsmill, K. Friedrich, and J. R. Minder

3:00 PM - Poster # 105 Evaluating collision–coalescence in WINTRE-MIX freezing drizzle forecasts
Megan R Schiede, University at Albany, Albany , NY; and J. R. Minder and J. R. French

3:00 PM - Poster # 106 Contrail Detection and Cloud Height Retrievals Using GOES-ABI Observations in Support of Field Campaigns
Jay P. Hoffman, CIMSS, Madison, WI; and T. Rahmes, A. J. Wimmers, W. Feltz, M. J. Foster, T. J. Wagner, C. Phillips, R. Miller, and D. Ochoa-Mendoza

3:00 PM - Poster # 107 Snow Sublimation–Driven Cold Pools in Marine Cold-Air Outbreaks: A Mechanism for Open-Cell Transition
Steven K. Krueger, Univ.of Utah, SALT LAKE CITY, UT; and A. Raudzens Bailey and C. A. Yang

3:00 PM - Poster # 108 Smoke and Cloud Impacts on Microphysical-Radiative Processes within Pyrocumulonimbus Anvils
Phoebe Lin, Colorado State University, Fort Collins, CO; and S. M. Saleeby, S. D. Miller, and S. C. Van Den Heever

3:00 PM WITHDRAWN Withdrawn: Impacts of Structural Deficiencies In Earth System Models on Predictions of Precipitation Extremes and Aerosol Radiative Effects
Po-Lun Ma, PNNL, Richland, WA; and A. Geiss, Y. Qin, M. Wu, S. Duan, M. Huang, L. Kang, V. Larson, N. Mahfouz, R. Marchand, H. Morrison, J. D. Neelin, C. J. Wall, R. A. Zaveri, and B. Zhao

Poster - POSTER From Lines to Complexes: Advances in Organized Convective System Research

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 136 Elucidating the Fine-Scale, Turbulent Structure of Convective Updrafts In Mesoscale Convective Systems Using Rapidly Collected RHIs
David J. Bodine, University of Oklahoma, Norman, OK; and K. L. Rasmussen, Ph.D., B. Dolan, P. Kirstetter, and H. B. Bluestein

3:00 PM - Poster 137 Phased-Array Radar Analysis of Downburst and Mesovortex Precursors in MCSs using Proxies of Vertical Motion
Willow Michael Phipps, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; The University of Oklahoma, School of Meteorology, Norman, OK; and A. A. Alford, K. Tuftedal, and T. J. Schuur

3:00 PM - Poster 138 Quasi-linear Convective Systems and their Hazards in the United States: 1996-2025

Alex Haberlie, Northern Illinois Univ., DeKalb, IL; and W. S. Ashley, M. Van Den Broeke, and N. D. Sonntag

3:00 PM - Poster 139 Low-Frequency Gravity Waves within Mesoscale Convective Systems: Characteristics and Simulated Origins
Hannah C. Vagasky, JANUS Research Group, LLC, Atmospheric and Environmental Research (AER), Evans, GA; and R. Adams-Selin, D. V. Blount, and C. Evans

3:00 PM - Poster 140 Evaluating HRRR Model Representation of Tornadic QLCS Mesovortices and Environments from PERiLS
Braelyn Long, University of Oklahoma, Norman, OK; and A. J. Hill

3:00 PM - Poster 141 The Role of the Environment Prior to QLCS Mesovortex Tornadogenesis
Mandy Voth, University of Oklahoma, Norman, OK; and A. J. Hill

3:00 PM - Poster 142 Damage Survey and Observational Study on Beijing "5.30" Extremely High Wind
Xiuming wang, China meteorological Administration Training Center, Beijing, Beijing, China; CMATC（China meteorological administration training center）, Beijing, China; and H. tan, F. xu, X. yu, and Y. zhang

3:00 PM - Poster 143 Characteristics of Tornado Outbreaks in Canadian QLCSs
Morgan E. Schneider, Canadian Severe Storms Laboratory, London, ON, Canada; and D. M. L. Sills and L. Schielicke

3:00 PM - Poster 144 A Study on the Nocturnal Evolution During PERILS 2023 IOP3's Supercells to QLCS Development.
Meredith Logan Bean, Skyview Weather, Madison, AL

3:00 PM - Poster 145 Mesovortexgenesis in an Idealized High-Shear, Low-CAPE QLCS
Geoffrey R. Marion, NSSL, Norman, OK; and M. C. Brown, M. C. Coniglio, and T. J. Galarneau

Poster - POSTER Severe Convective Storm Climatologies: Past, Present, and Future

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 146 Assessing Severe Storm Frequency and Risk Using Multi-Decadal Geostationary Infrared Satellite Data Records
Kristopher M. Bedka, LRC, HAMPTON, VA; NASA, Hampton, VA; and K. F. Itterly, D. Spangenberg, K. Khlopenkov, P. H. Groenemeijer, and F. Battaglioli

3:00 PM - Poster 147 Bow Echo Formation Features: An Analysis of 290 Events
Eliot Morrow, Iowa State University, Ames, IA; and W. A. Gallus Jr.

3:00 PM - Poster 148 Climate Change Impacts on Severe Thunderstorms in Italy: Results from MPAS PGW Simulations of two Extreme Events
William A. Gallus Jr., Iowa State Univ., AMES, IA; and E. M. Dougherty

3:00 PM - Poster 149 Investigating Recent Geographic and Seasonal Trends in United States Tornado Activity
Nathan Robert Harms, Purdue University, West Lafayette, IN; and D. Chavas and D. T. Dawson II

3:00 PM - Poster 150 A High-Resolution Severe Hail Climatology for the CONUS
Jacob Widanski, Cooperative Institute for Severe and High-Impact Weather Research and Operations, Univ. of Oklahoma, Norman, OK; and B. A. Schenkel and T. J. Galarneau

3:00 PM - Poster 151 A Comparison Of The Joplin, Missouri May 22, 2011 Tornado, The Moore, Oklahoma May 20, 2013 Tornado, And The May 4, 2007 Tornado That Struck Greensburg, Kansas To Gain A Better Understanding Of The Meteorological ingredients That Favor Or Minimize The Chances Of Stronger And Higher Rated Tornadic Potential And Longevity In The Tornado Lifecycle To Enhance Forecasting Skills And Weather Awareness In Severe Weather Situations And Develop Effective Warning Strategies With This Leatned Knowledge To Keep The Public Safe And Informed
Caitlin J. Lawrence, USA Weather, Inc, Carmichaels, PA

3:00 PM - Poster 152 Severe Convective Outbreaks and Heatwaves – A Large-scale Compound Event
Monika Feldmann, Institute for Atmospheric and Climate Science, ETH Zürich, Zürich, ZH, Switzerland; and D. I. V. Domeisen and O. Martius

3:00 PM - Poster 153 Historical Trends and Future Projections of Elevated Mixed Layers over the Contiguous United States
Margo S. Andrews, Northern Illinois University, DeKalb, IL; and V. A. Gensini, A. Haberlie, W. S. Ashley, and A. C. Michaelis

3:00 PM - Poster 154 Quantifying Severe Convective Storm Characteristic Uncertainty in Warming Climates
Rebecca Oh, University of Oklahoma, School of Meteorology, Norman, OK; and A. J. Hill

3:00 PM - Poster 155 A Climatology of Elevated Thunderstorms from Dynamically Downscaled Regional Climate Simulations
Kyle D Pittman, Northern Illinois University, DeKalb, IL; and K. Hawthorne, E. Mantia, L. Moeller, B. L. Weart, W. S. Ashley, A. Haberlie, and V. A. Gensini

3:00 PM - Poster 156 Improved Projection of Future Tornado Climatology Using Downscaled, Projected Global Climate Data
Shanshan Shi, Missouri University of Science and Technology, Rolla, MO; and G. Yan

3:00 PM - Poster 157 Projecting Shifts in U.S. Tornado Frequency and Intensity using Hierarchical Bayesian Modeling
Kaleb Raven Clover, Central Michigan Univ., Mount Pleasant, MI; and J. T. Allen

3:00 PM - Poster 158 Digitizing Purdue's Role in Tornado Science: Preserving Materials from the 20 March 1976 West Lafayette, Indiana Tornado
Evelyn Rose Ka'ilialoha Girardi, Purdue University, West Lafayette, IN; and L. E. Grose, M. Smith, and R. Tanamachi

3:00 PM - Poster 159 Demonstration of an Extreme-Event Attribution Methodology on a High-Impact Hailstorm
Robert J. Trapp, Univ. of Illinois, URBANA, IL; and S. Lasher-Trapp, D. claybrooke, and O. Martius

3:00 PM - Poster 160 Are the Trends in Convective Parameters from Reanalysis Data Reliable?
John T. Allen, Central Michigan University, Mt Pleasant, MI; and C. Cuervo-Lopez

3:00 PM - Poster 161 Long-Term Trends in Convective Weather and Environments in the Southeastern United States
Ryan Toomey, North Carolina State Univ., Raleigh, NC; and M. D. Parker, A. C. Michaelis, and G. M. Lackmann

3:00 PM - Poster 162 Spatiotemporal Trends in Convective Precipitation across Kentucky: A Radar-Based Climatology
Josh Sherman, University of Louisville, Louisville, KY; Univ. of Louisville, Louisville, KY; and J. A. Naylor

3:00 PM - Poster 163 A Preliminary Investigation of the Classification of Elevated Mixed Layer/Capped Environments Using Machine Learning Techniques
Haleigh Dyer, University of South Alabama, Alabaster, AL; and J. E. Wiley and J. M. Lanicci, Ph.D.

3:00 PM - Poster 164 Utilizing Gridded Radar and Sounding Data to Confirm the Proposed Link Between Vertical Wind Shear and Relative Humidity with Supercell Size
Alexander James Colgate, Univ. at Albany (SUNY), Pompton Plns, NJ; and J. P. Mulholland

3:00 PM - Poster 165 Statistical Analysis of GOES-R Derived Convective Parameters and ERA5 Reanalysis Meteorological Variables
Brianna Hauke, University of Wyoming, Laramie, WY; and M. Saito and T. W. Emmenegger

Poster - POSTER The In-Situ Collaborative Experiment for the Collection of Hail in the Plains (ICECHIP)

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 124 Coupling Multi-Doppler Wind Retrievals With Lagrangian Thermodynamic Analysis to Model Hail Trajectories in Severe Storms
Jaime L Herriott, University of Oklahoma, Norman, OK

3:00 PM - Poster 125 How Reliable Are Storm Kinematic Features from Single-Doppler Radar?
Francesco De Martin, School of Meteorology, University of Oklahoma, Norman, Oklahoma, Norman, OK; and C. R. Homeyer and D. J. Bodine

3:00 PM - Poster 126 Hail or Hail No; Machine Learning Used to Classify Hail Images
Jeana Mascio, AER, Lexington, MA; Janus Research Group, Evans, GA; and R. Adams-Selin, T. D. Kurtz, and A. D. Kennedy

3:00 PM - Poster 127 The First Surface-Based Isotopic Measurements of Hailstorm Inflows Using a Vehicle-Mounted Cavity Ring-Down Spectrometer during the 2025 ICECHIP Field Campaign
Anthony Crespo Bernal Ayala, AER, Lexington, MA; NCAR, Boulder, CO; and A. Raudzens Bailey, V. K. Thaxton, and R. Adams-Selin

3:00 PM - Poster 128 Observed Field Hailstone Strength Measurements and Implications for Laboratory Hail Strength Distributions
Marcus Cameron Watkins, Insurance Institute for Business & Home Safety, Richburg, SC; and B. Meisenzahl, J. Sorber, J. T. Allen, S. N. Servey, and I. M. Giammanco, PhD

3:00 PM - Poster 129 Multi-Radar 3D Wind Retrievals of Hail Producing Non-Supercell Storms During IOP5
Chase Calkins, Atmospheric and Environmental Research (AER), JANUS Research Group, LLC, Evans, GA; and H. C. Vagasky, R. Adams-Selin, and C. L. Ziegler

3:00 PM - Poster 130 Near-Surface Evolution of Cold Pool Properties over Hail Swaths Observed during ICECHIP
Barbara O'Neill, Central Michigan Univ., Mount Pleasant, MI; and J. M. Keeler

3:00 PM - Poster 131 Analysis of a Trajectory Model-Produced Hail Swath
Lydia Kate Spychalla, Pennsylvania State Univ., University Park, PA; The Pennsylvania State University, State College, PA; and M. Kumjian and K. Lombardo

3:00 PM - Poster 132 Preliminary Ground-Based Hailstone Observations during ICECHIP

Sabrina N. Servey, Central Michigan University, Mount Pleasant, MI; and G. Parkham, J. T. Allen, B. Meisenzahl, J. Sorber, L. Faulkner, I. M. Giammanco, PhD, A. C. Bernal Ayala, and R. Adams-Selin

3:00 PM - Poster 133 Evaluation of hailstone Terminal Velocity Parameterizations using In Situ Observations Collected During ICECHIP
Rebecca D. Adams-Selin, AER, Papillion, NE; and J. Brimelow, A. C. Bernal Ayala, E. A. Azriel, A. Bansemer, C. Bennett, J. Biemans, J. Boomgard-Zagrodnik, C. Calkins, M. Gartner, B. N. Guy, A. Habib, A. J. Heymsfield, A. D. Kennedy, T. D. Kurtz, J. Moore, T. Nordstrand, G. Reyna, M. Richer, S. N. Servey, and H. C. Vagasky

3:00 PM - Poster 134 Sub-cloud Fall Angles and Kinetic Energies of Hailstones
John T. Allen, Central Michigan University, Mt Pleasant, MI; and S. M. Ganson and M. Kumjian

3:00 PM - Poster 135 Evaluating Convection-Allowing Model Hail Size Forecasts Using MRMS MESH and ICECHIP Observations
Evelynn Mantia, Northern Illinois University, DeKalb, IL; and V. A. Gensini

Poster - Posters: Advances in 3D Radiative Transfer in Atmospheric Radiation and Remote Sensing

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Cochairs: Konrad Sebastian Schmidt / University of Colorado Boulder / Boulder, CO / Jake J Gristey / CIRES / Boulder, CO

3:00 PM - Poster 86 Modelling Light Pollution Over Fort Collins, Colorado, with 3D Radiative Transfer, Satellite Remote Sensing, and Surface Observations
Jackson Tobin, Colorado State University, Fort Collins, CO; and S. D. Miller, J. C. Chiu, J. Zhang, J. Frommeyer, M. Pagnutti, R. Ryan, H. Xu, C. Archuleta, J. White, and G. Sawyer

3:00 PM - Poster 87 Accelerating 3D radiative transfer via machine learning without loss of reliability and interpretability
Thu Le, University of Wisconsin-Madison, Madison, WI

3:00 PM - Poster 88 Towards Three-Dimensional Radiative Transfer Computations via Adaptive Mesh Refinement in both Space and Angle
Yassine Tissaoui, University of Wisconsin-Madison, Madison, WI; University of Wisconsin-Madison, Madison, WI; and H. Wang, S. Marras, R. Pincus, and S. N. Stechmann

3:00 PM - Poster 89 Investigating 3D Cloud Radiative Effects on High-Resolution Trace Gas Spectra to Elucidate the Mechanisms of Land-Ocean Retrieval Biases
Yu-Wen Chen, University of Colorado at Boulder, Boulder, CO; University of Colorado Boulder, Boulder, CO; and K. S. Schmidt, H. Chen, and S. Massie

3:00 PM - Poster 90 Measurements of Ångström exponent of aerosol by ozone Lidar with four wavelengths
Cao Nianwen, Nanjing University of Information Science & Technology, Nanjing City , 32, China

Poster - Posters: Surface Energy Budget, Albedo, and Atmosphere-Surface Coupling

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO / CoChair / Laura Riihimaki / LBNL / Lakewood, CO

3:00 PM - Poster 91 Cirrus and PBL Height Climatology and the Potential Effects of Cirrus on PBL Height in Metro Manila using Ground-based High Spectral Resolution Lidar
Shane Marie Aurelio Visaga, University of Wisconsin-Madison, Madison, WI; and R. Holz, L. M. P. Olaguera, M. Cambaliza, A. Bucholtz, E. W. Eloranta, J. T. Maquiling, I. Razenkov, E. Reid, J. S. Reid, K. S. Schmidt, and J. Simpas

Poster Viewing and Coffee Break

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

Joint 8 - Joint Session Honoring the Legacy of Roy Rasmussen II

4:30 PM - 6:00 PM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Cochairs: Lulin Xue / NCAR / Boulder, CO / Nicholas Dawson / NCAR / Boulder, CO

4:30 PM Joint 8.1
The Character and Causes of Elevation-Dependent Warming In High-Resolution Simulations of Andes Mountain Climate Change Justin R Minder, Univ. at Albany, Albany, NY; and A. Ridgway, E. Potter, and I. Acero-Ramirez

4:45 PM Joint 8.2
RM1.3: A 1.33 km Long-term Climate Reanalysis and Climate Change Dataset over the Rocky Mountain Region Bart Geerts, Univ. of Wyoming, Laramie, WY; and P. Adhikari, K. L. Rasmussen, S. Rahimi, T. L. Schneider, L. Xue, PhD, and K. Smith

5:00 PM Joint 8.3
Modeling the Effects of Glaciogenic Cloud Seeding on Precipitation and Atmospheric Moisture Processing in Complex Terrain Trude Eidhammer, NSF NCAR, Boulder, CO; NCAR, Boulder, CO; and S. A. Tessendorf, M. H. Stell, M. E. Frediani, PhD, C. Weeks, and J. K. Wolff

5:15 PM Joint 8.4
From Hawaiian Rainbands to Continental Hydroclimate: Perspectives on the Scientific Legacy of Roy Rasmussen Kristen Lani Rasmussen, Ph.D., Colorado State University, Fort Collins, CO

Joint Panel Discussion 8 - After the Crest: Reflections on the 2025 Southeastern Wisconsin Floods

4:30 PM - 6:00 PM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Zachary James Jon Riel / NV5 / Sun Prairie, WI

Chair: Zachary James Jon Riel / NV5 / Sun Prairie, WI

Panelists: Tim Halbach / NWS / Dousman, WI / Natalie Meier / Emergency Management Coordinator for Milwaukee County / Amanda Pulvermacher / Deputy Director for Dodge County Emergency Management

4:30 PM Joint Panel Discussion 8.1
The Future Path for Alerting and Communication in Severe Weather Events from a Multi-State Vortex USA Study Laura Myers, Univ. of Alabama, Tuscaloosa, AL; The University of Alabama Tuscaloosa, Tuscaloosa, AL; and J. M. Collins and D. B. Roueche

4:45 PM Joint Panel Discussion 8.2
Examining Extreme Convective Rainfall in Milwaukee, Wisconsin Using an Extremes-Preserving Pseudo-Global Warming Method Aaron Alexander, University of Wisconsin-Madison, Madison, WI; and E. M. Dougherty, D. Wright, and F. P. Sari

5:00 PM Panel Discussion

Panel Discussion 8 - What’s Next? The Future of Atmospheric Radiation Science, Modeling, and Observations

4:30 PM - 6:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Cochairs: Ryan Kramer / GFDL / Alexandria, MD / Laura Riihimaki / LBNL / Lakewood, CO / David Jeffrey Paynter / NOAA/GFDL / Princeton, NJ

Panelists: Martin Wild / ETH / Zurich, Zurich / Switzerland / Robert Pincus / Columbia University / New York, NY / J. Christine Chiu / Colorado State Univ. / Fort Collins, CO / Yunyan Zhang / LLNL / Livermore, CA / Odele M. Coddington / Laboratory for Atmospheric and Space Physics / Boulder, CO

4:30 PM Panel Discussion

Session 8 - Advances in Data Assimilation Techniques

4:30 PM - 6:00 PM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / Cincinnati, OH

Cochairs: Keenan Christopher Eure / CIRA / Boulder, CO / Therese Thompson Ladwig / NOAA Global Systems Laboratory / Boulder, CO

4:30 PM - 8.1 Towards Strongly Coupled Land–Atmosphere Data Assimilation for Numerical Weather Prediction and Earth System Models
Zhaoxia Pu, Univ. of Utah, Salt Lake City, UT; and H. Jing, M. C. Johncox, and D. Xu

4:45 PM - 8.2 Data Assimilation Development for the Rapid Refresh Forecast System
Shun Liu, EMC, College Park, MD; and M. Hu, D. Holdaway, S. K. Degelia, D. E. Lippi, T. Lei, X. Zhang, H. Liu, M. Oigawa, C. R. Martin, PhD, J. R. Carley, D. T. Kleist, M. E. Pyle, G. Ge, D. Dowell, C. Zhou, T. T. Ladwig, R. Li, and C. R. Alexander

5:00 PM - 8.3 Improving Process Understanding Using Nonlinear Ensemble Data Assimilation at Convective Scales
Derek J. Posselt, PhD, JPL, Pasadena, CA

5:15 PM - 8.4 Relationships between Assimilation Frequency and Localization Radii within an Idealized LETKF OSSE
Jessica M. McDonald, NSSL, Norman, OK; National Research Council, Washington, DC; and L. J. Wicker and D. R. Stratman

5:30 PM - 8.5 Preliminary Testing of Tomorrow.io Microwave Sounder Observations in a Prototype Rapid Refresh Forecast System version 2 (RRFSv2)
Keenan Christopher Eure, CIRA, Boulder, CO; and H. Lin, J. J. Guerrette, X. Zhang, R. Honeyager, B. C. Ruston, and M. Hu

5:45 PM - 8.6 Tropical and Extratropical Pacific Buoy Surface Pressure Observation Impacts
Carolyn A. Reynolds, NRL, Monterey, CA; and R. Stone, J. D. Doyle, and M. G. Fearon

Session 8 - R2O2R Activities: Processes & Paradigms

4:30 PM - 6:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Gina M. Eosco / NOAA Federal / Silver Spring, MD / Diana Rose Jesberg / ESRL / Boulder, CO

4:30 PM - 8.1 The Future of the National Blend of Models
Geoffrey S. Manikin, NOAA/NWS/OSTI/MDL, Silver Spring, MD; and D. E. Rudack, R. S. James, A. D. Schnapp, E. F. Engle, S. D. Scallion, M. N. Baker, C. S. Buxton, P. E. Shafer, K. M. Trinidad, F. G. Samplatsky, M. B. Allard-Lang, M. S. Antolik, S. L. Levine, G. G. Leone, S. M. Lower, B. J. Haynes, D. W. Plumb Jr., and C. Huang

4:45 PM - 8.2 Six Years of Atmospheric Physics Development in the UFS-R2O Project
Lisa Bengtsson, CIRES, Boulder, CO; and F. Yang

5:00 PM - 8.3 The National Windstorm Impact Reduction Program - A Cyclic Approach to Community Resilience
Joel William Cline, DOC / National Institute of Standards and Technology, Vienna, VA

5:15 PM - 8.4 From Theory to Practice: On Informing a Probabilistic Forecasting Philosophy
James Correia Jr., NWS/Weather Prediction Center and CU Boulder CIRES, College Park, MD; and S. M. Trojniak, W. M. Bartolini, and J. A. Nelson Jr.

5:30 PM - 8.5 Enhancing Operational Decision-Making Through a Unified Probabilistic Forecast Process
James A. Nelson Jr., NWS, College Park, MD

5:45 PM - 8.6 Bridging the Latency Gap: The Operational Impact of CSPP LEO Direct Broadcast Products on Local Decision Support
Kathleen I. Strabala, SSEC/CIMSS, Univ. of Wisconsin-Madison, Madison, WI; and D. Schumacher

Session 8 - Warm Clouds and Climate III

4:30 PM - 6:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Anita D. Rapp / College Station, TX / Mikael Kurt Witte / NPS / Monterey, CA / Christina S. McCluskey / NCAR / Boulder, CO

4:30 PM Investigating Convective Cold Pools and Aerosol Interactions at Bankhead National Forest
Angel Bright Chui, University of Washington, Seattle, WA; and A. Anderson-Frey and R. Wood

4:45 PM - 8.2 The Role of Mesoscale Cloud Morphology in Aerosol Radiative Forcing
Isabel L. McCoy, Colorado State University, Fort Collins, CO; and D. T. McCoy, J. J. Gristey, G. Feingold, and J. C. Chiu

5:00 PM - 8.3 Do Global Kilometer-Scale Models Reproduce the Diurnal Cycle of Cloud Spatial Patterns?
Gabrielle "Bee" R. Leung, University of Wisconsin-Madison, Madison, WI; and P. Stier

5:15 PM - 8.4 Marine Boundary Layer Convective Complexes over the Eastern North Atlantic: Observed Characteristics and Simulated Life Cycle
Qiuxuan Zheng, Rutgers University, HIGHLAND PARK, NJ; and M. A. Miller, S. Braddock, T. H. Hsu, P. J. van Leeuwen, and J. C. Chiu

5:30 PM - 8.5 Cloud-Precipitation-Turbulence Feedbacks Modulate Timing of the Stratocumulus to Cumulus Transition without Cold Pools
Mikael Kurt Witte, ; and M. Chinita, M. A. Smalley, and J. H. Jeong

5:45 PM - 8.6 Understanding Marine Stratocumulus Evolution through a Novel Causal Framework
Tzu-Han Hsu, Colorado State Univ., Fort Collins, CO; Colorado State Univ., Fort Collins, CO; and J. C. Chiu, P. J. van Leeuwen, S. Braddock, and M. A. Miller

Session 8A - Derechoes and Significant Convective Wind Events

4:30 PM - 6:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Joshua S. Ostaszewski / Texas Tech University / Lubbock, TX / Kate Olivia Stapleton / Norman, OK

4:30 PM - 8A.1 A Comparison of Wind-Producing Mechanisms across Four Extreme Derecho Events
Nicholas Markus, Iowa State University, Ames, IA; Iowa State University, Ames, IA

4:45 PM - 8A.2 Comparing Warm- and Cool-season Derecho-producing QLCSs: Dynamical Similarities and Differences
Garrett Travis Statum, University of Illinois Urbana-Champaign, Champaign, IL; and R. J. Trapp, K. J. Killion, and F. T. Lombardo

5:00 PM - 8A.3 A Possible Derecho in Extreme Humidity on the Northwestern Edge of the June-July 2021 Western Canada Heat Dome
Daniel M. Brown, EC, Edmonton, AB, Canada; and D. M. L. Sills, J. Achtymichuk, and M. Faramarzi

5:15 PM - 8A.4 Inflow Environments of Derechos and Non-Derechos
Andrew Wade, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and B. J. Squitieri and I. L. Jirak

5:30 PM - 8A.5 Testing the Sensitivity of Simulated, Idealized Derecho Intensity By Varying Ambient Conditions In CM1
Brian J. Squitieri, SPC, Norman, OK; and A. Wade and I. L. Jirak

5:45 PM - 8A.6 Challenges in Simulation of the 10 August 2020 Derecho: What can be learned from FV3, WRF, and MPAS?
William A. Gallus Jr., Iowa State Univ., AMES, IA; and N. Markus and R. Smith

Session 8B - Severe Convective Storm Climatologies: Past, Present, and Future II

4:30 PM - 6:00 PM | Hall of Ideas - HI (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Lauren E. Pounds / Norman, OK / Ethan B. Steward / Texas Tech University / Newburgh, IN

4:30 PM - 8B.1 A Climatology of the Role of Elevated Mixed Layers in Northeast United States Severe Convective Events
Alex D Kramer, Univ. at Albany (SUNY), Ballston Spa, NY; and J. P. Mulholland

4:45 PM - 8B.2 Evaluating the Predictability of Severe Convection in Current and Future Climate Scenarios
Andrea Benz, Texas Tech Univ., Lubbock, TX

5:00 PM - 8B.3 Extratropical Cyclone Track Regimes as a Control on the Spatial and Seasonal Climatology of U.S. Tornadoes
Peter A Forbin, Purdue Univ., Lafayette, IN; Purdue University, West Lafayette, IN; and D. R. Chavas, N. A. Goldacker, L. Wang, and D. T. Dawson II

5:15 PM - 8B.4 Seasonality of Hail Outbreaks in the United States and its Link to Weather Regimes
Matthew Graber, University of Illinois Urbana-Champaign, Champaign, IL; and R. J. Trapp and Z. Wang

5:30 PM - 8B.5 When and Where Does Large Hail Fall? An Evaluation of U.S. Hail Reports during 1975-2024
Jessica Meyer, The Univ. of Arizona, Tucson, AZ; and K. Wood

5:45 PM - 8B.6 Observing, Detecting, and Forecasting Large Hail across Europe: ESSL's 20th Anniversary
Brice Evan Coffer, European Severe Storms Laboratory, Wiener Neustadt, Lower Austria, Austria; and P. H. Groenemeijer, F. Battaglioli, T. Pucik, B. van't Veen, and A. M. Holzer

Event - SLS Reception

6:00 PM - 7:30 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Event - Social Science Social

6:00 PM - 8:30 PM | Grand Terrace (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

Submitter: Zachary James Jon Riel / NV5 / Sun Prairie, WI / View Related / Eighth Conference on Weather Warnings and Communication

Event - Early Career Networking Social for Cloud Physics

6:30 PM - 8:00 PM | Grand Terrace (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Keynote Speaker - CM1: Past, Present, and Future

7:30 PM - 8:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL / Chairs / Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL / Jennifer M. Laflin / NWS / Pleasant Hill, MO / Austin Coleman / NOAA

7:30 PM Keynote Speaker.1
Recent Developments and Future Directions of CM1 George H. Bryan, The Pennsylvania State University, University Park, PA

Panel Discussion - Severe Local Storms After Dark

8:00 PM - 9:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Chair: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Jennifer M. Laflin / NWS / Pleasant Hill, MO / Austin Coleman / NOAA

8:00 PM Panel Discussion

Wednesday, August 5, 2026

Joint 9 - Forecasting Severe Weather Outbreaks across Timescales

8:30 AM - 10:00 AM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitters: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO / Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Manda B. Chasteen / Leidos / Reston, VA / Thomas Galarneau / Norman, OK

8:30 AM Joint 9.1
Guidance for Forecasting Tornado Conditional Intensity at the Storm Prediction Center Andrew Wade, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and I. L. Jirak, C. Karstens, and R. L. Thompson

8:45 AM Joint 9.2
Assessment of Probabilistic Short-Term CAM Forecasts for Informing SPC Products in the Watch-to-Warning Timeframe Israel L. Jirak, SPC, Norman, OK; and M. E. Baldwin, T. A. Supinie, J. Picca, P. T. Marsh, R. L. Thompson, and R. S. Schneider

9:00 AM Joint 9.3
Assessing the Impact of Entrainment and Low-Level Buoyancy when Forecasting Tornadoes in Severe Storm Environments Daniel S Schwab, Purdue University, West Lafayette, IN; and D. Chavas, D. T. Dawson II, N. A. Goldacker, and J. M. Peters

9:15 AM Joint 9.4
AI and Physics Model Predictions for Severe Convection Rebecca Z Porter, Indiana University Bloomington, Bloomington, IN

9:30 AM Joint 9.5
Comparison Between Continuous Integration and Targeted Initialization for Medium Range Prediction of Severe Convective Storms Robert Clayton Fritzen, Northern Illinois Univ., DeKalb, IL; and V. A. Gensini

9:45 AM Joint 9.6
Real-Time Day 1–8 Convective Hazard Predictions Derived from Three AI NWP Ensembles Ryan A. Sobash, National Center for Atmospheric Research, BOULDER, CO; and Z. Hua, D. A. Ahijevych, and C. S. Schwartz

Joint 9 - Joint Session on Orographic Clouds, Precipitation, and Microphysics

8:30 AM - 10:00 AM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 25th Conference on Planned and Inadvertent Weather Modification

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Cochairs: Nicholas Dawson / NCAR / Boulder, CO / Samantha Turbeville / Seattle, WA

8:30 AM Joint 9.1
Microphysical Characteristics of Winter Orographic Clouds In Coastal and Interior Regions of the Western U.S. Sarah A. Tessendorf, NSF NCAR, Boulder, CO; and C. Weeks, E. M. Dougherty, T. Eidhammer, N. Dawson, PhD, S. Chen, M. A. Harrold, L. Xue, PhD, and J. K. Wolff

8:45 AM Joint 9.2
Investigation of Orographic Cloud Characteristics on Elk Mountain Utilizing High Resolution Numerical Simulations Nolan Thomas Robertson, University of Wyoming, Laramie, WY; and J. R. French, D. McCoy, and P. Adhikari

9:00 AM Joint 9.3
Ensemble Simulation of Winter Orographic Storms during SNOWIE: Understanding Model Uncertainty and Cloud Seeding Response Sisi Chen, NCAR, Longmont, CO; and L. Xue, PhD, S. A. Tessendorf, K. Friedrich, J. R. French, B. Geerts, C. Hohman, C. Weeks, T. J. Zaremba, J. K. Wolff, R. M. Rauber, N. Dawson, PhD, Z. Xie, Y. Zhang, and D. Blestrud

9:15 AM Joint 9.4
An Updraft-Based Framework for Organizing Orographic Snowfall and Cloud Structure during S2noCliME Troy Justin Zaremba, Mississippi State University, Starkville, MS; and L. McMurdie, P. Blossey, S. Vakhutinsky, C. Pettersen, A. Rowe, G. G. Mace, and B. Dolan

9:30 AM Joint 9.5
A Flexible Cloud-Identification Algorithm over Snow-Covered Terrain Clinton Alden, University of Washington, Seattle, WA; and J. D. Lundquist, S. Pestana, and E. D. Gutmann

Joint 9 - Marine Cold Air Outbreaks

8:30 AM - 10:00 AM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitters: Andrew Michael Dzambo / Univ. of Oklahoma - Norman / Norman, OK / Cooperative Institute of Severe and High-Impact Weather Research and Operations (CIWRO) / Norman, OK / Paquita Zuidema / Rosenstiel School, University of Miami / Miami, FL / Bart Geerts / Univ. of Wyoming / Laramie, WY / Manfred Wendisch / Univ. of Leipzig, Leipzig Institute for Meteorology / Leipzig Germany

Cochairs: Andrew Michael Dzambo / Cooperative Institute for Severe and High Impact Weather Research and Operations / Norman, OK / Paquita Zuidema / Rosenstial School / Miami, FL / Bart Geerts / Univ. of Wyoming / Laramie, WY / Manfred Wendisch / Leipzig University / Leipzig Germany

8:30 AM Joint 9.1

Arctic Clouds and Precipitation Observed In Moisture Space By the G-Band Radar for Water Vapor and Arctic Clouds (GRaWAC) Sabrina Schnitt, University of Cologne, Cologne, Germany; and B. Aydin, L. M. Bruder, K. Ebell, M. Klingebiel, M. Maahn, M. Mech, V. Schemann, and L. van Gelder

8:45 AM Joint 9.2

Radiative Impact of Cold Air Outbreak (CAO) Clouds: ERA5 Climatology and Airborne Validation Marcus Klingebiel, University of Leipzig, Leipzig, SN, Germany; and A. Luebke, S. Rosenburg, A. Ehrlich, M. Mech, H. Sundermann, H. M. Deneke, and M. Wendisch

9:00 AM Joint 9.3

Challenging Models to Predict MCAO Cloud Radiative Effects: Insights from COMBLE-MIP Florian Tornow, Columbia University and NASA GISS, New York, NY; and T. W. Juliano, A. M. Fridlind, A. Ackerman, G. S. Elsaesser, B. Geerts, C. Lackner, I. Silber, M. Ovchinnikov, G. Svensson, M. Tjernstrom, P. Wu, P. A. Bogenschutz, D. Chechin, K. K. Chandrakar, PhD, J. Chylik, A. Debolskiy, A. Ekman, A. Gupta, L. Ickes, R. Fadeev, A. Perez, M. Karalis, M. Köhler, B. Kosovic, W. Li, E. Mortikov, H. Morrison, R. Neggers, A. Possner, T. Raatikainen, P. Kuma, S. Romakkaniemi, N. Schnierstein, S. I. Shima, L. Xue, PhD, M. Zhang, X. Zheng, L. Raillard, N. Silin, M. Tolstykh, and E. Vignon

9:15 AM Joint 9.4

Aerosol Scavenging in Arctic Marine Stratocumulus Clouds during Cold Air Outbreaks Lintong Cai, University of California at Riverside, Riverside, CA; and J. R. Snider, S. Mahant, E. Weissburg, and M. D. Petters

9:30 AM Joint 9.5

Cloud and Precipitation Evolution and the Role of Sublimation in Arctic Marine Cold-Air Outbreaks from CAESAR Ching An Yang, University of Michigan, Ann Arbor, MI; and A. Raudzens Bailey, M. Sarker, E. Rosky, and S. K. Krueger

9:45 AM Joint 9.6

Drivers of Break-Up: A Satellite Perspective on Closed-to-Open-Cell Transitions in North Atlantic Cold Air Outbreaks Filip Severin von der Lippe, Univ. of Oslo, Oslo, Norway; and T. carlsen, T. Storelvmo, and R. O. David

Session 9 - Model Development and Application: MPAS Updates Part I

8:30 AM - 10:00 AM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / CIRES / WPC / Boulder, CO

Cochairs: Jack Lind / NWS / Lenexa, KS / Jeff Beck / NOAA/GSL and DTC / Boulder, CO

8:30 AM - 9.1 NOAA/GSL Model Development and Forecasting Activities using MPAS for Future NOAA Operational Modeling Systems

Clark Evans, NOAA/OAR/Global Systems Laboratory, Boulder, CO; and C. R. Alexander, L. Bernardet, T. T. Ladwig, M. Hu, D. Dowell, R. Ahmadov, and G. Ge

8:45 AM - 9.2 Integrating the Model Prediction Across Scales into the Unified Forecast System to Improve High-Impact Weather Forecasts

Ligia Bernardet, NOAA, Boulder, CO; Developmental Testbed Center (DTC), Boulder, CO; and J. R. Carley, E. A. Aligo, Y. Jung, D. J. Swales, G. Firl, D. Jovic, S. Rasmussen, L. Xue, PhD, M. Huang, C. R. Kondragunta, J. E. Ten Hoeve III, and K. Garrett

9:00 AM - 9.3 Characterizing Landfalling Atmospheric Rivers in the Western U.S. Using Radiosonde Observations and MPAS Simulations

Minghua Zheng, Center for Western Weather and Water Extremes, Scripps Institution of Oceanography, Univ. of California San Diego, La Jolla, CA; and D. Hooker, J. Wang, A. Wilson, Z. Zhang, B. Kawzenuk, A. Dooley, L. Delle Monache, D. Lavers, X. Wu, V. S. Tallapragada, and F. M. Ralph

9:15 AM - 9.4 The NCAR MPAS Program

Gretchen L. Mullendore, NSF NCAR, Boulder, CO; and K. Fossell and J. O'Brien

9:30 AM - 9.5 Best Practices in Using MPAS for Weather and Forecasting Research

Abigail Jaye, NSF NCAR, Boulder, CO

9:45 AM - 9.6 Assessing Upstream Sources of Medium-Range Forecast Errors in MPAS-A

May WS Wong, NCAR, Boulder, CO; and A. Lojko

Session 9 - Modeling and Remote Sensing of Earth's Radiation Budget, Climate and Relevant Processes I

8:30 AM - 10:00 AM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Cochairs: Cheng Dang / UCAR / Boulder, CO / Yan Feng / ANL / Hinsdale, IL / Ryan Kramer / GFDL / Alexandria, MD

8:30 AM - 9.1 How Historical and Future Radiative Forcing Shapes the Evolution and Spatial Structure of Earth’s Energy Imbalance

Lynn Aria Hirose, Princeton University, Princeton, NJ; and D. J. Paynter

8:45 AM - 9.2 Surface and Atmospheric Spectral Longwave Radiative Fluxes from Satellite Based Observations over 35 years: NASA Surface Radiation Budget Release 5.0 at 1/2 Degree

Paul Wesley Stackhouse Jr., NASA, Hampton, VA; and J. C. Mikovitz, S. J. Cox, and T. Zhang

9:00 AM - 9.3 Shortwave Results From NASA Surface Radiation Budget Release 5.0: Global Half Degree Resolution Over 35 Years
Stephen J. Cox, Analytical Mechanics Associates, Hampton, VA; and P. W. Stackhouse Jr., J. C. Mikovitz, and T. Zhang

9:15 AM - 9.4 Earth Hemispheric Albedo Symmetries and Their Implication for ERB
Jianhao Zhang, Univ. of Colorado/CIRES, Boulder, CO

9:30 AM - 9.5 Using Multi-Model Ensemble Experiments to Advance the Scientific Understanding of Observed Top-of-Atmosphere Albedo Phenomena Outside of the Current Observational Record
Daniel Feldman, LBNL, Berkeley, CA; and L. Cabral-Pelletier, J. J. Gristey, M. Z. Hakuba, S. Kahn, and D. Hellinger

9:45 AM - 9.6 Advances in Simulating Earth's Radiation Budget in GFDL's Next-Generation Atmospheric Model, AM5
David Jeffrey Paynter, GFDL, Princeton, NJ; and V. Naik, B. Xiang, R. Kramer, C. Fan, H. Guo, P. Lin, Z. Tan, and M. Zhao

Session 9 - Supercell Dynamics and Hazards I

8:30 AM - 10:00 AM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Brice Evan Coffer / European Severe Storms Laboratory / Wiener Neustadt, Lower Austria / Austria / Monika Feldmann / Institute for Atmospheric and Climate Science, ETH Zürich / Zürich, ZH / Switzerland

8:30 AM - 9.1 Deviant Supercell Storm Motion Across Ensembles of CM1 Simulations
Ronald Herman Kouski Jr., North Carolina State University, Raleigh, NC; and M. D. Parker

8:45 AM - 9.2 Sensitivity of Left-Moving Supercells to Varying Hodograph Structures
Aaron W. Zeeb, Central Michigan University, Mount Pleasant, MI; and J. T. Allen and C. Davenport

9:00 AM - 9.3 A Source-Layer Hypothesis to Explain the Uniqueness of Tornadic Supercell Updrafts
Andrew James Muehr, Colorado State University, Fort Collins, CO; and S. C. Van Den Heever and C. J. Nixon

9:15 AM - 9.4 Idealized Simulations Exploring Environmental Controls on Supercell Longevity
Casey Davenport, PhD, University of North Carolina at Charlotte, Charlotte, NC

9:30 AM - 9.5 Evolution and Dynamics of a Supercell's Mixed Phase Region and Implications for Hail Production
Lydia Kate Spychalla, Pennsylvania State Univ., University Park, PA; The Pennsylvania State University, State College, PA; and K. Lombardo and M. Kumjian

9:45 AM - 9.6 What Determines Updraft Width? Reconciling Continuity-Based and Edge-Propagation Perspectives
Paul M. Markowski, Penn State University, University Park, PA; and Y. P. Richardson

Coffee Break

10:00 AM - 10:45 AM | Grand Terrace (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Joint 10 - Joint Session on Severe Local Storms and Terrain Interactions I

10:45 AM - 12:00 PM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Cochairs: Anthony Lyza / NOAA/OAR/NSSL / Norman, OK / Kaitlyn R. Jesmonth / Univ. of Illinois Urbana-Champaign / Urbana, IL

10:45 AM Joint 10.1
Successive Quasi-linear Convective Systems Triggered by Collision of a Surface Convergence Line and Dryline over North China Complex Terrain in a Cold Vortex Background Rudi Xia, State Key Laboratory of Severe Weather Meteorological Science and Technology, Chinese Academy of Meteorological Sciences, Beijing, China; and Y. Zhang, J. Yin, J. Sun, and L. Zheng

11:00 AM WITHDRAWN
Joint 10.2 Withdrawn: From Nocturnal Warm-Rain Processes to Afternoon Mixed-Phase Precipitation: A Diurnal Transition in the Dynamical–Microphysical Pathways of Heavy Rainfall over North China Yuanchun Zhang, IAP, Beijing, Beijign, China; and T. Wang and J. Sun Meeting Rooms KLOP (Monona Terrace Community and Convention Center) Cancelled

11:15 AM Joint 10.3
Heavy Rain Events from Convective Storms in Taiwan: Ingredients, Storm Scale, Storm Motion, and the Role of Orography Alison D. Nugent, Univ. of Hawaii at Manoa, Honolulu, HI; and T. Zuo

11:30 AM Joint 10.4
How Do East African Rift Convective Storms and Environmental Conditions Influence Convection over the Congo Basin Rainforest? Juliet Pilewskie, Colorado State University, Fort Collins, CO; and A. J. Muehr and S. C. Van Den Heever

11:45 AM Joint 10.5
The Need for a Field Campaign to Improve Scientific Understanding and Predictability of Moisture Transport and Deep Convection in Semi-Arid Complex Terrain Curtis N. James, PhD, Embry-Riddle Aeronautical University, Prescott, AZ; and S. E. Koch, B. Geerts, A. Rowe, M. I. Biggerstaff, S. F. J. De Wekker, J. C. Knievel, and A. Raudzens Bailey

Joint 10 - Quantifying, Communicating, and Visualizing Ensemble Forecast Information

10:45 AM - 12:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Tomer Burg / WindBorne Systems / Palo Alto, CA / Patrick S. Skinner / University of Oklahoma / Norman, OK

10:45 AM Joint 10.1
A Deep Dive into Ensemble Spread Performance: A Serious Deficiency for Longer-Range Weather Forecasts Jun Du, EMC, College Park, MD

11:00 AM Joint 10.2
Ensemble Tools in NWS Operations: Results from the First NOAA WPC HMT Ensemble Clustering and Sensitivity Analysis Testbed Experiment Austin A. Coleman, ; and J. A. Nelson Jr. and M. Klein

11:15 AM Joint 10.3
Identifying Early Indicators of Mesoscale Rotation in Quasi-Linear Convective Systems with Ensemble Sensitivity Analysis and Sensitivity-Based Ensemble Subsetting Michael L Brown, Texas Tech Univ., Lubbock, TX; and C. C. Weiss and D. Dowell

11:30 AM Joint 10.4
Probabilistic Wind Forecasting for Improved PSPS Decisions at Idaho Power Company Jared A. Lee, Ph.D., NSF NCAR, Boulder, CO; and S. Alessandrini, M. Eghdami, P. A. Jimenez, J. H. Kim, P. McCarthy, S. A. Tessendorf, and J. K. Wolff

11:45 AM Joint 10.5
Incorporating Forecast Uncertainty into Tropical Cyclone Wind Speed Probability Products Alan Brammer, CIRA, Fort Collins, CO; and J. Martinez, M. DeMaria, K. Musgrave, P. Santos, W. A. Hogsett, and M. Onderlinde

Session 10 - Cloud Seeding Agents & Laboratory Studies

10:45 AM - 12:00 PM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | 25th Conference on Planned and Inadvertent Weather Modification

Chair: Michelle A. Harrold / NCAR / Boulder, CO / Corey Bois / University of Utah / Salt Lake City, UT

10:45 AM - 10.1 On the Nucleation Pathways of Common and Some Emerging Seeding Agents
Tom DeFelice, ASCE/EWRI AWM SC, Sheboygan, WI

11:00 AM - 10.2 Chemical-Free Cloud Seeding: Operational Results from Ground-Based Ionization Systems in Utah
Scott Morris, Rain Enhancement Technologies, Naples, FL; and J. Chagnon, R. Dongre, T. Gresham, and E. Powell

11:15 AM - 10.3 Hygroscopic Seeding with Powders: Insights from the Laboratory
Will Cantrell, Michigan Technological Univ., HOUGHTON, MI; and J. Simmons, H. Fahandezh sadi, S. P. Singh, A. Vakhtin, K. Hibert, B. A. Boe, Y. Wehbe, C. Bois, S. K. Krueger, J. C. Anderson, and R. A. Shaw

11:30 AM - 10.4 Optimizing the Efficiency of Hygroscopic Burn-In-Place (H-BIP) Flares Through a Size-Resolved Analysis
Swafuva Varappillikudy Sulaiman, Michigan Technological University, Houghton, MI; and K. Hibert, J. Simmons, H. Fahandezh sadi, S. P. Singh, A. Vakhtin, B. A. Boe, J. C. Anderson, C. Harper, R. A. Shaw, and W. Cantrell

Session 10 - Cloud and Precipitation Physics: Results from Laboratory, Field, Modeling, and Theoretical Studies I

10:45 AM - 12:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Sisi Chen / NCAR / Longmont, CO / Kamal Kant Chandrakar / NCAR / Boulder, CO

10:45 AM - 10.1 A Virtual Laboratory for Studying Cloud Microphysical and Dynamical Processes
Hugh Morrison, NCAR, Boulder, CO; and K. K. Chandrakar, PhD, M. Rehme, G. H. Bryan, S. Chen, W. W. Grabowski, P. Lawson, G. M. McFarquhar, E. M. Rosky, R. A. Shaw, and L. Xue, PhD

11:00 AM - 10.2 Probing kappa-Köhler Equivalence in the Pi Convection-Cloud Chamber
Kristin Joy Swartz-Schult, University of Notre Dame, South Bend, IN; and J. C. Anderson, W. Cantrell, H. Fahandezh sadi, R. A. Shaw, D. H. Richter, K. Flore GALI, J. Simmons, and S. P. Singh

11:15 AM - 10.3 Solutions to the Population Balance Equation for Cloud Hydrometeors
Antonio Torregrosa Abellan, Johannes Gutenber Universität, Mainz, RP, Germany; and P. Spichtinger

11:30 AM - 10.4 Conservation Laws for Moist Potential Vorticity with Clouds
Parvathi M. Kooloth, PNNL, Richland, WA; and L. M. Smith and S. N. Stechmann

11:45 AM - 10.5 A Climatology of Active Cold Pools Distinct from Background Variability at the Eastern North Atlantic Observations Site
Mark A Smalley, Pasadena, CA; and M. K. Witte, J. H. Jeong, and M. Chinita

Session 10 - Model Development and Application: MPAS Updates Part II

10:45 AM - 12:00 PM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / CIRES / WPC / Boulder, CO

Cochairs: Clark Evans / NOAA/OAR/Global Systems Laboratory / Boulder, CO / Randy A Graham / NWS / Kansas City, MO

10:45 AM - 10.1 The New RRFS Era: MPAS plus JEDI Development and 2026 Convective Season Performance
Therese Thompson Ladwig, NOAA Global Systems Laboratory, Boulder, CO; and M. Hu, C. Zhou, and D. Dowell

11:00 AM - 10.2 Progress Toward WoFSv2: Storm-scale Ensemble Development with MPAS-JEDI and Initial Experimental Results
Kent H. Knopfmeier, Cooperative Institute for Severe and High-Impact Weather Research and Operations, University of Oklahoma, Norman, OK; and C. Liu, Y. Wang, N. Yussouf, C. Schwartz, S. Ha, J. Park, T. A. Jones, E. R. Mansell, and D. Dowell

11:15 AM - 10.3 Investigating Resolution Tradeoffs within the MPAS-WoFS Prediction System
Louis J. Wicker, NOAA/National Severe Storms Laboratory, Norman, OK; and B. Skamarock and A. J. Clark

11:30 AM - 10.4 Assessing the Value of Remote Sensing Data Assimilation in JEDI for MPAS Winter Storm Simulations
Ethan Michael Schaefer, Pennsylvania State University, Glen Allen, VA; and S. J. Greybush, Y. Zhang, PhD, and D. J. Stensrud

11:45 AM - 10.5 Comparing the Performance of MPAS and HRRR Forecasts of High-Impact Fog Events
Matthew Wilson, NSF NCAR, Boulder, CO; and J. O. Pinto, Phd and J. Colavito

Session 10 - Modeling and Remote Sensing of Earth's Radiation Budget, Climate and Relevant Processes II

10:45 AM - 12:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Cochairs: Cheng Dang / UCAR / Boulder, CO / Yan Feng / ANL / Hinsdale, IL / Ryan Kramer / GFDL / Alexandria, MD / Colten Peterson / GSFC / Greenbelt, MD

10:45 AM - 10.1 Trend of Global Mean Net Atmospheric Shortwave and Longwave Irradiances Derived from Cloud and the Earth’s Radiant Energy System Data
Seiji Kato, NASA, Hampton, VA; NASA Langley Research Center, Hampton, VA; and T. Thorsen, F. Rose, N. G. Loeb, S. H. Ham, and D. A. Rutan

11:00 AM - 10.2 CERES Edition-5 MODIS and VIIRS Cloud Properties for Understanding and Monitoring Earth's Energy Budget
William Leo Smith Jr., AMA = Analytical Mechanics Associates, Hampton, VA; and S. Sun-Mack, Q. Trepte, C. R. Yost, Y. Chen, G. Hong, P. Minnis, and D. Painemal

11:15 AM - 10.3 Microphysics of Clouds and Aerosols, Vertical Air Motion and Radiative Properties from EarthCARE Observations
Hajime Okamoto, Kyushu University, Kasuga, Japan; Research Institute for Applied Mechanics, Kyushu University, Kasuga, Japan; and K. Sato, T. Nishizawa, Y. Jin, K. Suzuki, and H. Ishida

11:30 AM - 10.4 Comparison of CCCma-computed upwelling SW fluxes with EarthCARE BBR, and CERES FM1/FM6 SSF observations
Baike Xi, Univ. of Arizona, Tucson, AZ; and J. Brendecke, X. Dong, H. Barker, M. Kacimi, J. Cole, and Z. Qu

11:45 AM - 10.5 PATMOS-x v7.0: Updated Satellite-Based Climate Cloud Products since 1981.
Jongjin Seo, CIMSS, Madison, WI; Univ. of Wisconsin, Madison, WI; and M. J. Foster, C. Phillips, and A. Heidinger

Session 10 - Novel Field Observations of Severe Convective Storms I

10:45 AM - 12:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Madeline Diedrichsen / NOAA / OAR / NSSL / Norman, OK / Adam L. Houston / Univ. of Nebraska-Lincoln / Lincoln, NE

10:45 AM - 10.1 The Low-Level Internal Flow of Tornadoes Experiment: Overview and Highlights from the 2024-2026 Field Campaigns
Christopher C. Weiss, Texas Tech Univ., Lubbock, TX; and E. B. Steward, J. Buhrman, and J. J. Melke

11:00 AM - 10.2 The Low-Level Internal Flow of Tornadoes (LIFT) Field Campaign: Highlights from Three Years of NSSL Operations
Michael C. Coniglio, NOAA/OAR/NSSL, Norman, OK

11:15 AM - 10.3 Comparing High-Frequency Radar Measurements of Spatiotemporally Varying Low-Level Vertical Wind Shear Near Tornadic and Nontornadic Supercells from the TORUS and LIFT Campaigns
Julia Buhrman, Texas Tech University, Lubbock, TX; and C. C. Weiss

11:30 AM - 10.4 Surface Rear-Flank Downdraft Observations in Supercells: Insights From Recent Field Campaigns
Tyler James Pardun, Cooperative Institute for Severe and High-Impact Weather Research and Operations, Norman, OK

11:45 AM - 10.5 TTUKa Mobile Doppler Radar Analysis of Tornado Structure from the Low-Level Internal Flow of Tornadoes (LIFT) Experiment
Ethan B. Steward, Texas Tech University, Lubbock, TX; and C. C. Weiss

Lunch Break

12:00 PM - 1:45 PM | View Related | 34th Conference on Weather Analysis and Forecasting

Joint 11 - Joint Session on Severe Local Storms and Terrain Interactions II

1:45 PM - 3:00 PM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Cochairs: Sean Robert Ernst / IPPRA / Pepperell, MA / Steve Nesbitt / Department of Climate, Meteorology & Atmospheric Sciences / URBANA, IL

1:45 PM Joint 11.1
Characterizing Environments Associated with Mountain-Crossing Mesoscale Convective Systems in Argentina Using CACTI Data John J DelPizzo, The Pennsylvania State Univ., University Park, PA; and K. Lombardo, K. A. Bowley, X. Chen, Z. Feng, and A. Varble, PhD

2:15 PM Joint 11.3
Why are Tornadoes Less Frequent in the Central Appalachian Mountains? Samuel Jeffrey Porter, The Ohio State University, Medina, OH; and J. B. Houser, PhD

Session 11 - Cloud Modification & New Instrumentation

1:45 PM - 3:00 PM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | 25th Conference on Planned and Inadvertent Weather Modification

Chair: Katja Friedrich / Univ. of Colorado / Boulder, CO / Lynnlee Storm Bestul / University of North Dakota / Grand Forks, ND

1:45 PM - 11.1 Hygroscopic Seeding to Trigger a Natural Secondary Ice Process and Potential for Rain Enhancement
Paul Lawson, SPEC Inc., Boulder, CO; and S. Calderón and S. Romakkaniemi

2:00 PM - 11.2 Parcel and Particle Pathways to Warm-Phase Cloud Seeding
Corey Bois, University of Utah, Salt Lake City, UT; University of Utah, Salt Lake City, UT; and S. K. Krueger, W. Cantrell, and J. Simmons

2:15 PM - 11.3 Leveraging GOES-R and HRRR for Cloud Seeding Operations: A Data-Driven Approach to Cloud Microphysical Analysis
Caleb Steele, Weather Modification International, Fargo, ND; and Y. Wehbe and J. Ceratto

2:30 PM - 11.4 Electrostatic Enhancement of Droplet Coalescence in Radiation Fog
Taylor Gresham, Rain Enhancement Technologies, Naples, FL; Rain Enhancement Technologies, Naples, FL; and S. Morris, J. Chagnon, R. Dongre, and E. Powell

Session 11 - Cloud Plenary

1:45 PM - 3:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Chair: Adel L. Igel / University of California, Davis / Davis, CA

Cochairs: Julia A Shates / SIO / La Jolla, CA / Mikael Kurt Witte / NPS / Monterey, CA / Fan Yang / Brookhaven National Laboratory / Mastic, NY

1:45 PM - 11.1 The Influence of Facet Re-Development on the Growth Rates of Hollowed Columnar Ice Crystals
Jerry Y. Harrington, The Pennsylvania State University, University Park, PA; The Pennsylvania State Univ., University Park, PA; and A. M. Moyle

2:00 PM - 11.2 From Fire to Rain: Microphysical Pathways Linking Wildfire Aerosols to Southern Ocean Clouds and Precipitation
Jennifer D. Small Griswold, University of Hawaii at Manoa, Honolulu, HI

2:15 PM - 11.3 Association between High Cloud Droplet Number over the summer Southern Ocean and Airmass Trajectories over the Continental Ice Sheets.
Gerald G. Mace, University of Utah, Salt Lake City, UT; and S. Benson

2:30 PM - 11.4 Accelerating Cloud Microphysics Scheme Development with Machine Learning
Kara D. Lamb, Columbia University, New York, NY

2:45 PM Discussion

Session 11 - Modeling and Remote Sensing of Earth's Radiation Budget, Climate and Relevant Processes III

1:45 PM - 3:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Cochairs: Cheng Dang / UCAR / Boulder, CO / Yan Feng / ANL / Hinsdale, IL / Ryan Kramer / GFDL / Alexandria, MD / Jianhao Zhang / Univ. of Colorado/CIRES / Boulder, CO

1:45 PM - 11.1 On the Use of Geostationary Imagers as Transfer Radiometers to Inter-calibrate CERES Broadband Radiometers
David R. Doelling, NASA, Hampton, VA; and N. G. Loeb, K. F. Itterly, J. Daniels, and C. Haney

2:00 PM - 11.2 Evaluation of HRRR and ERA5 with observations of surface radiation and clouds
Kelly A. Balmes, CIRES, Boulder, CO; NOAA GML, Boulder, CO; and J. Sedlar, L. D. Riihimaki, D. D. Turner, S. G. Benjamin, and E. P. James

2:15 PM - 11.3 Penetrating Clouds: A Spectral-Spatial Deep Learning Model Framework to Retrieve Temperature and Humidity Profiles below Clouds
Qikai Hu, University of Michigan, Ann Arbor, Ann Arbor, MI; and X. Chen, Q. Yue, N. G. Loeb, S. Kato, and X. Huang

2:30 PM - 11.4 Diurnal Variations of Outgoing Longwave Radiation from Tropical Deep Convective Clouds
Ana Paulina Castaneda Montoya, University of Michigan, Ann Arbor, MI; and A. Merrelli and C. Pettersen

2:45 PM - 11.5 Is Radiative Convective Equilibrium Applicable Over theTibetan Plateau?
Meng Zuo, Chinese Academy of Meteorological Sciences, Beijing, Beijing, China; and J. Li

Session 11 - Testing, Metrics, and Validation of Artificial Intelligence for Numerical Weather Prediction (AI4NWP) Part II

1:45 PM - 3:00 PM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitters: Austin A. Coleman / CIRES / WPC / Boulder, CO / Christina E. Kumler / CIRES / Boulder, CO / Aaron J. Hill / University of Oklahoma, School of Meteorology / Norman, OK

Cochairs: Kathryn M. Newman / DTC / Boulder, CO / Christina E. Kumler / CIRES / Boulder, CO

1:45 PM - 11.1 Beyond Bulk Statistics: Diagnostic Validation of AI-NWP Models Using Interactive Visualization
Charlie Becker, NCAR, Boulder, CO

2:00 PM - 11.2 Recent Enhancements to METplus Verification Capabilities for AI-Driven Forecasts
Christina P. Kalb, NCAR, Boulder, CO; National Center for Atmospheric Research and Developmental Testbed Center, Boulder, CO; and K. M. Newman, J. Halley Gotway, V. Vargas Jr., G. Ketefian, M. A. Harrold, M. Win-Gildenmeister, and G. P. McCabe Jr.

2:15 PM - 11.3 Evaluating AI and Hybrid Numerical Weather Predication Models against Operational Benchmarks using METplus
Kathryn M. Newman, DTC, Boulder, CO; NCAR, Boulder, CO; and V. Vargas Jr., C. P. Kalb, M. A. Harrold, G. Ketefian, and J. Halley Gotway

2:30 PM - 11.4 Storm-scale Verification of WindBorne's HIRES WeatherMesh 6
Larissa Joy Reames, WindBorne Systems, Redwood City, CA; WindBorne Systems, Palo Alto, CA; and T. Hutchinson, J. T. Radford, H. Du, A. Shetty, J. Creus-Costa, J. Michaels, J. Dean, and D. Sorvisto

2:45 PM - 11.5 Systematic and High-Impact Case Study Evaluation of WindBorne’s WeatherMesh Model
Tomer Burg, Windborne Systems, Inc, Arlington, MA; and T. Hutchinson, C. Riedel, PhD, and H. Du

Session 11 - WFIP3 Field Campaign

1:45 PM - 3:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Joseph B. Olson

Cochairs: Joseph B. Olson / Matthew Wilson

1:45 PM - 11.1 HRRR/RRFS Physics Development for Improved Marine Cloud Forecasts in WFIP3
Joseph B. Olson, NOAA-Global Systems Laboratory, BOULDER, CO; and X. Sun, B. Adler, L. Bianco, A. A. Jensen, D. D. Turner, C. Evans, J. Sedlar, and K. A. Balmes

2:00 PM Characterizing Sea Breeze Events Using Observations and Model Simulations during WFIP3
Xia Sun, CIRES, Boulder, CO; and J. Olson, B. Adler, L. Bianco, C. Evans, and D. D. Turner

2:15 PM - 11.3 Improved Forecasts of Coastal Low-Level Jets with an Updated MYNN Turbulence Parameterization Scheme
Bianca Adler, NOAA, Boulder, CO; and J. Olson, X. Sun, L. Bianco, J. Sedlar, and D. D. Turner

2:30 PM - 11.4 Impact of HRRRv4 Model Physics Development on the Forecast of Low-Level Winds and Near-Surface Variables during WFIP3

Laura Bianco, CIRES, Boulder, CO; CIRES, Boulder, CO; NOAA, Boulder, CO; and J. Olson, X. Sun, I. V. Djalalova, B. Adler, J. Sedlar, K. A. Balmes, and D. D. Turner

2:45 PM Multi-Doppler Wind Retrievals of Nor'easter Cases during the 3rd Wind Forecast Improvement Project (WFIP3)
Robert C. Jackson, Argonne National Laboratory, Argonne, IL; ANL, Chicago, IL; and P. Muradyan, V. P. Ghate, J. Wang, J. Stock, PhD, C. Jung, D. Fytanidis, W. J. Pringle, H. Tan, S. Sethuram, and R. Kotamarthi

Session 11A - Supercell Dynamics and Hazards II

1:45 PM - 3:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Matthew Graber / University of Illinois Urbana-Champaign / Champaign, IL / Kristen L Axon / Purdue University / West Lafayette, IN

1:45 PM - 11A.1 Improved Understanding of Drivers of Upper-Tropospheric Updrafts and Their Relationship to Supercell Thunderstorm Overshooting Tops
Melinda Berman, University of Illinois Urbana-Champaign, Urbana, IL; and R. J. Trapp and S. W. Nesbitt

2:00 PM - 11A.2 The Dual Effect of Urban Areas on Supercell Storms
Francesco De Martin, School of Meteorology, University of Oklahoma, Norman, OK; Department of Physics and Astronomy, University of Bologna, Bologna, BO, Italy; U.S. National Science Foundation, National Center of Atmospheric Research, Boulder, CO; and C. M. Rozoff, PhD, A. Zonato, S. Alessandrini, and S. Di Sabatino

2:15 PM - 11A.3 How Does Vertical Wind Shear Distribution Affect Supercell Storms?
Yuzhu Lin, NCAR, Boulder, CO

2:30 PM - 11A.4 The Effects of a Veer-Back-Veer Wind Profile and Effective Inflow Layer Depth on Supercell Evolution
Martin Satrio, ., Norman, OK; and B. Saba, T. J. Pardun, J. C. Snyder, M. D. Flournoy, and M. Coniglio

2:45 PM - 11A.5 A Doppler Radar Climatology of Vertical Updraft and Mesocyclone Alignment in Supercells
Erik J Creighton, Stony Brook University, Stony Brook, NY; and M. M. French and D. M. Kingfield

Session 11B - Tornado Climatology and Near-Surface Wind Characteristics

1:45 PM - 3:00 PM | Hall of Ideas - HI (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitters: Franklin T. Lombardo / Univ. of Illinois Urbana-Champaign / Urbana, IL / Anthony Lyza / NOAA/OAR/NSSL / Norman, OK / CoChair / Elizabeth Tirone / CIWRO / Norman, OK

1:45 PM - 11B.1 Investigating the Effects of Idealized and Real-World Terrain on a Violent Tornado Using Ultra-Fine Scale CFD Simulations
Jana B. Houser, PhD, The Ohio State University, Columbus, OH; and J. DANG, G. Yan, and L. Orf

2:00 PM - 11B.2 The Application of Eddy Recycling on Turbulent Tornadic Flow Through Urban Environments
Alex Schueth, University of Oklahoma, Norman, OK; and D. J. Bodine, A. E. Reinhart, F. T. Lombardo, J. A. Gibbs, D. Candela, and S. M. Moon

2:15 PM - 11B.3 Organized Surface Layer Rolls in Tornado-Like Vortices: Properties and Influencing Factors
Dominic Candela, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and D. J. Bodine

2:30 PM - 11B.4 Photogrammetric Measurements of a High Plains Tornado Exhibiting EF4-5 Velocities near the Surface within Ten Minutes of Tornadogenesis
Anton Seimon, Bard College, Annandale-on-Hudson, NY; and S. J. Talbot

2:45 PM - 11B.5 Considerations for the Future of Estimating Tornado Intensity
Anthony W. Lyza, NOAA/OAR/NSSL, Norman, OK; and F. T. Lombardo, A. A. Alford, M. D. Flournoy, E. Tirone, and J. G. LaDue

Joint Poster - POSTER Convection-Allowing Models and Forecasting

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 259 Greenness and Growth: Vegetation Impacts on Convective Storm Development Across the Southeastern U.S.
David S Henderson, Space Science and Engineering Center, Madison, WI; and J. A. Otkin, PhD and J. R. Mecikalski

3:00 PM WITHDRAWN
Poster # 260 Withdrawn: Grid Spacing Impacts on Forecasted Tornadic Supercell Evolution in the Warn-On Forecast System Austin Dixon, CIWRO, Norman, OK; and P. Skinner and W. L. Faletti Jr. Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

Poster - 30th Conference on Numerical Weather Prediction Wednesday Posters

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Lisa Harris / Boston, MA

3:00 PM - Poster 213 Evaluation of Convective Vertical Transport Properties and Their Sensitivity to Parameterization Characteristics
Rebecca D. Adams-Selin, AER, Papillion, NE; and C. Duzgun, H. E. Fuelberg, N. Heath, and J. D. Hegarty

3:00 PM - Poster 214 Comparing the Impact of Differing Environmental Characteristics on MCS Dissipation Using CM1
Sam Ritter, Iowa State University, Ames, IA; and W. A. Gallus Jr.

3:00 PM - Poster 215 Data Assimilation Errors Due to Incomplete Accounting of Balanced Versus Unbalanced Moisture
H. Reed Ogrosky, Virginia Commonwealth University, Richmond, VA; and A. N. Wetzel, L. M. Smith, and S. N. Stechmann

3:00 PM - Poster 216 Effects of Supercooled Liquid Water on ATMS Brightness Temperatures using the VIIRS cloud product
Aditya Kumar, CIMSS, Madison, WI; and J. A. Jung, A. Heidinger, S. E. Nebuda, H. Sun, I. Genkova, S. Nadiga, L. Soulliard, and Q. Zhao

3:00 PM - Poster 217 Jexpresso: User-Friendly, Modular Software with Capability for Generic Atmospheric Dynamics, Including 3D Adaptive Mesh Refinement, Spectral-Bin Cloud Microphysics and 3D Radiative Transfer
Simone Marras, New Jersey Institute of Technology, Newark, NJ; and Y. Tissaoui, H. Wang, and S. N. Stechmann

3:00 PM - Poster 218 Three-Dimensional Adaptive Mesh Refinement (3D AMR) for Simultaneously Resolving the Cloudy Boundary Layer, Cirrus, and Larger-Scale Deep Convection
Hang Wang, New Jersey Institute of Technology, Newark, NJ; and Y. Tissaoui, M. Li, D. H. Marsico, S. Du, S. N. Stechmann, and S. Marras

Poster - 34th Conference on Weather Analysis and Forecasting Wednesday Posters

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Lisa Harris / Boston, MA

3:00 PM - Poster 254 Analysis of the Google DeepMind (GDM) Tropical Cyclone Intensity Forecasts for the 2025 Atlantic Hurricane Season
Jacob Vile, State University of New York at Albany, Albany, NY; and N. Elgalad and B. H. Tang, PhD

3:00 PM - Poster 255 Multi-Month Verification of HRRRv4 Dynamic and Thermodynamic Variables In the Midwestern United States By In-Situ and Ground Based Remote Sensing Observations
Laura Bianco, CIRES, Boulder, CO; and I. V. Djalalova, B. Adler, T. P. Meyers, T. R. Lee, J. Sedlar, D. D. Turner, and V. Caicedo

3:00 PM - Poster 256 Optimizing NWP Model Background Profiles of Atmospheric Thermodynamics Based on Multi-Satellite Radiance Data Fusion
Anthony DiNorscia, Analytical Mechanics Associates, Inc., Hampton, VA; and W. L. Smith Jr.

3:00 PM WITHDRAWN
Poster # 257 Withdrawn: Real-Time Global Variable-Resolution Hurricane Forecasts using the Model for Prediction Across Scales during the 2025-26 Hurricane Forecast Improvement Program Real-Time Experiments Clark Evans, NOAA/OAR/Global Systems Laboratory, Boulder, CO; and S. Trahan, C. R. Holt, B. W. Green, and L. Bernardet Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM - Poster 258 SatCORPS Global Cloud Composite: Enabling Rapid Transfers of Knowledge Between Research and Operations
Thad Chee, LaRC - NASA Langley Research Center, Bethesda, MD; and L. Nguyen, W. L. Smith Jr., D. Painemal, J. Hicks, D. Spangenberg, R. Palikonda, C. W. Fleeger, K. Khlopenkov, A. A. Vakhnin, and K. Birkland

Poster - Cloud Physics Poster III

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

3:00 PM - Poster 181 Ice Nucleating Properties of Secondary Organic Aerosol
Christopher Nathan Rapp, Purdue, West Layfayette, IN; Purdue University, West Lafayette, IN; Purdue University, West Lafayette, IN; and D. Cziczo

3:00 PM - Poster 182 A Simple Observational Constraint for Precipitation Phase Change with Warming
Jennifer Kay, CIRES, Boulder, CO; and K. Friedrich, J. M. Theriault, T. S. L'Ecuyer, and H. Chepfer

3:00 PM - Poster 183 A Detailed Comparison of Cloud Properties During a Marine Cold Air Outbreak in Two Convection Permitting Numerical Weather Prediction Models

Mari Steinslid, University of Bergen, Bergen, Norway; Bjerknes Centre for Climate Research, Bergen, Norway; and M. Kähnert, A. Miltenberger, and H. Sodemann

3:00 PM - Poster 184 Characterization of Digital In-line Holographic Microscopy for Measurement of Droplets and Ice Particles in a Rapid Expansion Aerosol Chamber (REACh)
Samuel Koblensky, Princeton University, Princeton, NJ; and G. Pokrifka, C. Sagan, M. Weichman, and L. Deike

3:00 PM - Poster 185 Evaluation of the Water Content Meter (WCM) Hot Wire Probe for In-Situ Water Content Measurements
Summer Marie Coleman, University of North Dakota, Grand Forks, ND; and D. J. Delene, A. Neumann-Skow, B. Rickbeil, and S. Wagner

3:00 PM - Poster 186 Using HOLODEC to Improve Optical Array Probe Estimates of Small-Particle Concentration Measurements
Sarah Woods, NSF NCAR, Broomfield, CO; and A. Bansemer

3:00 PM - Poster 187 Tying in High Resolution E3SM with ARM Data (THREAD) – Project Overview and Recent Progress
Yunyan Zhang, LLNL, Livermore, CA

3:00 PM - Poster 188 3D Lagrangian Tracking of Cloud Droplets In a Convection-Cloud Chamber
Rose O'Brien Jude, Michigan Technological University, Houghton, MI; and S. P. Singh, J. C. Anderson, R. Grosz, and R. A. Shaw

3:00 PM - Poster 189 A Cloud Droplet Population Model for Warm Rain Initiation
Elisa Lucia Ponte, Princeton University, Princeton, NJ; and G. Pokrifka, S. Koblensky, L. Deike, and M. Mueller

3:00 PM - Poster 190 Next-Generation Optical Instrumentation for Comprehensive Aerosol and Cloud Characterization
Matt Freer, CloudSci LLC, Fort Collins, CO

3:00 PM - Poster 191 Microphysical Response of a Mixed Phase Cloud to Combined Glaciogenic and Hygroscopic Seeding
Jesse C Anderson, Michigan Technological University, Houghton, MI; Michigan Technological Univ., Houghton, MI; and W. Cantrell, H. Fahandezh Sadi, S. P. Singh, and R. A. Shaw

3:00 PM - Poster 192 Moisture Transport, Cloud Structure, and Precipitation Microphysics During a February 2025 Snowfall Event Observed in the Snow Sensitivity to Clouds in a Mountain Environment (S2noCliME) Campaign
Evan Barrett, University of Michigan, Ann Arbor, MI; and C. Pettersen, A. Raudzens Bailey, and C. A. Castillo

3:00 PM - Poster 193 Tracing the Isotopic Signature of Extreme CAO Evaporation In Precipitation
Adriana Raudzens Bailey, University of Michigan, Ann Arbor, MI; and E. Rosky and M. Sarkar

3:00 PM - Poster 194 Understanding the Transition from Shallow to Deep Open-Cellular Convection In a Marine Cold Air Outbreak: A Case Study from CAESAR
Pablo Velt, University of Wyoming, Laramie, WY; and B. Geerts, S. R. Martrich, J. R. French, P. Zuidema, S. Ephraim, G. M. McFarquhar, Z. Xia, and J. D. Doyle

3:00 PM WITHDRAWN
Poster # 195 Withdrawn: Cloud Microphysics and Precipitation Intensity: How Partitioning Between Convective and Large-Scale Cloud Processes Sets Model-Simulated Precipitation Travis W Aerenson, NA, NA, WY; University of Wyoming, Laramie, WY; and D. McCoy, H. Y. Brown, D. A. Caulton, and J. M. Nugent Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM WITHDRAWN
Poster # 196 Withdrawn: Recent Developments in ARM Science Data Products for Advancing Aerosol, Cloud, and Precipitation Research Jennifer M. Comstock, PNNL, RICHLAND, WA; and J. E. Shilling, G. Kulkarni, J. Tian, D. Zhang, I. Silber, S. Collis, S. Gupta, L. Ma, A. Zhou, C. Sivaraman, and M. Levin Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM - Poster 197 Characterization of Precipitation Microphysics on Alaska's North Slope using Long-Term MASC Observations
Carl G. Schmitt, University of Alaska, Fairbanks, Fairbanks, AK; and M. Stuefer and W. Powers

3:00 PM - Poster 198 Evaluation of AROME-Arctic Sensible and Latent Heat Fluxes During the 16 March 2024 Norwegian Sea Marine Cold Air Outbreak
Andrew Michael Dzambo, Cooperative Institute for Severe and High Impact Weather Research and Operations, Norman, OK; and M. Kähnert, Z. Xia, G. M. McFarquhar, and H. Sodemann

3:00 PM - Poster 199 Beyond Better Hardware: A Connected Radar Framework for Studying Convective Storm Dynamics
Nithin Allwayin, Stony Brook University, Stony Brook, NY; and O. Omitusa, J. Yan, K. McKeown, E. Luke, M. Oue, B. Puigdomenech, and P. Kollias

3:00 PM - Poster 200 New Methods to Retrieve Near-Cloud Aerosol Optical Depth and Size Characteristics from Geostationary Satellite Observations
Olivia Pierpaoli, Colorado State University, Fort Collins, CO; and J. C. Chiu, J. J. Gristey, and G. Feingold

3:00 PM - Poster 201 Quantify Ice Generation of Cold Air Outbreak Clouds with Combined Airborne Lidar and Radar Observations

Zhien Wang, Stony Brook University, Stony Brook, NY; and B. Geerts, P. Zuidema, C. D. Grasmick, and O. Cruikshank

3:00 PM - Poster 202 Inferring Processes Governing Cloud Transition during Mid-Latitude Marine Cold-Air Outbreaks from Satellite
Jianhao Zhang, Univ. of Colorado/CIRES, Boulder, CO

3:00 PM - Poster 203 Synoptic Influences on Snowfall Regimes during a Multi-day S2 noCliME Event
Marian Mateling, Univ. of Wisconsin-Madison, MADISON, WI; and A. Rowe, C. Pettersen, G. G. Mace, L. A. McMurdie, B. Dolan, A. G. Hallar, M. Oue, and S. Benson

3:00 PM - Poster 204 Inferring Aerosol Properties from Droplet Size Modes in Steady-State Convection-Cloud Chambers
Grant Daniels, University of Utah, Salt Lake City, UT; and S. K. Krueger and C. Bois

3:00 PM - Poster 205 Along-Track Radar Perturbations Reveal Dynamical Structure within the Melting Layer of Extratropical Cyclones
Troy Justin Zaremba, University of Washington, Seattle, WA; and O. B. Samolej, R. M. Rauber, P. Blossey, and L. McMurdie

3:00 PM - Poster 206 Quantifying the Evolution of Cloud Street Structure During Arctic Marine Cold Air Outbreaks Using Satellite Observations
Hannah Sundermann, Leibniz Institute for Tropospheric Research, Leipzig, SN, Germany; and M. Klingebiel, A. Ehrlich, and H. M. Deneke

3:00 PM - Poster 207 Measurements of Saturation Ratio, Water Droplets, and Ice Crystals In a Rapid Expansion Chamber
Gwenore Pokrifka, Princeton University, Princeton, NJ; and C. Sagan, S. Koblensky, M. Weichman, and L. Deike

3:00 PM - Poster 208 Modeling Isotopic Fractionation with Particle-Based Microphysics Using the Bolin Number
Agnieszka Żaba, AGH University of Krakow, Krakow, Poland; and M. Zimnoch and S. Arabas

3:00 PM WITHDRAWN
Poster # 209 Withdrawn: Numerical Simulation of Artificially Initiated Deep Moist Convection via Concentrated Surface Heating Willoughby Winograd, Convective Technology Company, San Jose, CA; and D. Anderson Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM - Poster 210 Enhancement Above Expectation: Rigorous Statistical Evaluation of Weather Modification Performance
Rutuja Dongre, Rain Enhancement Technologies, Naples, FL; and S. Morris, J. Chagnon, T. Gresham, and E. Powell

3:00 PM - Poster 211 Electrostatic Fog Dispersal: Deployment and Optimization of Ground-Based Ionization System
Taylor Gresham, Rain Enhancement Technologies, Naples, FL; and S. Morris, J. Chagnon, R. Dongre, and E. Powell

3:00 PM - Poster 212 Utilization of MAPNet Ground Based Remote Sensing Instruments for Improved Understanding of Cloud and Boundary Layer Interactions
Kevin Knupp, Univ. of Alabama in Huntsville, Huntsville, AL; and S. W. Freeman and P. T. Pangle

Poster - POSTER General Contributions to Severe Local Storms Research

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 221 Discriminating Between Tornadic and Nontornadic Supercell Synoptic Environments Using Self-Organizing Maps
Brandon Antonio Garcia, Pennsylvania State University, University Park, PA; and K. A. Bowley, P. Markowski, and M. Gervais

3:00 PM - Poster 222 Case Study of 25 June 2024 "Whitman, NE" Tornadic Supercell
Josie Pettis, University of Nebraska - Lincoln, Lincoln, NE; and A. L. Houston

3:00 PM - Poster 223 Creating C++ Code to Efficiently Compute Sounding-Derived Parameters
Daniel S Schwab, Purdue University, West Lafayette, IN; and D. Chavas, D. T. Dawson II, J. M. Peters, and N. A. Goldacker

3:00 PM - Poster 224 A Track-Based Dataset of Tornadoes, Hail, and Wind
Cameron J. Nixon, CIWRO, Norman, OK; and I. L. Jirak and N. A. Wendt

3:00 PM - Poster 225 Developing an Updated Analog-Based Severe Weather Forecast System
Cameron J. Nixon, CIWRO, Norman, OK; and I. L. Jirak, R. E. D. Jewell, and A. D. Moore

3:00 PM - Poster 226 A Multi-Scale Analysis of Tropical Cyclone Supercells and Their Environments
Benjamin A. Schenkel, Cooperative Institute for Severe and High-Impact Weather Research and Operations, The Univ. of Oklahoma, Norman, OK; and J. H. Ruppert Jr., M. C. Brown, T. Sandmael, K. M. Calhoun, C. R. Homeyer, T. J. Galarneau, and R. Edwards

3:00 PM - Poster 227 Expansion and Reanalysis of the Pre-1950 U.S. Tornado Record
Blue Ayanami, Tornado Archive, Salt Lake City, UT; and J. Finch, N. Wilkes, Z. Esch, and A. Berrington

3:00 PM - Poster 228 Investigating Climatological Relationships Influencing Tropical Cyclone Tornado Environmental Favorability
Kailey R Jones, Purdue University, West Lafayette, IN; and D. R. Chavas

3:00 PM - Poster 229 Comparing Size-Sorting Signature Algorithm Performance in Distinguishing Tornadic and Nontornadic Supercells
Jacob Segall, CIWRO, Oklahoma City, OK; and S. Loeffler and M. Wilson

3:00 PM - Poster 230 Warning Performance Rates of Low-Buoyant Supercells
Josue Chamberlain, Pennsylvania State University, State College, PA; and J. M. Peters

3:00 PM - Poster 231 Evaluating Impact-Based Warning Guidelines for Tornado Emergency Issuance
Evan Gustafson, Valparaiso Univ., Valparaiso, IN; and E. Conklin, L. Blind-Doskocil, E. C. Wolff IV, J. Goodnight, and J. E. Trujillo-Falcón

3:00 PM - Poster 232 The Sensitivity of the Impact of Cell Mergers on Supercell Thunderstorms Before vs. After Sunset
Ethan Wanley ONeill, Central Michigan Universtiy, Mount Pleasant, MI

3:00 PM - Poster 233 Rivers and Rotation: Characteristics of Tornado Path Length and Width Near the Upper Mississippi River
Nathan Makowski, University of Illinois, Urbana, IL; and R. J. Trapp and A. M. Klees

3:00 PM - Poster 234 Potential of Quantum Computing for Modeling Severe Convective Weather
Oliver Isaac Jackson, ASCE, Rolla, MO

3:00 PM WITHDRAWN
Poster # 235 Withdrawn: Evolution and Microphysical Structures of a Narrow-Core Muticellular Hail-Bearing Storm over the Pingtung Plain in South Taiwan Rong-Guang Hsiu, National Taiwan University, Taipei, Taiwan; and J. D. B. Jou Sr. Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM - Poster 236 The Diurnal Cycle of Tornadoes, Rainfall, and Precipitating Cloud Types within Landfalling Tropical Cyclones
Kayla Marie Wheeler, CIWRO, Univ. of Oklahoma, Norman, OK; and B. A. Schenkel, J. H. Ruppert Jr., and M. C. Brown

3:00 PM - Poster 237 Closing the Radar Gap for Western Wisconsin Tornadoes
Elijah Conklin, Valparaiso University, Pittsburgh, PA

3:00 PM - Poster 238 An Integrated Meteorological and Damage Analysis of the Historic Tornado Outbreak in Southern Brazil on 7 November 2025
Mauricio Oliveira, Instituto Nacional de Pesquisas da Amazônia (INPA), Manaus, Amazonas, Brazil; Universidade Federal de Santa Maria (UFSM), Santa Maria, Rio Grande do Sul, Brazil; and M. M. Lopes, R. J. C. Santana, V. Goede, B. Z. Ribeiro, G. Schild, L. Santos, L. D. O. dos Santos, E. de Lima Nascimento, I. Costa, Y. Cherbonnier, and V. Ferreira

3:00 PM - Poster 239 Development of a Neural Network-Based T-Matrix Emulator for Polarimetric Radar Forward Modeling
Jacob T. Carlin, NSSL, Norman, OK; and M. Johnson

Poster - POSTER Novel Field Observations of Severe Convective Storms

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 240 Close-Range Radar Observations of QLCS Tornadogenesis Failure Processes during PERiLS
Morgan E. Schneider, Canadian Severe Storms Laboratory, London, ON, Canada; and M. D. Flournoy, M. C. Coniglio, and D. J. Bodine

3:00 PM - Poster 241 Near Inflow Environmental Evolution in PERiLS
Rachel L. Miller, CIWRO - Cooperative Institute for Severe and High-Impact Weather Research and Operations, Norman, OK; and L. E. Pounds, M. D. Flournoy, and M. C. Coniglio

3:00 PM - Poster 242 Leveraging Monocular Depth-Anchored Reconstruction to Mitigate Perspective Distortion in Crowdsourced Hailstone Reports
Hesam Fallahian, Insurance Institute for Business & Home Safety (IBHS), Richburg, SC; and I. M. Giammanco

3:00 PM - Poster 243 Cold Pool Moisture Structure and Relationship to Storm Morphology from US High Plains Radiosonde Observations
Leah D. Grant, Colorado State Univ., FORT COLLINS, CO; and B. D. Ascher, T. W. Barbero, C. M. Davis, J. A. Escobedo, N. M. Falk, S. W. Freeman, C. A. Neumaier, G. R. Leung, A. C. Mazurek, E. A. Sherman, D. S. Veloso-Aguila, B. K. Heffernan, M. Nieto-Caballero, S. M. Kreidenweis, R. J. Perkins, E. A. Stone, and S. C. Van Den Heever

3:00 PM - Poster 244 Variability in Northern Colorado Cold Pool Thermodynamic Profiles as Observed by Drones
Christine A. Neumaier, Colorado State Univ., Fort Collins, CO; and L. D. Grant, B. D. Ascher, T. W. Barbero, C. M. Davis, J. A. Escobedo, N. M. Falk, S. W. Freeman, B. K. Heffernan, G. R. Leung, A. C. Mazurek, M. Nieto-Caballero, E. A. Sherman, D. S. Veloso-Aguila, S. M. Kreidenweis, R. J. Perkins, E. A. Stone, and S. C. van den Heever

3:00 PM - Poster 246 Insights on Near-Surface Vorticity, Tornadogenesis, and Tornadoes from Mobile Doppler Lidar Observations
Joshua G. Gebauer, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and M. C. Coniglio and R. A. Saba

Poster - POSTER Supercell Dynamics and Hazards

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM WITHDRAWN
Withdrawn: Vector Vorticity Forcing of Tornado-Cyclonic Rotation in the Radar-Observed 13-14 April 2018 Calhoun-Sterlington Supercell during VORTEX-SE Conrad L. Ziegler, NSSL, Norman, OK; NSSL, Norman, OK; and T. A. Murphy, Z. Wang, G. Lin, and M. J. Hosek Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM - Poster 247 Analyzing Simulated ZDR Column Characteristics in Supercell Thunderstorms Across Differing Environmental Regimes
Forest Garty, Univ. at Albany (SUNY), Albany, NY; and J. P. Mulholland

3:00 PM - Poster 248 Anything from 10 to 686km, What Influences the Hail Swath Length In Supercells?
Tomas Pucik, European Severe Storms Laboratory - Science and Training, Wiener Neustadt, Austria; and B. E. Coffer, M. Taszarek, P. H. Groenemeijer, and F. Battaglioli

3:00 PM - Poster 249 How Does Extratropical Transition Impact Tropical Cyclone Supercell Characteristics?
Gabriel Cenker, University of Oklahoma School of Meteorology CIWRO, Norman, OK; and M. C. Brown, B. A. Schenkel, and J. H. Ruppert Jr.

3:00 PM - Poster 250 Low-level Storm-relative Flow in Tornadic and Nontornadic Supercells: Using Great Plains Environments to Model Impacts on Supercell Behavior
Luke Anderson, NSSL, Norman, OK; and M. D. Flournoy and R. A. Saba

3:00 PM - Poster 251 Comparing the Impacts of Cell Mergers on the 23 May 2022 Morton, TX Tornadic Supercell at Different Life Stages
William Louis Faletti Jr., The University of Oklahoma, Norman, OK; and P. Skinner, K. H. Knopfmeier, T. A. Jones, and M. Coniglio

3:00 PM - Poster 253 Volatility of Large Hail Growth in a Supercell “Ensemble of Ensembles”
Matthew D. Parker, North Carolina State University, Raleigh, NC; North Carolina State Univ., Raleigh, NC; and M. Kumjian and K. Lombardo

Poster - Posters: Modeling and Remote Sensing of Earth's Radiation Budget, Climate and Relevant Processes

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / Jet Propulsion Laboratory, California Institute of Technology / Pasadena, CA

3:00 PM - Poster 1 Investigating the Impact of Nonlinear Interactions on Radiative Forcing from CO2
Abigail Marie McDonnell, University of Miami, Miami, FL

3:00 PM - Poster 2 Using Dual-Regression and MODIS to Produce Month Averaged Polar Hyperspectral Soundings
Anthony DiNorscia, Analytical Mechanics Associates, Inc., Hampton, VA; and S. Kato, X. Huang, and X. Chen

3:00 PM - Poster 3 Analysis of CCCma RTM Surface and TOA fluxes at ARM SGP and ENA sites
Jordann Brendecke, University of Arizona, Tucson, AZ; Univ. of Arizona, Tucson, CO; and X. Dong, B. Xi, and H. Barker

3:00 PM - Poster 4 Investigating Dust Scattering Models for GEO and LEO AOD Retrievals of Saharan Dust
Min Oo, CIMSS, Madiosn, WI; and R. Holz, J. S. Reid, A. Gumber, M. J. Foster, and S. Wanzong

3:00 PM - Poster 5 Evaluating Surface Radiation and Precipitation Uncertainties in Atmospheric Energy Balance
Zhujun Li, Analytical Mechanics Associates, Hampton, VA; and W. S. Olson, J. B. Roberts, S. Kato, P. W. Stackhouse Jr., and M. G. Bosilovich

3:00 PM - Poster 6 EarthCARE Observations of Cloud Radiative Properties and Vertical Velocity
Kaori Sato, Research Institute for Applied Mechanics, Kyushu University, Kasuga, Japan; and H. Okamoto, T. Nishizawa, and Y. Jin

3:00 PM - Poster 7 The Community Radiative Transfer Model (CRTM): From research to operations
Cheng Dang, UCAR, Boulder, CO; UCAR, Boulder, CO; and B. Johnson, Q. Liu, and Y. Ma

3:00 PM - Poster 8 Investigating Reduced-Latency Surface Radiative Hourly Fluxes Using Global Cloud Composite Data and Radiative Transfer Modeling
PC Sawaengphokhai, ADNet System, Inc, Hampton, VA; and P. W. Stackhouse Jr., F. L. Chang, A. Gopalan, H. Winecoff, W. L. Smith Jr., D. R. Doelling, and K. M. Xu

3:00 PM - Poster 9 Improvement of Radiative Fluxes for the Edition 5 CERES FluxByCldTyp Data Product
Moguo Sun, Analytical Mechanics Associates, Inc, Hampton, VA; LRC, Hampton, VA; and D. R. Doelling and J. Wilkins

3:00 PM - Poster 10 Fusion of VIIRS and CrIS Observations to Generate MODIS-Like Infrared Absorption Band Radiances with Pixel Level Uncertainty Estimates
E. Eva Borbas, CIMSS, Madison, WI; and E. Weisz and W. P. Menzel

Poster Viewing and Coffee Break

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Joint J12 - NWA Research Operations Nexus (RON) at the AMS 2026 Joint Summit

4:30 PM - 6:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitters: Jack Lind / NWS / Lenexa, KS / Austin A. Coleman / CIRES / WPC / Boulder, CO / Janice Bunting / National Weather Association (NWA) / Norman, OK / Alyssa V. Bates / CIWRO / Norman, OK / Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL / View Related / 34th Conference on Weather Analysis and Forecasting / 30th Conference on Numerical Weather Prediction / 32nd Conference on Severe Local Storms

Panel Discussion 12 - What’s Next? Navigating Careers and Resilience in Atmospheric Radiation Science

4:30 PM - 6:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Cochairs: Daniel Feldman / LBNL / Berkeley, CA / Yan Feng / ANL / Hinsdale, IL

Panelists: Shaima L. Nasiri / DOE / Washington, DC / Edward P. Nowottnick / NASA / Washington, CO / Jean-François Lamarque / NCAR / Boulder, CO / Laura D. Riihimaki / Boulder, CO

4:30 PM Panel Discussion

Session 12 - Advances in Instrumentation and Analytical Techniques for Cloud and Aerosol Research

4:30 PM - 6:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Sarah Woods / NSF NCAR / Broomfield, CO

Cochairs: Fan Mei / PNNL / Richland, WA / Justin Jacquot / NSF NCAR / Broomfield, CO

4:30 PM - 12.1 Quantifying the Giant Sea Salt Aerosol Size Distribution with the Mini-GNI
Alison D. Nugent, Univ. of Hawaii at Manoa, Honolulu, HI; and K. L. Ackerman

4:45 PM - 12.2 Recent Advances in Phase-Doppler Interferometry for Efficient Cloud Characterization in Airborne, Wind Tunnel, and Laboratory Applications
Mason Douglas Leandro, Artium Technologies Inc., Sunnyvale, CA; Univ. of California, Santa Cruz, Santa Cruz, CA; and W. Bachalo, K. Ibrahim, G. Payne, P. Chuang, and M. K. Witte

5:00 PM - 12.3 UAS, Remote Sensing, and Ground-Based In-Situ Observations of Stratiform Clouds in Finnish Subarctic
Marie Lou Hirschy, University of Helsinki, Helsinki, Uusimaa, Finland; and K. Doulgeris, D. Brus, and D. Moisseev

5:15 PM - 12.4 Same Particles, Different Answers: Algorithmic Uncertainty in Airborne Optical Array Probe Microphysical Retrievals
Yongjie Huang, Univ. of Oklahoma, Norman, OK; and G. M. McFarquhar, A. Bansemer, M. Wolde, L. Nichman, and S. Woods

5:30 PM - 12.5 Identifying Ice Crystal Chain Aggregates in Cold-Season Storms Using Machine Learning to Map Occurrence and Distribution
Christian M. Nairy, University of North Dakota, Grand Forks, ND; and D. J. Delene, S. Wagner, J. A. Finlon, and J. E. Yorks

5:45 PM - 12.6 Artificial Intelligence (AI)-Ready and Machine Learning-Powered Cloud and Precipitation Value-Added Products Synthesizing ARM Observations
Israel Silber, PNNL, Richland, WA; and J. M. Comstock, C. R. Williams, A. Theisen, S. Giangrande, M. Kieburtz, Z. Zhu, and J. Kyrouac

Session 12 - Techniques for Evaluating Weather Modification & Societal Implications

4:30 PM - 6:00 PM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | 25th Conference on Planned and Inadvertent Weather Modification

Cochairs: Jonathan Jennings / Utah Division of Water Resources / Salt Lake City, UT / Sisi Chen / NCAR / Longmont, CO

4:30 PM - 12.1 Increasing Awareness of the Role of the National Oceanographic and Atmospheric Administration in Weather Modification Recording and Publishing
Linden Wolf, OAR, Silver Spring, MD; NOAA, Silver Spring, MD

4:45 PM - 12.2 A Radar-GOES Fusion Method for Evaluating Hail Suppression in Western North Dakota
Lynnlee Storm Bestul, University of North Dakota, Grand Forks, ND; and M. Majdi and M. Chrit

5:00 PM - 12.3 Deciphering Seeding Signals from Noise In Model Simulations
Lulin Xue, PhD, ; and W. Li, S. Chen, N. Dawson, PhD, M. A. Harrold, M. E. Frediani, PhD, T. Eidhammer, C. Weeks, E. M. Dougherty, S. A. Tessendorf, and J. K. Wolff

5:15 PM - 12.4 Analog Ensembles as a Precipitation Benefit Estimation Tool

Nicholas Dawson, PhD, ; and S. Alessandrini, C. Weeks, S. A. Tessendorf, D. Blestrud, J. Russell, PhD, C. Keene, and F. Gariglio

5:30 PM Linking Precipitation Enhancement to Streamflow: A Physically Based Hydrologic Approach
Yongxin Zhang, NCAR, Boulder, CO; and A. Dugger, A. Rafieei Nasab, and N. Dawson, PhD

Session 12A - Novel Field Observations of Severe Convective Storms II

4:30 PM - 6:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Robby M. Frost / Univ. of Oklahoma / Norman, OK / Leanne Blind-Doskocil / Valparaiso Univ. / Valparaiso, IN

4:30 PM - 12A.1 Retrieving the Four-Hour Evolution of the 23 May 2022 Morton, TX Tornadic Supercell Using Recent Innovations in Storm-Scale Data Assimilation
William Louis Faletti Jr., The University of Oklahoma, Norman, OK; and P. Skinner, K. H. Knopfmeier, T. A. Jones, M. Coniglio, and M. Satrio

4:45 PM - 12A.2 An Examination of Unique Lightning Hole Evolution during PERiLS
Vanna C. Chmielewski, NSSL, NORMAN, OK; and A. Adderley and S. M. Stough

5:00 PM - 12A.3 Ensemble Kalman Filter Data Assimilation of VORTEX-SE P3 Compact Raman Lidar and Tail Doppler Radar Data for a 13 April 2013 Tornadic Supercell
Jonathan Tyler DeGraw, CAPS, Norman, OK; Univ. of Oklahoma, Norman, OK; and M. Xue, X. M. Hu, R. Kong, N. Du, C. L. Ziegler, Z. Wang, and G. Lin

5:15 PM - 12A.4 Simulated Convection in Tornadic and Non-Tornadic Environments from Collaborative Field Campaigns in the Southeast U.S.
Bobby Saba, OU/CIWRO, Norman, OK; and M. D. Flournoy and M. Coniglio

5:30 PM - 12A.5 Insights into the Low-Level Kinematics of Tornado Vortices from Mobile-Radar-Observed Polarimetric Doppler Spectra
Ameya Naik, Univ. of Oklahoma, Norman, OK; and H. B. Bluestein, D. J. Bodine, D. Schvartzman, B. Cheong, T. Y. Yu, and L. M. Swinney

5:45 PM - 12A.6 Tornadogenesis Observations from the RaXPol: Elucidating the Rapid Evolution of Near-Surface Vorticity Structure and Updraft Evolution in Two Cyclic Tornadic Supercells
David J. Bodine, University of Oklahoma, Norman, OK; and H. B. Bluestein, R. M. Frost, A. Schueth, A. Naik, H. W. Hsu, S. Emmerson, D. Candela, J. G. Gebauer, and M. C. Coniglio

4:30 PM WITHDRAWN
Withdrawn: Preliminary results from the BEST 2024 and 2026 low-level tornado wind studies Joshua Wurman, Flexible Array of Radars and Mesonets (FARM), UAH, Boulder, CO Madison Ballroom A (Monona Terrace Community and Convention Center) Cancelled

Session 12B - Emerging Datasets and Climatologies for SLS Research

4:30 PM - 6:00 PM | Hall of Ideas - HI (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Luke Justin LeBel / The Pennsylvania State University / State College, PA / Rachael Nicole Cross / University of Oklahoma / Norman, OK

4:30 PM - 12B.1 Know Your Data: A Comprehensive Severe Weather Report Climatology and Analysis of Dataset Source Impacts
Thea Sandmael, Cooperative Institute for Severe and High-Impact Weather Research and Operations, Univ. of Oklahoma, Norman, OK

4:45 PM - 12B.2 A Climatology of the Conditional Probability of Significant Severe Hazards and Its Operational Implications
David R. Harrison, Cooperative Institute for Severe and Hazardous Weather Research and Operations, University of Oklahoma, Norman, OK; and I. L. Jirak and P. T. Marsh

5:00 PM - 12B.3 An Observational Analysis Evaluating Drivers of Severe Weather Production in Supercell Thunderstorms Near Surface Boundaries
Jasen Greco, University of North Carolina at Charlotte, Charlotte, NC; and C. E. Davenport and C. J. Nixon

5:15 PM - 12B.4 Meteorologically Based 30-Year Climatologies of U.S. Tropical Cyclone Tornadoes
Roger Edwards, Supercell Scientific and Consulting, LLC, Norman, OK; and R. M. Mosier

5:30 PM - 12B.5 Democratizing Severe Storm Science: A Natural Language AI Interface for High-Resolution Historical Data Access and Analysis
Kaitlyn N. Fons, Athenium, Madison, WI; and J. G. Fairman Jr, M. Dyanati, and A. Vacek

5:45 PM - 12B.6 Comparison of Observed Southeast U.S. Environments to SFCOA and HRRR Modeled Environments
Lauren E. Pounds, Cooperative Institute for Severe and High-Impact Weather Research and Operations, Norman, OK; Univ. of Oklahoma, Norman, OK; and M. D. Flournoy, M. C. Coniglio, A. Wade, and R. L. Miller

Event - Women & Early Career Reception

7:00 PM - 9:00 PM | Description | Events

Submitter: Jennifer Walton / Girls Who Chase / Broomfield, CO / View Related / Events

Thursday, August 6, 2026

Session 13 - Cloud and Precipitation Physics: Results from Laboratory, Field, Modeling, and Theoretical Studies II

8:30 AM - 10:00 AM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Kamal Kant Chandrakar / NCAR / Boulder, CO / Fan Yang / Brookhaven National Laboratory / Mastic, NY / Sisi Chen / NCAR / Longmont, CO

8:30 AM - 13.1 Is the Freezing Mode Important for the Growth Rate of Frozen Solution Particles?
Dixuan Rui, Penn State University, University Park, PA; and H. Zangara, A. M. Moyle, and J. Y. Harrington

8:45 AM - 13.2 How Do Secondary Ice Production Processes Impact Precipitation Formation in Deep Convection?
Zachary J. Lebo, Univ. of Oklahoma, Pasadena, CA; and Y. Hu

9:00 AM - 13.3 Assessing Ice-Phase Microphysical Parameterizations for Winter Storms in the United States using In-Situ Observations
Brian Filipiak, University of Connecticut, Storrs, CT; Eversource Energy Center, Storrs, CT; and D. Cerrai and M. Astitha

9:15 AM - 13.4 Can Studying Habits and PSDs Together Improve Understanding of Growth Processes Affecting Snow Ratio, Onset, and Amount?
Peter Anthony Brechner, MS, Univ. of Oklahoma/CIWRO, Norman, OK; and G. M. McFarquhar, J. C. Douglas, J. C. Schima, D. J. Delene, C. Nairy, K. L. Thornhill, J. Finlon, and D. Toohey

9:30 AM - 13.5 Reducing Uncertainty In Ice Crystal Size Distributions through Multi-Frequency Radar Perspectives
Julia A Shates, SIO, La Jolla, CA; and D. J. Posselt, PhD, M. D. Lebsock, and J. A. Finlon

9:45 AM - 13.6 Ice Microphysics in Stratiform and Convective Precipitation Regions during the IMPACTS Field Campaign
Jack Ryan Richter, University of Michigan, Ann Arbor, MI; and C. Pettersen, L. McMurdie, A. Raudzens Bailey, and J. A. Shates

Session 13 - Forecast Verification Methods and Performance: Model Evaluation Techniques Part I

8:30 AM - 10:00 AM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / CIRES / WPC / Boulder, CO

Cochairs: Jun Du / NOAA / James A. Nelson / NWS / College Park, MD

8:30 AM - 13.1 Verification of GEFS Precipitation for Atmospheric River-Associated Northeast Extreme Precipitation Events
Jackson Lane Powers, Western Kentucky University, Bowling Green, KY; and R. D. Torn and K. L. Corbosiero

8:45 AM - 13.2 Predictability of Mediterranean Cyclones: A Process-Based Analysis of Upper-Level Potential Vorticity and Precipitation Features.
Tali Sarit Gens, Weizmann institute of science, Rehovot, Israel; and L. Magaritz-Ronen and S. Raveh Rubin

9:00 AM - 13.3 Analysis on TC forecast of CMA-TYM (GRAPES_TYM) driven by CMA-GFS(GRAPES-GFS)
Suhong Ma, CEMC (CMA Earth System Modelling and Prediction Centre), BEIJING, China

9:15 AM - 13.4 Revisiting the Fractions Skill Score: Sensitivity to Multi-scale Structure
Michael E. Baldwin, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and I. L. Jirak, H. E. Brooks, and T. A. Supinie

9:30 AM - 13.5 Linking Land Surface and Boundary Layer Errors with Land-Atmosphere Process Representation in the High-Resolution Rapid Refresh (HRRR)
Julia M. Simonson, CIRES, Univ. of Colorado, Boulder, CO; Developmental Testbed Center, Boulder, CO; NOAA/ESRL/GSL, Boulder, CO; and D. D. Turner, T. R. Lee, T. P. Meyers, I. V. Djalalova, B. Adler, K. A. Balmes, L. Bianco, V. Caicedo, J. Sedlar, L. D. Riihimaki, K. O. Lantz, C. Bernacchi, T. Pederson, M. C. Anderson, W. P. Kustas, and C. R. Hain

9:45 AM - 13.6 Comparison of the January 2026 Cold Air Outbreak Forecast Performance in Artificial Intelligence Weather and Traditional Physics-Based Weather Models
William Y. Cheng, NSF NCAR, BOULDER, CO; and L. Xue, PhD, J. C. Knievel, R. S. Sheu, and J. S. Shaw

Session 13 - Ice Nucleation and Glaciogenic Cloud Seeding Studies

8:30 AM - 10:00 AM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | 25th Conference on Planned and Inadvertent Weather Modification

Cochairs: Binod Pokharel / Utah State University / Logan, UT / Lynnlee Storm Bestul / University of North Dakota / Grand Forks, ND

8:30 AM Evaluating Precipitation Impacts of Cloud Seeding over the Central Colorado Mountains River Basins (CCMRB) Region using WRF-WxMod®
Michelle A. Harrold, NCAR, Boulder, CO; and W. Li, S. Chen, C. Weeks, S. A. Tessendorf, J. K. Wolff, and T. Blitz

8:45 AM Numerical Investigation of the Snowpack Enhancement Program In Colorado's Front Range: From Evaluation to Optimization of Glaciogenic Seeding
Sisi Chen, NCAR, Longmont, CO; and S. A. Tessendorf, J. K. Wolff, T. Blitz, J. A. Grim, and S. Griebling

9:00 AM Dynamic Responses due to Cloud Seeding in an Orographic Wintertime Cloud during SNOWIE
Christopher Hohman, Univ. of Wyoming, Laramie, WY; and J. R. French, L. Xue, PhD, S. Chen, K. Friedrich, S. A. Tessendorf, B. Geerts, R. M. Rauber, and C. D. Grasmick

9:15 AM - 13.4 Seasonal Simulations of Wintertime Orographic Cloud Seeding in the Sierra Nevada
Alex Adams, University of California, Davis, Davis, CA; and H. E. Pimperl, A. L. Igel, S. Chen, L. Xue, PhD, and S. A. Tessendorf

9:30 AM - 13.5 Identifying Dynamical and Microphysical Drivers of Activation and Snow Growth from Ground-Based Seeding in Complex Terrain
Joshua Gooch, University of Colorado - Boulder, Lafayette, CO

Session 13 - R2O2R Activities: Testbed Activities Part I

8:30 AM - 10:00 AM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Austin Coleman / NOAA / Tomer Burg / WindBorne Systems / Palo Alto, CA

8:30 AM - 13.1 The NOAA Hazardous Weather Testbed Severe Weather Extended-Range Forecasting and Verification Experiment
Thomas Galarneau, NSSL, Norman, OK; and I. L. Jirak, A. J. Clark, and K. A. Hoogewind

8:45 AM - 13.2 Toward WoFSv2: Evaluating an MPAS-Based Storm-Scale Ensemble Prediction System in NOAA Testbeds
Nusrat Yussouf, Cooperative Institute for Severe and High-Impact Weather Research and Operations, Univ. of Oklahoma, Norman, OK; and C. S. Schwartz, S. Ha, Y. Wang, K. H. Knopfmeier, T. A. Jones, J. Martin, B. C. Matilla, L. J. Wicker, P. Skinner, E. Mansell, M. K. Silcott, H. Obermeier, D. R. Stratman, C. A. Kerr, A. J. Clark, C. K. Potvin, P. C. Burke, C. Liu, I. L. Jirak, and J. A. Nelson Jr.

9:00 AM - 13.3 Ensembles, AI, and CAMs, Oh My! Insights from the 2026 Precipitation Extremes for Atmospheric Rivers Experiment
Emily Nicole Tinney, NOAA/CIRES, Boulder, CO; and J. Correia Jr., S. M. Trojniak, W. M. Bartolini, K. J. Harnos, and J. A. Nelson Jr.

9:15 AM - 13.4 Comparison of HREF and REFS Neighborhood Probabilities for QPF Exceedances
Sarah Marie Trojniak, CIRES, NOAA/NWS/WPC/HMT, College Park, MD; and J. Correia Jr., D. R. Jesberg, J. L. Bytheway, and K. M. Mahoney

9:30 AM - 13.5 Results and Findings of CAPS CAM Forecasts during the 2024 and 2025 HMT Flash Flood and Intense Rainfall (FFaIR) Experiments
Nathan A. Snook, CAPS, Norman, OK; and A. Berrington, K. A. Brewster, and M. Xue

9:45 AM - 13.6 Limitations of Existing NWP Precipitation-Type Algorithms in Sleet and Freezing Rain Cases: Results from the 2025-2026 Winter Weather Experiment
W. Massey Bartolini, CIRES, College Park, MD; and J. Correia Jr., S. M. Trojniak, and J. A. Nelson Jr.

Session 13 - The Spectral Dimension of Atmospheric Shortwave and Longwave Radiation

8:30 AM - 10:00 AM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / Jet Propulsion Laboratory, California Institute of Technology / Pasadena, CA

Cochairs: David Jeffrey Paynter / NOAA/GFDL / Princeton, NJ / Daniel Feldman / LBNL / Berkeley, CA

8:30 AM - 13.1 Using a Physics-Informed Neural Network to Retrieve Vertical Cloud Droplet Profiles from Hyperspectral Measurements: Beyond the Optimal Estimation Method
Andrew John Buggee, University of Colorado Boulder, Boulder, CO; Laboratory for Atmospheric and Space Physics, Boulder, CO; and V. Nataraja and P. Pilewskie

8:45 AM - 13.2 Global Trends In Earth's Spectrally Resolved Outgoing Longwave Radiation
Mareya Saba, Princeton University, Princeton, NJ; and D. J. Paynter, R. Kramer, and V. Ramaswamy

9:00 AM - 13.3 Spectral Fingerprinting of E-AERI Observations Highlights the Role of Cloud Longwave Feedback on Arctic Amplification
Benjamin Riot Bretecher, McGill University, Department of Atmospheric and Oceanic Sciences, Montreal, QC, Canada; and Y. Huang

9:15 AM - 13.4 CLARREO Pathfinder: The First Earth-observing SI-Traceable Satellite Sensor (SITSat)
Yolanda L. Shea, Hampton, VA; and R. Bhatt, P. Smith, and P. Pilewskie

9:30 AM - 13.5 How Clear Skies Shape Radiation In Cloudy Atmospheres
Paulina Czarnecki, Columbia University, New York, NY; and R. Pincus

9:45 AM - 13.6 Infrared Hyperspectral Observations and Climate: An Unfolding Trilogy
Xianglei Huang, Univ. of Michigan, Ann Arbor, MI

Session 13 - Tornadogenesis: New Insights from Observations and Models

8:30 AM - 10:00 AM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Laura Shedd / Norman, OK / A. Addison Alford / Norman, OK

8:30 AM - 13.1 Digging For Worms: Characterizing Near-Ground Vertical Vorticity During Supercell Tornadogenesis with Doppler Radar
Robby M. Frost, University of Oklahoma, Norman, OK; and D. Bodine, J. Gebauer, and M. Coniglio

8:45 AM - 13.2 Dissecting Vorticity Worms: An Analysis of TTUKa Radar Inferred Vertical Vorticity Observations in Supercell Thunderstorm Inflow Environments
Josiah John Melke, Texas Tech Univ., Lubbock, TX; and C. C. Weiss

9:00 AM - 13.3 "Immense" ZDR Columns: A Marker for Tornadic Supercells
Michael M. French, Stony Brook Univ., Stony Brook, NY; and D. M. Kingfield

9:15 AM - 13.4 Vortex Merging and Intensification of Tornado Like Vortices in Idealized CM1 Simulations.
Divya Viswanath, Texas Tech University, Lubbock, TX; and J. M. L. Dahl and J. Fischer

9:30 AM - 13.5 Dependence of the In-and-up Vorticity Mechanism on Tornado Structure
Johannes Dahl, Texas Tech University, Lubbock, TX

9:45 AM - 13.6 Sensitivity of Simulated Vortex Development to Near-Surface Temperature Perturbations
Matthew D. Flournoy, NSSL, Norman, OK

Session 13 - Waves, Wakes, Channeling, and Terrain-Influenced Flows

8:30 AM - 10:00 AM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 22nd Conference on Mountain Meteorology

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Cochairs: William O. J. Brown / NCAR / Boulder, CO / Kristen L. Rasmussen / Colorado State University / Longmont, CO

8:30 AM - 12.1 Diurnal Characteristics of the Island Wake Marine Atmospheric Boundary Layer Downstream of Barbados
James Anthony Hlywiak, NRL, Monterey, CA; and D. Flagg, J. D. Doyle, J. S. Reid, J. Ruiz-Plancarte, C. Jackson, R. Yamaguchi, Q. Wang, and E. Melvin

8:45 AM - 12.2 Unrepresented Tropospheric Drag in Global NWP due to Gravity Wave Trapping
Hette Houtman, University of Reading, Reading, RDG, United kingdom; and M. A. C. Teixeira, S. Gray, S. Vosper, P. Sheridan, and A. van Niekerk

9:00 AM - 12.3 Synoptic Forcing and Vertical Structure of Pressure-Driven Winds in the Snake River Plain
Nicholas Dawson, PhD, NCAR, Boulder, CO; and M. A. Harrold and S. A. Tessendorf

9:15 AM - 12.4 Identifying Control Parameters of Wind Channeling along the Saint Lawrence River Valley using Numerical Simulations
Dustin Roy Douglas Fraser, ATDD, Prince William, NB, Canada; McGill University, Montreal, QC, Canada; and D. J. Kirshbaum and J. R. Gyakum

9:30 AM - 12.5 Comparing FastEddy Simulations and Fire Decision Support Tool Wind Predictions with Observed Doppler LiDAR Data of Thermally Driven Circulations in the Salt Lake Valley
Aaron Caleb McCutchan, USFS Northern Research Station, East Lansing, MI; and B. M. Welch, S. W. Hoch, and J. J. Charney

9:45 AM - 12.6 Evidence of Diurnal Winds In Highly Complex Terrain: The EMRTC 2019 Field Experiment
Patrick Stofanak, LANL, Santa Fe, NM; and Z. Ruble, M. A. Nelson, and S. Brambilla

Coffee Break

10:00 AM - 10:45 AM | Grand Terrace (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Session 14 - Cloud and Precipitation Physics: Results from Laboratory, Field, Modeling, and Theoretical Studies III

10:45 AM - 12:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Sisi Chen / NCAR / Longmont, CO / Kamal Kant Chandrakar / NCAR / Boulder, CO

10:45 AM - 14.1 The IMPACTS Legacy: Airborne Observations of Banded Precipitation Structures in Winter Storms
Lynn McMurdie, Univ. of Washington, Seattle, WA

11:00 AM - 14.2 Snowbands and Vertical Motions: Assessing the Variability of Cloud Properties across Scales in the 23 January 2023 Winter Storm
Jonathan Connor Douglas, The University of Oklahoma/CIWRO, Norman, OK; and G. M. McFarquhar, P. A. Brechner, MS, D. J. Delene, K. L. Thornhill, and C. N. Helms

11:15 AM - 14.3 Multi-Radar 3D Tracking of a Single Cloud-Top Generating Cell and Preliminary Microphysical Interpretation
Jairo Michael Valdivia Prado Sr., University of Colorado Boulder, Boulder, CO

11:30 AM - 14.4 Observation-Constrained Optimization of Hydrometeor Parameters in a Cloud Microphysics Scheme: Impacts on Precipitation and Cloud Vertical Structure
Kyo-Sun Lim, Seoul National University, Seoul, South korea; and K. B. Kim, H. Wang, Y. Qian, K. Kim, and G. LEE

11:45 AM - 14.5 Flow influences on snowfall processes in mountainous terrain: Early insights from the S2noCliME campaign
Angela Rowe, Univ. of Wisconsin-Madison, Madison, WI; and S. Phillips, M. Mateling, T. J. Zaremba, B. Dolan, C. Pettersen, G. G. Mace, L. McMurdie, A. G. Hallar, S. Benson, and M. Oue

Session 14 - Forecast Verification Methods and Performance: Model Evaluation Techniques Part II

10:45 AM - 12:00 PM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / CIRES / WPC / Boulder, CO

Cochairs: Michelle A. Harrold / NCAR / Boulder, CO / Kathryn M. Newman / DTC / Boulder, CO

10:45 AM - 14.1 An Overview of Verification and Evaluation Activities at the NOAA Modeling and Development Center
Jason J. Levit, NOAA/NWS/NCEP/EMC, College Park, MD

11:00 AM - 14.2 METplus: NWP Evaluation Across Scales and Systems
Michelle A. Harrold, NCAR, Boulder, CO; and M. B. Smith, J. H. Gotway, G. P. McCabe Jr., J. Prestopnik, H. Soh, M. Win-Gildenmeister, D. R. Adriaansen, T. J. Hertneky, C. P. Kalb, M. J. Kavulich Jr., G. Ketefian, W. Mayfield, K. Newman, J. Opatz, V. Vargas Jr., and J. L. Vigh

11:15 AM - 14.3 DTC’s METplus Verification Framework: A Unified Approach Towards Accelerating Testing and Evaluation
Michael J. Kavulich Jr., NSF-NCAR, DTC, Boulder, CO; and G. Ketefian, G. P. McCabe Jr., and K. M. Newman

11:30 AM - 14.4 Toward Unified UFS Physics through a Hierarchical Testing Framework
Man Zhang, NOAA Global Systems Laboratory, Boulder, CO; and W. Li, E. Grell, J. Dudhia, T. J. Hertneky, J. M. Simonson, PhD, F. Yang, X. Zhou, L. Bengtsson, and L. Bernardet

11:45 AM - 14.5 Building Community Capabilities for Noah-MP Focused Hierarchical System Development and Evaluation within the Unified Forecast System
Andrew J. Newman, NCAR, Boulder, CO

Session 14 - General Contributions to Severe Local Storms Research

10:45 AM - 12:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Kylee Matousek / Prairie Research Institute / Champaign, IL / Aaron W. Zeeb / Central Michigan University / Mount Pleasant, MI

10:45 AM - 14.1 Impacts of Phased Array Radar on Forecaster Understanding, Confidence, and Communication Practices in the Hazardous Weather Testbed
Charles M. Kuster, NOAA/OAR/NSSL, Norman, OK; and A. A. Alford, K. Tuftedal, J. Madden, J. Monroe, M. Krocak, T. Maciag, J. B. Boettcher, B. R. Smith, T. J. Schuur, E. Blumenauer, J. T. Carlin, K. L. Berry, and A. E. Reinhart

11:00 AM - 14.2 Tornado Emergency! Exploring Their Characteristics and Associated Tornadoes
Scott Gunter, Univ. of Louisville, Louisville, KY; and L. White and C. Ledbetter

11:15 AM - 14.3 A Climatology of Tornado Warning Skill In Landfalling Tropical Cyclones
Benjamin A. Schenkel, Cooperative Institute for Severe and High-Impact Weather Research and Operations, The Univ. of Oklahoma, Norman, OK; and T. Sandmael, K. M. Calhoun, A. A. Alford, H. E. Brooks, R. Edwards, and R. M. Mosier

11:30 AM - 14.4 Spatiotemporal Analyses of Lightning and Tornado Exposure to Large Outdoor Gatherings in the Conterminous United States
Stephen M. Strader, Villanova Univ., Villanova, PA

11:45 AM - 14.5 On the Road Again: An Overview of U.S. Higher Education Storm Observation Programs
Leanne Blind-Doskocil, Valparaiso University, Valparaiso, IN; and K. A. Barber and R. Stenz

Session 14 - Observations of the Far Infrared: from PREFIRE to FORUM

10:45 AM - 12:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / Jet Propulsion Laboratory, California Institute of Technology / Pasadena, CA

Cochairs: Xianglei Huang / University of Michigan / Ann Arbor, MI / Kyle S Mattingly / University of Wisconsin–Madison / Madison, WI

10:45 AM - 14.1 Novel Insights from Multiple CubeSats: An Overview of Orbit Resampling During NASA PREFIRE
Natasha Lyne Vos, University of Wisconsin-Madison, Minneapolis, MN; Univ. of Wisconsin-Madison, Madison, WI; and T. S. L'Ecuyer, T. Michaels, K. S. Mattingly, and C. Schmear

11:00 AM - 14.2 Validation of PREFIRE L2 Geophysical Products: Interdependence, Cross-Platform Synergy, and Ground-Based Validation
Qing Yue, California Institute of Technology, Pasadena, CA; NASA JPL, Pasadena, CA; and B. Kahn, B. Drouin, K. S. Mattingly, N. Miller, T. I. Michaels, L. Borg, T. S. L'Ecuyer, A. Merrelli, X. Chen, and X. Huang

11:15 AM - 14.3 Impact of Spectral Coverage, Resolution and Noise on Infrared Cloud Retrievals: A Simulation Study for IASI, IASI-NG and FORUM
Tiziano Maestri, Univ. of Bologna, Bologna, BO, Italy; and E. Fabbri, M. Martinazzo, G. Masiello, and G. Liuzzi

11:30 AM - 14.4 A Latent-Twin Architecture for Scene Recognition and Fast All-Sky Retrievals from FORUM Far-Infrared Spectra
Cristina Sgattoni, CNR, Sesto Fiorentino, FI, Italy; and M. Chung, L. Sgheri, and M. Martinazzo

11:45 AM - 14.5 Impact of Temperature-Dependent Ice Cloud Optical Properties on Longwave Radiative Fluxes
Elisa Fabbri, University of Bologna, Bologna, BO, Italy; University of Bologna, Bologna, BO, Italy; and T. Maestri, M. Martinazzo, G. Proietti Pelliccia, C. Serio, G. Masiello, G. Liuzzi, T. Ren, and P. Yang

Session 14 - Plumes and Transport over Complex Terrain

10:45 AM - 11:45 AM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 22nd Conference on Mountain Meteorology

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Cochairs: Claire Pettersen / Univ. of Michigan / MADISON, WI / David Robinson / LANL / Santa Fe, NM

10:45 AM - 13.1 Meteorological Impacts on Wintertime Air Quality and Human Health In the Western United States
Heather Holmes, ; and K. Kaur, C. Riss, J. Boomsma, C. Phelan, J. Krall, C. Ivey, L. Darrow, K. Kelly, and M. Strickland

11:00 AM - 13.2 Impact of Wind Flow in Mountainous Terrain on Long-Distance Plume Transport
Matthew A. Nelson, Los Alamos National Laboratory, Los Alamos, NM; and R. L. Beal, S. Brambilla, and H. Boukhalfa

11:15 AM - 13.3 METEX21: Multiscale Observations and Simulations of Plume Behavior across the Turbulence Gray-Zone in Mountain-Valley Terrain
Sonia Wharton, LLNL, Livermore, CA; and D. Wiersema, R. Newsom, W. Schalk, and D. Dexheimer

11:30 AM - 13.4 Advective Carbon Dioxide Transport over Sloped Terrain in a California Coastal Redwood Forest
Corrin Grace Clemons, University of California, Davis, Davis, CA; LBNL, Berkeley, CA; and H. Chu, W. S. Chan, K. T. Paw U, K. Suvočarev, D. Mengering, and H. J. Oldroyd

Session 14 - R2O2R Activities: Testbed Activities Part II

10:45 AM - 12:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Jack Lind / NWS / Lenexa, KS / Kent H. Knopfmeier / NOAA

10:45 AM - 14.1 An Overview of the 2026 Hazardous Weather Testbed Spring Forecasting Experiment
Adam J. Clark, NSSL, Norman, OK; and I. L. Jirak, T. J. Galarneau, T. A. Supinie, K. H. Knopfmeier, D. R. Harrison, J. Vancil, M. K. Silcott, D. E. Jahn, C. Karstens, A. Wade, J. Milne, K. A. Hoogewind, C. K. Potvin, N. Yussouf, S. R. Ernst, J. Picca, M. E. Baldwin, J. A. Duarte, J. Martin, and B. C. Matilla

11:00 AM - 14.2 Advancing Current and Future Geostationary Satellite Capabilities in the Hazardous Weather Testbed
Kevin Thiel, Cooperative Institute for Severe and High-Impact Weather Research and Operations, Norman, OK; and J. Monroe and J. Madden

11:15 AM - 14.3 Evaluation of the RRFS Ensemble Forecast System during the 2026 NOAA Hazardous Weather Testbed Spring Forecasting Experiment
Jacob Vancil, CIWRO, Norman, OK; and I. L. Jirak

11:30 AM - 14.4 WoFSCast Performance in the 2026 HWT Spring Forecasting Experiment
Corey K. Potvin, NSSL, Norman, CO; and J. Martin, M. K. Silcott, J. A. Duarte, A. J. Clark, B. C. Matilla, and P. C. Burke

11:45 AM - 14.5 Lessons from Three Springtime Demonstrations of Real-Time, MPAS-Based, Medium-Range, Convection-Allowing Ensemble Forecasts for NOAA's Hazardous Weather Testbed Spring Forecasting Experiments
Craig Schwartz, NCAR, Boulder, CO; and R. A. Sobash and D. A. Ahijevych

Session 14 - Recent Field Campaign Results on Planned or Inadvertent Weather Modification

10:45 AM - 12:00 PM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | 25th Conference on Planned and Inadvertent Weather Modification

Cochairs: Jonathan Jennings / Utah Division of Water Resources / Salt Lake City, UT / Joshua Gooch / University of Colorado - Boulder / Lafayette, CO

10:45 AM - 14.1 Overview and Early Results from the SNOWSCAPE Winter Orographic Precipitation Field Campaign
Nicholas Dawson, PhD, ; and P. Brooks, A. Dobracki, A. Dugger, P. Golden, J. D. Horel, J. Jennings, G. G. Mace, J. D. D. Meyer, B. Pokharel, M. Skiles, A. Stansfield, J. Steenburgh, K. Suski, S. A. Tessendorf, P. Veals, and D. Yorty

11:00 AM - 14.2 Performance of Forecast Model on Cloud Liquid Water Prediction over the Mountains in Utah: Implications for Cloud Seeding Operations
Binod Pokharel, Utah State University, Logan, UT; Utah State University, LOGAN, UT; Utah State University, Logan, UT; Utah State University, Logan, UT; and C. S. Olson, J. D. D. Meyer, and J. Jennings

11:15 AM - 14.3 Radar and Satellite Seeding Signatures from Drone-Based Glaciogenic Cloud Seeding Operations
Kaitlyn Suski, Rainmaker Technology Corporation, El Segundo, CA; and M. Satrio, E. F. De Biasio, J. Aikins, A. Dixon, A. Dobracki, M. Switzer, C. Wessel, M. Decker, and K. Kaushal

11:30 AM - 14.4 Evolution of Radar Seeding Signatures: Insights from the 2025–2026 Rainmaker Season
Martin Satrio, Rainmaker Technology Corporation, Norman, OK; and J. Aikins, A. Dixon, K. Suski, C. Wessel, and M. Decker

11:45 AM - 14.5 First North American Field Evaluation of Ground-Based Ionization for Wintertime Precipitation Enhancement
Jeffrey Chagnon, Rain Enhancement Technologies, Naples, FL; and S. Morris, R. Dongre, and T. Gresham

Lunch Break

12:00 PM - 1:45 PM | View Related | 34th Conference on Weather Analysis and Forecasting

Session 15 - Advances Facilitated By New Observational Platforms and Capabilities Part I

1:45 PM - 3:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Thea Sandmael / Cooperative Institute for Severe and High-Impact Weather Research and Operations, Univ. of Oklahoma / Norman, OK / Leanne Blind-Doskocil / Valparaiso University / Valparaiso, IN

1:45 PM - 15.1 Evaluation of 2026 AR Recon Observations for the Analysis and Forecasting of Atmospheric Rivers and Downstream Winter Storms Using MPAS-JEDI
Minghua Zheng, Center for Western Weather and Water Extremes, Scripps Institution of Oceanography, Univ. of California San Diego, La Jolla, CA; and L. Delle Monache, G. Casaretto, J. Wang, X. Wu, A. Wilson, V. S. Tallapragada, and F. M. Ralph

2:00 PM - 15.2 WSF-M MWI Calibration and Validation and Impacts with the Navy's Global Model
Hui Christophersen, NRL, Monterey, CA; and S. Swadley, E. Simon, G. Poe, and A. Thorpe

2:15 PM The Role of Stratocumulus Clouds In Modulation of Thermodynamic Profiles within the Cool Season Pre-QLCS Environment
Kevin Knupp, Univ. of Alabama in Huntsville, Huntsville, AL; and P. T. Pangle

2:30 PM - 15.4 Investigating the Capability of the Warn-on-Forecast System (WoFS) to Diagnose Processes that led to Rapid Environmental Destabilization in a High-Shear Low-CAPE QLCS Event
Madeline Diedrichsen, CIWRO, Norman, OK; Univ. of Oklahoma, Norman, OK; and M. Coniglio, W. L. Faletti Jr., P. S. Skinner, and K. H. Knopfmeier

2:45 PM - 15.5 Observations and Realistic Modeling of Convection-Generated Gravity Waves and Breaking during the CGWAVES Field Campaign
Christopher G. Kruse, NCAR, Broomfield, CO; and D. C. Fritts, D. S. Nolan, B. P. Williams, T. S. Lund, Y. Dai, J. A. Guerrero, and L. Hoffmann

Session 15 - Aerosol-Cloud Indirect Effects I

1:45 PM - 3:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Isabel L. McCoy / Florida State University / Tallahassee, FL / Siddhant Gupta / ANL / Lemont, IL

1:45 PM - 15.1 Clarifying Cloud Liquid Response to Aerosol by Accounting for Precipitation
August Mikkelsen, Univ. of Wyoming, Laramie, WY; and D. McCoy, J. M. Nugent, H. Gordon, and J. Mülmenstädt

2:00 PM - 15.2 A New Interpretation of Precipitation Suppression in Shallow Clouds
Fabian Hoffmann, Freie Universität Berlin, Berlin, Berlin, Germany; and T. Goren, G. Choudhury, G. Feingold, and T. Yamaguchi

2:15 PM - 15.3 The Effect of Aerosol Properties on Cloud Droplet Activation and Optical Properties in a Convective Cloud Chamber.
Kadja Flore GALI, Michigan Technological University, Houghton, MI; and H. Fahandezh sadi, J. C. Anderson, P. A. Beeler, A. Wang, D. H. Richter, R. A. Shaw, F. Yang, W. Cantrell, and L. M. Fierce

2:30 PM - 15.4 Surface Observations From Atmospheric Radiation Measurement Sites Constrain the Anthropogenic Contribution to Cloud Droplet Number
Danielle Jones, Univ. of Wyoming, Laramie, WY; and H. Y. Brown, J. M. Nugent, A. Mikkelsen, C. Song, D. Zhang, S. M. Burrows, H. Gordon, A. Kirby, and D. McCoy

2:45 PM - 15.5 Estimating Effective Radiative Forcing from Aerosol-Cloud Interactions from Satellite Observations Through Scaling Cloud Adjustments at Mesoscale to Near Global Scale
Tianle Yuan, UMBC GESTAR-II/NASA GSFC, Greenbelt, MD; and R. Wood, H. Song, L. Yuan, and K. Meyer

Session 15 - Machine Learning/AI Advances in Mountain Meteorology (Joint with the Committee on Artificial Intelligence Applications to Environmental Science)

1:45 PM - 3:00 PM | Meeting Rooms KLOP (Monona Terrace Community and Convention Center) | 22nd Conference on Mountain Meteorology

Submitter: Jason C. Knievel / NSF NCAR / Boulder, CO

Chair: Heather Holmes / University of Utah / Salt Lake City, UT

1:45 PM Examining a Kilometer Scale AI Atmospheric Model for Mountain Meteorology
Ethan D. Gutmann, NCAR, Boulder, CO; and Y. Sha, R. R. McCrary, A. J. Newman, S. McGinnis, T. J. Hertneky, L. Xue, PhD, K. M. Newman, J. Schreck, and D. J. Gagne II

2:00 PM - 14.2 Prediction of Derived and Crowdsourced Precipitation Phase Observations of the Western CONUS via Machine Learning
Savanna Wolvin, University of Utah, Murray, UT; and C. Strong and S. Brewer

2:15 PM Simulation-Driven Machine Learning for Wind Flow Patterns over Complex Terrain
Shane Coffing, LANL, Los Alamos, NM; LANL, Santa Fe, NM; and J. Slaten, D. Robinson, A. Mohan, and S. Brambilla

2:30 PM - 14.4 Short-Term Forecasting of Foehn Events in Complex Terrain: A 3-Hour Lead Time Machine Learning Approach
Jeff (Jafar) Sepehri, PhD Candidate., York University (Student), Toronto, ON, Canada; York University (PhD Candidate), Toronto, ON, Canada; and A. Hakimi

2:45 PM - 14.5 Improving HRRR 1-h APCP with a U-Net and Rain-Aware Sampling
Oscar V Chimborazo, PhD, Howard University, Washington, DC; and S. Chiao

Session 15 - Other Topics on Numerical Weather Prediction

1:45 PM - 3:00 PM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Curtis R. Alexander / NOAA / Boulder, CO / Austin Coleman / NOAA

1:45 PM - 15.1 Can Second-order Numerical Accuracy be Achieved in Cloud-resolving Models?
David H Marsico, CIRES, Boulder, CO; and S. N. Stechmann

2:00 PM - 15.2 The Relative Impacts of Assimilating Uncrewed Aircraft Systems versus Additional Surface Stations in Regional NWP within an OSSE
Shawn Murdzek, CIRES/CU Boulder and NOAA/OAR/GSL, BOULDER, CO; and T. T. Ladwig, A. L. Houston, E. P. James, K. A. Swan, and M. Tsey

2:15 PM - 15.3 Observing System Simulation Experiments for Uncrewed Aircraft Systems: The Relative Impact of Quasi-Horizontal Profiles Versus Vertical Profiles
Adam L. Houston, Univ. of Nebraska-Lincoln, Lincoln, NE; and S. Murdzek, K. A. Swan, T. T. Ladwig, and E. P. James

2:30 PM - 15.4 Practical Application: Quantum-Classical Forecasts of Turbulence Parameters at the Savannah River Site
Joseph E. Wermter, Savannah River National Laboratory (SRNL), Aiken, SC; Savannah River National Laboratory (SRNL), Aiken, SC

2:45 PM - 15.5 Role of Dynamic Vegetation in Noah-MP on the Simulation of Dry and Wet Conditions

David Lorenz, Univ. of Wisconsin-Madison, Space Science and Engineering Center, Madison, WI; and J. A. Otkin, PhD and A. J. Newman

Session 15 - Surface-Atmosphere Interactions and the Surface Energy Budget of the Cryosphere from Satellite and Sub-Orbital Observations

1:45 PM - 3:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Patrick C. Taylor / NASA Langley Research Center / Hampton, VA

Cochairs: Paquita Zuidema / Rosenstiel School, University of Miami / Miami, FL / Kerry Meyer / GSFC / Greenbelt, MD / K. Sebastian Schmidt / Laboratory for Atmospheric and Space Physics / Boulder, CO / Amy B. Solomon / CIRES and NOAA/PSL / Boulder, CO / Patrick C. Taylor / NASA Langley Research Center / Hampton, VA

1:45 PM 1

Summer Surface and Subsurface Energetic Processes in the Percolation Zone of Southwest Greenland Anne Sledd, Univ. of Colorado-Boulder CIRES, Boulder, CO; and M. D. Shupe, M. Gallagher, C. J. Cox, H. P. Marshall, M. Town, C. Pettersen, R. Hawley, R. R. Neely III, H. Guy, D. Pickell, E. Olson, C. Hebson, V. P. Walden, and A. Martin

2:00 PM 2

Quantifying the Contributions of Clouds and Surface Albedo Change to Sea Ice Melt In 2024 during ARCSIX Jason Brant Dodson, NASA, Yorktown, VA; and P. C. Taylor and D. Kim

2:15 PM 3

Bridging the Gap between Shortwave Imager-derived and Directly Measured Cloudy-Sky Surface Radiative Fluxes over the Arctic Ocean using NASA ARCSIX Observations Colten Peterson, Goddard Earth Science Technology and Research II, Baltimore, MD; and K. Meyer, T. Arnold, G. Wind, and N. Amarasinghe

2:30 PM 4

Evolution of Arctic Sea Ice BRDF/Albedo Using MODIS and ARCSIX in a Lagrangian Framework Vikas Hanasoge Nataraja, Laboratory for Atmospheric and Space Physics at the University of Colorado Boulder, Boulder, CO; Department of Atmospheric and Oceanic Sciences, University of Colorado Boulder, Boulder, CO; and K. Hirata, H. Chen, Y. W. Chen, K. Meyer, C. Peterson, and K. S. Schmidt

2:45 PM 5

Spring- and Summertime Airborne Observations in the Arctic Suggest Prolonged Cloud-Induced Surface Melt Yu-Wen Chen, University of Colorado at Boulder, Boulder, CO; University of Colorado Boulder, Boulder, CO; and K. S. Schmidt, K. Hirata, V. Nataraja, and H. Chen

Session 15A - Toward Improved Understanding and Prediction of Heavy Rainfall and Flash Floods

1:45 PM - 3:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Jaime L Herriott / University of Oklahoma / Norman, OK / Melinda Berman / University of Illinois Urbana-Champaign / Urbana, IL

1:45 PM - 15A.1 Convection-Permitting Ensemble Simulations of Extreme Rainfall in Wisconsin: Implications for Probable Maximum Precipitation Assessment

Fitria Puspita Sari, University of Wisconsin-Madison, Madison, WI; and A. Alexander and D. B. Wright

2:00 PM - 15A.2 Medium-Range Excessive Rainfall Prediction with Machine Learning

Aaron J. Hill, University of Oklahoma, School of Meteorology, Norman, OK; and R. S. Schumacher

2:15 PM - 15A.3 The 50th Anniversary of the Big Thompson Flood: Reviewing Past Research and Modern Advances in Predicting Localized Flash Floods

Russ S. Schumacher, Colorado State Univ., Fort Collins, CO; and K. M. Mahoney, R. Kleyla, A. C. Mazurek, K. L. Rasmussen, Ph.D., P. T. Schlatter, and E. A. Sherman

2:30 PM - 15A.4 Coupled WoFS-FLASH System Prediction of the July 2025 Texas Hill Country Flash Flood Disaster

Derek R. Stratman, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and J. J. Gourley, N. Yussouf, P. C. Burke, A. J. Clark, P. Heinselman, and L. J. Wicker

2:45 PM - 15A.5 Evaluating Precipitation Characteristics in Different Storm Structures Using EXPRESS Stations

Ian Earley, University of Louisville, Louisville, KY; and S. Gunter

Session 15B - Insights into Severe Storms through Remote Sensing

1:45 PM - 3:00 PM | Hall of Ideas - HI (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Joshua Gebauer / Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO) / Norman, OK / Matthew D. Flournoy / NSSL / Norman, OK

1:45 PM - 15B.1 Spatiotemporal Evolution of Polarimetric Radar Signatures and Lightning Metrics in Supercells: Observations from the Horus All-Digital Phased Array Radar on 3 June 2025
Vitor Goede, School of Meteorology, Norman, OK; Advanced Radar Research Center, Norman, OK; and D. Schvartzman, D. J. Bodine, and V. C. Chmielewski

2:00 PM - 15B.2 How Different Lightning Measurements Portray Convective Intensity
Sarah M. Stough, Cooperative Institute for Severe and High-Impact Weather Research and Operations, The University of Oklahoma, Norman, OK; and V. C. Chmielewski and K. M. Calhoun

2:15 PM 15B.3 is now Poster 331A

2:30 PM - 15B.4 Three-Dimensional Thunderstorm Wind Retrievals using a Multistatic Passive Weather Radar Network
Patrick Skinner, CIWRO, Norman, OK; and S. Emmerson, B. H. Moose, D. J. Bodine, R. D. Palmer, D. Schvartzman, P. Kirstetter, and B. Cheong

Poster - 30th Conference on Numerical Weather Prediction Thursday Posters

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Lisa Harris / Boston, MA

3:00 PM - Poster 319 Evaluating Storm-Scale Machine Learning Weather Prediction Using Idealized Convective Simulations
Lawrence Jing-Yueh Liu, University of Illinois Urbana-Champaign, Urbana, IL; and K. Droegemeier, J. Gao, and C. K. Potvin

3:00 PM - Poster 320 A Multi-Model Investigation into the Sensitivity of Urban Heat Island Structure to Small Variations in Land Surface Representation
Jason A. Naylor, University of Louisville, Louisville, KY

3:00 PM - Poster 321 Comparing Thunderstorm Nowcasts in Numerical Weather Prediction and Data Driven Models
Jorge Alberto Duarte, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and P. S. Skinner, C. K. Potvin, and M. L. Flora

Poster - 34th Conference on Weather Analysis and Forecasting Thursday Posters

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Lisa Harris / Boston, MA

3:00 PM WITHDRAWN
Poster # 341 Withdrawn: Medium Range Forecasting for Wind-Forced Expansion and Retreat of the Wintertime Bering Sea Ice Edge Using AI/ML and Linear Inverse Modeling Christopher James Cox, NOAA Physical Sciences Laboratory, Westminster, CO Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM WITHDRAWN
Poster # 342 Withdrawn: Bridging the Nighttime Visibility Gap: Validation of Synthetic Lunar Reflectance for Fog Detection Alexxis Brown, U.S. Air Force, Dayton, OH Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM - Poster 343 Success in Simulating CAD: UGA WRF and MPAS Simulations of the January 23-25 Southeast U.S. Ice Storm
Scout Carlson, Univ. of Georgia, Athens, GA; and J. A. Knox

3:00 PM - Poster 344 The Human Side of AI-Assisted Forecasting
Jonathan Fairman Jr., Athenium, Cleveland Heights, OH; and K. N. Fons, A. Vacek, and J. Vivola

3:00 PM - Poster 345 Expanding Synchronous Training for NWP and Machine Learning Literacy to Earth System Science Audiences Anytime and Anywhere
Andrea M. Smith, UCAR, Boulder, CO; and B. Guarente, D. J. Gagne II, and M. G. Cains

3:00 PM WITHDRAWN
Poster # 346 Withdrawn: Assessing Climatic Changes to Lake-Effect Snow in Northern Michigan using WRF-ARW and PGW Tyson Stewart, Northern Illinois University, DeKalb, IL; and V. A. Gensini, W. S. Ashley, A. Haberlie, and A. C. Michaelis Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) Cancelled

Poster - Cloud Physics Poster IV

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

3:00 PM - Poster 269 Positive and Negative CCN-drizzle Relationships
James G. Hudson, Desert Research Institute, Reno, NV; and S. Noble

3:00 PM - Poster 270 Low Cloud Dispersion Effects By Anthropogenic Aerosols In Polluted Air
Xiaomi Teng, Zhejiang Univerisity, Hangzhou, Zhejiang, China; Zhejiang University, Hangzhou, Zhejiang, China; and X. Liang, F. Zhang, Y. Cheng, M. Gao, X. Chen, J. Li, S. Zhou, S. Shu, S. Zhu, D. Liu, C. Zhao, P. P. Ching, and W. Li

3:00 PM - Poster 271 Re-Examination of Aerosol-Cloud-Interaction (ACI) By Accounting for Cloud-PBL Coupling and By Sorting out Influential Factors Using ML Approach
Zhanqing Li, Univ. of Maryland, College Park, MD

3:00 PM - Poster 272 The Other Indirect Effect: Cloud Influence Aerosol Direct Radiative Effect
Eshkol Eytan, CIRES, Boulder, CO

3:00 PM - Poster 273 The power and pitfall of invisible and virtual ship-tracks as opportunistic experiments in studying aerosol-cloud interactions
Tianle Yuan, UMBC JCET/NASA GSFC, Greenbelt, MD

3:00 PM - Poster 275 Isolated Convection Characteristics In Different Aerosol and Wind Regimes during TRACER
Manikandan Rajagopal, University of Oklahoma, NORMAN, OK; and Z. J. Lebo

3:00 PM - Poster 277 Meteorological and Aerosol Controls on Deep Convective Cell Properties: Insights from TRACER Observations and Ensemble Large-eddy Simulations
Dhwanit J Mise, CAPS, Norman, OK; and Y. Huang, G. M. McFarquhar, M. Xue, A. V. Ryzhkov, J. C. Snyder, L. Xue, PhD, W. Y. Cheng, M. Oue, P. Kollias, M. P. Jensen, M. Taszarek, C. Kuang, and T. Subba

3:00 PM - Poster 278 Insights from Tracking Deep Convection within Satellite and Radar Retrievals and a Novel Framework for Adaptive Scanning of Deep Convection
Siddhant Gupta, ANL, Lemont, IL; and B. A. Raut, D. Wang, C. P. Lackner, T. Hahn, S. Giangrande, M. P. Jensen, A. Theisen, and N. L. Hickmon

3:00 PM - Poster 279 Where the Air in Tropical Updrafts Came From, How it Got There, and its Thermodynamic Implications
Andrew James Muehr, Colorado State University, Fort Collins, CO; Colorado State University, Fort Collins, CO; and C. Davis, J. H. Ruppert Jr., and S. C. Van Den Heever

3:00 PM - Poster 280 Systematic Relationships Between Overshooting Top Characteristics and Anvil Life Cycle Evolution Observed with Geostationary Satellites
Rachael Auth, Colorado State University, Fort Collins, CO; and K. M. Bedka, J. Cooney, S. W. Freeman, and S. C. van den Heever

3:00 PM - Poster 281 Exploring the Pathways of Aerosol-Induced Modifications on Lightning Enhancement
Hallie Elizabeth Pimperl, University of California, Davis, Davis, CA

3:00 PM - Poster 282 Analyzing Microphysical Characteristics of Low-Ice Regions in Convective Cells During ESCAPE
Amanda Richter, Cooperative Institute for Severe and High-Impact Weather Research and Operations, The University of Oklahoma, Norman, OK; and G. M. McFarquhar and R. Patnaude

3:00 PM - Poster 283 How Do Deep Convective Cloud Properties and their Environments Differ Between Tropical Africa and South America?
Juliet Pilewskie, Colorado State University, Fort Collins, CO; and A. Drager, B. Pan, B. Dolan, T. S. L'Ecuyer, and S. C. Van Den Heever

3:00 PM - Poster 284 Regional Variations in Relationships among CAPE, Shear and Deep Convective Clouds and Implications for Empirical Studies of Aerosol Impacts on Convective Clouds
Eric M. Wilcox, Desert Research Institute - Reno, RENO, NV; and E. Zubar and T. Yuan

3:00 PM - Poster 285 Reprocessing the GOES GVAR Archive to Generate a Long-Term Geostationary Record of Cirrus Cloud Properties
Andrew Heidinger, NOAA, Madison, WI; and S. T. Wanzong, D. T. Lindsey, J. Robaidek, M. J. Foster, J. Seo, and C. Phillips

3:00 PM - Poster 286 Towards the Practical Application of Tomography for the Measurement of 3D Cloud Microphysical Properties
John Lundstrom, University of Illinois Urbana Champaign, Urbana, IL; and J. R. Loveridge and L. Di Girolamo

3:00 PM - Poster 287 Application of Trajectory Analysis to Understand Microphysical Signatures In Radar and Lidar Observation during a Cyclonic Snowfall Event
Tempei Hashino, Kochi University of Technology, Kami, Kochi, Japan; and A. Takahata and N. Wood

3:00 PM - Poster 288 Dominant Ice Cloud Microphysics Characterized by Simultaneous Observations of Radar Reflectivity and Doppler Velocity
Tatsuya Seiki, JAMSTEC, Yokohama, 14, Japan; and A. Noda, Y. Hagihara, and H. Horie

3:00 PM - Poster 289 Assessing Continuity Between AIRS and CrIS for Cloud Thermodynamic Phase Classification and Ice Cloud Properties
Nathaniel Miller, CIMSS, Madison, WI; and B. Kahn, K. S. Mattingly, T. Ochoa Peterson, F. Irion, and T. S. L'Ecuyer

3:00 PM - Poster 290 Leveraging AI Cloud Detection Techniques to Investigate the Impacts of Decision-Making on MAIA's Aerosol Retrievals
Joseph Nied, University of Illinois Urbana-Champaign, Urbana, IL; and M. V. De Vera, G. Zhao, and L. DiGirolamo

3:00 PM - Poster 291 A New High Spatial Resolution Cloud Mask as part of the NASA Continuity MODIS/VIIRS Products
Paolo Veglio, CIMSS, Madison, WI; and R. Holz, K. Meyer, S. Platnick, A. Heidinger, and E. Zhan

3:00 PM - Poster 292 Constraining CloudSat Radar Reflectivity Profiles From Passive Cloud Properties and Environmental Context Using a Mixture Density Network
Kevin M. Smalley, LLNL, Livermore, CA; and E. K. de Jong, H. Beydoun, A. Donahue, P. M. Caldwell, N. Gunawardena, and B. Perfect

3:00 PM - Poster 293 Quantifying the 4D Evolution of Cumulus Updrafts and Deep Convection Initiation Using the Ka-band Rapid Volume Imaging Radar (KaRVIR)
David J. Bodine, University of Oklahoma, Norman, OK; and Z. J. Lebo, C. R. Homeyer, D. Schvartzman, G. M. McFarquhar, S. M. Loria-Salazar, J. L. Salazar, and R. D. Palmer

3:00 PM - Poster 294 The NASA Langley Global Cloud Composite (GCC) Project: Integrating Geostationary and Sun-Synchronous Imagers for the Generation of Homogenized Cloud Properties with High Spatiotemporal Resolution
David Painemal, NASA Langley Research Center, Hampton, VA; and W. L. Smith Jr., L. Nguyen, T. Chee, R. Palikonda, K. Khlopenkov, D. Spangenberg, C. W. Fleeger, and J. Hicks

3:00 PM - Poster 295 Multi-Decadal Trends In Cloud Properties and Derived Radiation Datasets from the Latest NASA MODIS and VIIRS Cloud Products
Kerry Meyer, GSFC, Greenbelt, MD; and R. Holz, E. Borbas, S. Platnick, G. Wind, N. Amarasinghe, C. Peterson, P. Hubanks, P. Veglio, D. Fu, Y. Li, and A. Heidinger

3:00 PM - Poster 296 Structure and Radiative Effect of Cloud Regimes from Synergistic Remote Sensing
Johannes Happich, Leibniz Institute for Tropospheric Research, Leipzig, Germany; and A. Macke, C. Frangipani, C. Knist, M. Stengel, V. Tzallas, S. Wacker, and H. M. Deneke

3:00 PM - Poster 297 Assessment of Cloud Properties for CERES Edition 5 and Progress Towards Cross-Platform Consistency
Christopher Rogers Yost, AMA = Analytical Mechanics Associates, Hampton, VA; and W. L. Smith Jr., Q. Trepte, R. Palikonda, and D. Painemal

3:00 PM - Poster 298 Lake Effect Cloud Evolution over the Great Lakes from a Satellite Perspective
Leila Jean Gabrys, CIMSS, Madison, WI

3:00 PM - Poster 299 Evaluating Lidar and Radar Retrievals in Mixed-Phase Clouds: Case Studies from the M-PACE Campaign
Keying Li, University of Wyoming, Laramie, WY; and M. Saito, O. Cruikshank, and C. Grasmick

3:00 PM - Poster 300 Surface Seeding Triggers Self-Sustaining Glaciation in Arctic Stratocumulus
Zeen Zhu, Brookhaven National Laboratory, Upton, NY; and F. Yang, E. Luke, P. Kollias, D. Zhang, M. Oue, and G. de Boer

3:00 PM - Poster 301 Leveraging Hyperspectral Imaging to Study the Small-Scale Variability of Cloud Phase
Lukas J. Monrad-Krohn, Univ. of Oslo, Oslo, Norway; and R. O. David, T. Storelvmo, J. Henneberger, and T. Carlsen

3:00 PM - Poster 302 Characterization of the Fine-scale Structure of Mixed-Phase Clouds from University of Wyoming King Air In-situ and Radar-Lidar Observations
Masanori Saito, University of Wyoming, Laramie, WY; and K. Li, C. D. Grasmick, O. Cruikshank, and A. Majewski

3:00 PM - Poster 303 Exploring the Influence of Refreezing after Partial Melting in Cumulus Congestus Clouds with a Lagrangian Mixed-Phase Model
Marco Wurth, Free University of Berlin, Berlin, Berlin, Germany; and F. Hoffmann

3:00 PM - Poster 304 Using Direct Numerical Simulation to Study the Influence of Mixed-Phase Cloud Spatial Heterogeneity on Microphysical Evolution
Elise Marie Rosky, NCAR, Boulder, CO; NCAR, Boulder, CO; and S. Chen

3:00 PM - Poster 305 Processes Driving the Springtime Arctic Fog Lifecycle
Sonya Rauschenbach, University of California, Davis, Sacramento, CA; and A. L. Igel

3:00 PM - Poster 306 Development of a Lagrangian Balloon and Holographic Imager for a New Perspective on Cloud Microphysics
Sebastian Käser, ETH Zürich, Zürich, Switzerland; ETH Zürich, Zürich, Switzerland; and L. Kamber, C. Fuchs, and J. Henneberger

3:00 PM - Poster 307 Below-Cloud Scavenging for Polydisperse Aerosols In the Arctic
Seungmee Oh, Ewha Womans University, Seoul, Korea, Republic of (South); and J. Lee, K. M. Han, J. Um, Y. J. Yoon, and C. H. Jung

3:00 PM - Poster 308 A Satellite-Based Perspective on Atmospheric River Precipitation and Clouds in the Greenland-North Atlantic Region
Alanna Emma Wedum, University of Michigan-Ann Arbor, Ann Arbor, MI; University of Michigan, Ann Arbor, MI; and C. Pettersen and M. Mateling

3:00 PM - Poster 309 Aerosol Effects on the Microphysical Characteristics of Cold Air Outbreak-Embedded Maritime Stratocumulus over the Yellow Sea
Gaeun Lee, Kongju National University, Gongju, Chungcheongnam-do, South korea; and H. LEE

3:00 PM - Poster 310 Observed Macrophysical and Microphysical Characteristics of Marine Stratocumulus Clouds over the East Sea / Sea of Japan during Wintertime Cold Air Outbreaks
Yunjeong Lee, Kongju National University, Gongju, Chungcheongnam-do, South korea; and H. LEE

3:00 PM - Poster 311 Spectral-Bin Cloud Microphysics: Saving Computational Cost with Adaptive Bin Size Refinement
Jinbo Zhang, University of Wisconsin-Madison, Madison, WI; UW-Madison, Madison, WI; and Y. Tissaoui and S. N. Stechmann

3:00 PM - Poster 312 Quantifying Processing-Induced Variability in Optical Array Probe Measurements: A Stepwise Error Decomposition Approach
Mastooreh Ameri, Cooperative Institute of Severe and High-Impact Weather, Research and Operations (CIWRO), University of Oklahoma, Norman, OK; and G. M. McFarquhar, A. Bansemer, S. U. Patil, Y. Huang, S. Kirschler, A. V. Korolev, A. Majewski, L. Nichman, A. Takeishi, P. Rosenberg, L. Jaffeux, and D. G. Baumgardner

3:00 PM - Poster 313 Assessing the Spatial Scales Resolved by High-Rate Nevzorov Hot-Wire Probe Measurements During CAESAR
Shane Richard Martrich, Univ. of Wyoming, Laramie, WY; and S. Woods, P. Zuidema, G. M. McFarquhar, B. Geerts, and J. R. French

3:00 PM - Poster 314 Impacts of Mixed-phase Clouds on Surface Radiative Fluxes over the Southern Ocean: Insights from CAPE-k Observations and Simulations
Zirui Du, School of Meteorology, University of Oklahoma, Norman, OK; Cooperative Institute for Severe and High Impact Weather Research and Operations (CIWRO), Norman, OK; and Y. Huang, G. M. McFarquhar, Z. Xia, L. Lin, and P. A. Bogenschutz

3:00 PM - Poster 315 Generalized Double-Moment Scaling Framework for Drop Size Distributions: Parameterization Using 2DVD and MPS measurement
GyuWon Lee, Kyungpook National

3:00 PM - Poster 316 High-resolution Numerical Simulations of Tropical and Subtropical Convection for the NASA INCUS Mission
Itinderjot Singh, Colorado State University, Longmont, CO; and J. Bukowski, P. J. Marinescu, L. D. Grant, R. L. Storer, G. R. Leung, B. Dolan, R. Schulte, S. Prasanth, D. J. Posselt, K. L. Rasmussen, and S. C. Van Den Heever

3:00 PM - Poster 317 Tracking Storms from Space: The INCUS AUX-GEOIR Product
Sean W. Freeman, University of Alabama in Huntsville, Huntsville, AL; and Z. J. Luo, H. Takahashi, Y. Selevich, P. Kollias, and S. C. van den Heever

3:00 PM - Poster 318 Sensitivity of Observations and Simulations of Deep Convection to Ice Cloud Microphysical Assumptions
Derek J. Posselt, PhD, JPL, Pasadena, CA

Poster - POSTER Insights into Severe Storms through Remote Sensing

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 322 Examining Visible Cloud-Top Patterns of Severe Thunderstorms In Real Time Using Stereo Images Created Using Near-Simultaneous Images from GOES-East and GOES-West
Daniel M. Brown, EC, Edmonton, AB, Canada

3:00 PM - Poster 323 Investigating the Effects of Entrainment on Thunderstorm Outflows
Kylee Matousek, Univ. of Illinois at Urbana–Champaign, Champaign, IL; and D. A. R. Kristovich and J. Wang

3:00 PM - Poster 324 A Multi-Year Study of Inferred Hail Probabilities using Environmental and Remotely Sensed Parameters
Tyler Peterson, University of Wisconsin - Madison, Madison, WI; and A. Rowe, L. Heuscher, J. Cintineo, J. Sieglaff, and T. S. L’Ecuyer

3:00 PM - Poster 325 Sythesizing WSR-88D Storm Cell Tracking and Tornado Damage Surveys to Explore Tornadic Storm Life Cycles
Robert W Moore III, Univ. of Oklahoma, Norman, OK

3:00 PM - Poster 326 Integrating Radar Observations and Post-Event UAS Imagery to Assess the 26 February 2023 Norman, OK Tornado
Elizabeth Tirone, CIWRO, Norman, OK; and A. W. Lyza, E. N. Smith, and L. Bunting

3:00 PM - Poster 327 Potential Issues with Passive Microwave Satellite-Based Hail Climatology Estimates
Rebecca D. Adams-Selin, AER, Papillion, NE

3:00 PM - Poster 328 Improving Radar-Based Hail Detection: Recent Efforts at NSSL
Jeffrey C. Snyder, NSSL, Norman, OK; and J. M. Krause, A. Witt, J. T. Carlin, and A. A. Alford

3:00 PM - Poster 329 A Refined Polarimetric and Multi-Doppler Analysis of the 26 February 2023 Norman, Oklahoma Tornado
Vitor Goede, School of Meteorology, Norman, OK; Advanced Radar Research Center, Norman, OK; and D. Schvartzman, D. J. Bodine, M. A. Wagner, and E. N. Rasmussen

3:00 PM - Poster 330 Dual-Polarization Comparison of Descending Reflectivity Cores in Tornadic and Non-Tornadic Supercells
Ashlee Nicholle Ziegler, Custar, OH; and J. B. Houser, PhD

3:00 PM - Poster 331 Investigating Environmental Controls on Lightning Flash Rates in Isolated Ordinary Convection and Supercell Thunderstorms
David Topping, Texas A&M Univ., College Station, TX; and C. J. Nowotarski and M. Sharma

3:00 PM WITHDRAWN
Withdrawn: Develop a Multi-Doppler Wind Retrieval Algorithm to Examine Turbulent Wind Structures in Convection - An LES-Based Radar Simulator Framework S. M. Shajedul Karim, Hampton University, Hampton, VA; and S. R. Guimond Lakeside Commons / Exhibit

Hall A (Monona Terrace Community and Convention Center) Cancelled

3:00 PM - Poster 331A EVAD-Derived Low-Level Wind Field Evolution in Supercell Thunderstorms
Alec Prosser, CIWRO - Cooperative Institute for Severe and High-Impact Weather Research and Operations, Norman, OK; and R. L. Miller and A. W. Lyza

Poster - POSTER Tornadogenesis: New Insights from Observations and Models

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 332 Damage Analysis of the 18 May 2025 Arnett, Oklahoma, Tornado
Anthony W. Lyza, NSSL, Norman, OK; and E. Tirone, D. Candela, E. N. Smith, J. T. Carlin, Z. B. Wienhoff, S. M. Waugh, and M. C. Coniglio

3:00 PM - Poster 333 Application of Statistical Mechanics of Vortex Gases to Tornadogenesis
Douglas P. Dokken, University of St.Thomas, St. Paul, MN; and P. Belik, M. M. Shvartsman, and K. Scholz

3:00 PM - Poster 334 Impacts of Boundary Layer Turbulence on Supercell Morphology and Embedded Tornadoes within Tornado-Resolving CM1 Simulations
Andrew Berrington, CAPS, Norman, OK; and N. A. Snook, M. Xue, C. K. Potvin, A. McGovern, A. Fagg, and R. Martz

3:00 PM - Poster 335 The Impact of CM1 Microphysics Schemes on the Dynamics and Polarimetric Radar Signatures in Simulated Pre-Tornadic Supercells
Rachael Nicole Cross, University of Oklahoma, Norman, OK; and D. J. Bodine and L. Orf

3:00 PM - Poster 336 Applying Real-World Multi-Dimensional Data to Study Terrain Effects on Tornado Intensity
Grace Yan, Missouri Univ. of Science and Technology, ROLLA, MO

Poster - POSTER Toward Improved Understanding and Prediction of Heavy Rainfall and Flash Floods

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

3:00 PM - Poster 337 Excessive Rainfall Forecasting with the Ensemble Framework for Flash Flood Forecasting (EF5)
William Stewart Samson, Iowa State University, Ames, IA; and K. J. Franz and W. A. Gallus Jr.

3:00 PM - Poster 338 Analysis of a Heavy rainfall Event over Western Sichuan Basin under the Influence of Inertial Oscillation and Topography
Yun Chen, National Meteorological Centre□China Meteorological Administration, Beijing, China

3:00 PM - Poster 339 Mesoscale Forecast Evaluations During the Rio Grande Valley Extreme Precipitation Event of March 2025
Anastasia Joy Tomanek, Colorado State University, Fort Collins, CO; and R. S. Schumacher

3:00 PM - Poster 340 Environmental Regimes Associated with Spaceborne Observations of Heavy Convective Rainfall
Miles Harmala, Colorado State University, Fort Collins, CO; and K. L. Rasmussen and S. C. Van Den Heever

Poster - Posters: Aerosol-Cloud-Radiation Interactions and Feedbacks on Weather and Climate

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Cochairs: Xiquan Dong / The Univ. of Arizona / Tucson, AZ / Graham Feingold / NOAA / Boulder, CO / Ryan Kramer / GFDL / Alexandria, MD

3:00 PM - Poster 261 Cloud Modulation of Aerosol Direct Radiative Effects
Fei Yang, Princeton University, Princeton, NJ; and D. J. Paynter and V. Ramaswamy

Poster - Posters: In-situ Measurements of Radiation, Clouds and Aerosol Properties

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Chair: Xiquan Dong / The Univ. of Arizona / Tucson, AZ / Kelly A. Balmes / NOAA GML / Boulder, CO

3:00 PM - Poster 262 Characterizing Dust Aerosol Vertical Structure Using CALIOP Lidar and IIR Observations

Jayce Bradley Crayne, Texas A&M University, College Station, TX; and P. Yang and D. Li

3:00 PM - Poster 263 Linking Observed Droplet Microphysics to Aerosol Sources during the Eastern Pacific Cloud Aerosol Precipitation Experiment (EPCAPE)
Rachel Chang, Dalhousie University, Halifax, NS, Canada; and L. Robinson, J. L. Dedrick, S. Han, A. S. Williams, L. M. Russell, M. Wheeler, J. Liggio, and X. Zhou

Poster - Posters: Surface-atmosphere interactions and the surface energy budget of the cryosphere from satellite and sub-orbital observations

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Cochairs: Patrick C. Taylor / NASA / Virginia Beach, VA / Amy B. Solomon / CIRES and NOAA/PSL / Boulder, CO / Kerry Meyer / Greenbelt, MD

3:00 PM - Poster 264 Impact of Surface Heterogeneity on Spectral Reflectance and Albedo of Arctic Sea Ice
Ken Hirata, University of Colorado Boulder, Boulder, CO; and V. Nataraja and K. S. Schmidt

3:00 PM - Poster 2 The Influence of Atmosphere-Surface Interactions on Sea Ice Survivability
Patrick C. Taylor, NASA, Virginia Beach, VA; and R. Boeke, M. A. Webster, L. Boisvert, and C. Parker

3:00 PM - Poster 3 Analysis of ARCSIX Albedo Observations for Improvements to CERES
Emily Monroe, ADNET Systems, Inc., Hampton, VA; and D. A. Rutan, S. H. Ham, S. Kato, P. C. Taylor, and F. Rose

Poster - Posters: The Spectral Dimension of Atmospheric Shortwave and Longwave Radiation

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Cochairs: Daniel Feldman / LBNL / Berkeley, CA / David Jeffrey Paynter / NOAA/GFDL / Princeton, NJ

3:00 PM - Poster 267 Analytical solutions for longwave fluxes in radiative-convective equilibrium with water vapor
Yi Zhang, New York University, New York, NY

3:00 PM - Poster 268 Profiling the Marine Boundary Layer on the Tasmanian Coast with the M-AERI during CAPE-k
Jonathan Gero, University of Wisconsin - Madison, Madison, WI; and R. Knuteson

Poster Viewing and Coffee Break

3:00 PM - 4:30 PM | Lakeside Commons / Exhibit Hall A (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Joint 16 - Community Modeling Efforts at Convection-Allowing Scales: Part I (Joint between the 30th Conference on Numerical Weather Prediction and the 32nd Conference on Severe Local Storms)

4:30 PM - 6:00 PM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitters: Austin A. Coleman / CIRES / WPC / Boulder, CO / Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Jessica M. McDonald / National Research Council / Washington, DC / Jacob T. Carlin / NSSL / Norman, OK

4:30 PM Joint 16.1
Evaluation of a Multi-Model, Convection-Allowing Ensemble of Opportunity for Severe Weather Prediction: SSEOv2 Israel L. Jirak, SPC, Norman, OK; and T. A. Supinie and J. Vancil

4:45 PM Joint 16.2
Severe Weather Forecast Accuracy For In-development, Convection-Allowing Ensemble Prediction Systems Josue Chamberlain, NCAS, Washington, DC; Howard University, Washington, DC; and A. Back, D. Dowell, A. A. Jensen, and K. C. Eure

5:00 PM Joint 16.3
Will Partial Cycling Soon be a Relic of the Past? Initial Condition Blending In the Mpas-Jedi RRFSv2X System Jeff Beck, NOAA/GSL and DTC, Boulder, CO; and R. Li, M. Hu, C. Zhou, T. T. Ladwig, C. S. Schwartz, A. T. Johnson, M. L. Larson, W. Mayfield, M. A. Harrold, V. Vargas Jr., and X. Wang

5:15 PM Joint 16.4
Stochastic Physics Implementation and Testing in MPAS and Transition to the UFS Will Mayfield, NCAR, Eugene, OR

5:30 PM Joint 16.5
Working Towards Rapid-Refresh Regional Modeling using MPAS-JEDI at the U.S. Air Force's 16th Weather Squadron Matthew Thomas Vaughan, 16th Weather Squadron, Offutt AFB, NE; and S. I. Baker, S. Childs, S. Rentschler, M. Searls, W. T. Sedlacek, C. J. Melick, PhD,

S. Schulze, R. Strickler, W. Cassell, B. T. Gallo, J. Martinelli, and E. L. Kuchera

5:45 PM Joint 16.6

Assessments of Explicit Lightning Threat Predictions from the United States Air Force Mesoscale Ensemble Prediction Suite Christopher J. Melick, PhD, U.S. Air Force, Offutt AFB, NE; and B. T. Gallo, M. T. Vaughan, W. T. Sedlacek, G. R. Brooks, S. Rentschler, M. Searls, S. J. Childs, PhD, A. J. Elliott, G. Creighton, and E. L. Kuchera

Joint 16A - Controls on Convective Mass Flux and Connections to Severe Weather and Cloud Anvils

4:30 PM - 6:00 PM | Meeting Rooms MNQR (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitters: Derek J. Posselt / JPL / Pasadena, CA / Kristen Lani Rasmussen / Colorado State University / Fort Collins, CO / Peter J. Marinescu / Colorado State Univ. / Fort Collins, CO / Juliet Pilewskie / Colorado State University / Fort Collins, CO

4:30 PM Joint 16B.1

Linking Convective Mass Flux to Storm Dynamics through the NASA INCUS Mission Susan C. Van Den Heever, Colorado State Univ., Fort Collins, CO; and D. J. Posselt, PhD, S. Tanelli, P. Kollias, K. L. Rasmussen, Ph.D., P. Partain, G. L. Stephens, R. Auth, J. Bukowski, A. Burzynski, B. Dolan, N. M. Falk, S. W. Freeman, P. N. Gatlin, L. D. Grant, G. J. Huffman, G. R. Leung, Z. J. Luo, G. G. Mace, P. J. Marinescu, M. G. Morris, J. Pilewskie, S. Prasanth, S. C. Reising, R. Schulte, C. Schumacher, I. Singh, and H. Takahashi

4:45 PM Joint 16B.2

Using High Temporal Resolution Radar Observations from the TIME-SLICE AL Field Campaign to Understand Convective Updraft Evolution Ethan Isaac Ebbert, The Univ. of Alabama in Huntsville, Huntsville, AL; and S. W. Freeman, P. Kollias, P. N. Gatlin, E. Luke, C. G. Amiot, C. Kwinta, K. Knupp, P. T. Pangle, and B. Dolan

5:00 PM Joint 16B.3

Comparison of Entrainment, Dilution, and Source Regions of Air in Three Deep Convective Storm Morphologies Charles Davis, Colorado State University, Fort Collins, CO; and S. C. van den Heever

5:15 PM Joint 16B.4

An Investigation of Convective Mass Flux Characteristics by Storm Mode in Kilometer-Scale Climate Simulations Justin Tyler Hassel, Colorado State University, Fort Collins, CO; and K. L. Rasmussen, Ph.D.

5:30 PM Joint 16B.5

Using the Temporal Evolution of Overshooting Tops Characteristics as a Proxy for Hail Size Rachael Auth, Colorado State University, Fort Collins, CO; Colorado State University, Fort Collins, CO; and S. C. van den Heever

5:45 PM Joint 16B.6

A Storm Model Study on the Satellite-observed Pancake Clouds Atop Deep Convective Storms Developed in Weak Upper-level Wind Shear Pao K. Wang, Univ. of Wisconsin-Madison, Madison, WI

Joint 16B - In-situ Measurements of Radiation, Clouds and Aerosol Properties

4:30 PM - 6:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Adam Ahern / Cooperative Institute for Research in Environmental Sciences, University of Colorado Boulder / Boulder, CO / NOAA / Boulder, CO

Cochairs: Adam Ahern / NOAA / Boulder, CO / Xiquan Dong / The Univ. of Arizona / Tucson, AZ / Kelly A. Balmes / NOAA GML / Boulder, CO

4:30 PM Joint 16.1

Observing the Atmosphere in Detail: Thirty Years of Cloud, Aerosol, and Radiation Measurements from ARM Adam Theisen, ANL, Lemont, IL; and N. L. Hickmon, S. Collis, S. Gupta, J. H. Mather, J. M. Comstock, G. Prakash, O. Mayol-Bracero, and J. E. Shilling

4:45 PM Joint 16.2

Putting the CAPS on Filter-Based Absorption Measurements: Overcoming Accuracy Limitations from the AMICE Campaigns Connor Joseph Flynn, Univ. of Oklahoma, Norman, OK; and A. Sedlacek, T. Onasch, S. Smith, E. Lewis, A. A. Fakoya, B. F. Lamkin, E. D. Lenhardt, L. T. Mitchell, E. West, O. Enekwizu, A. Dobracki, A. McMahon, L. Mueller, A. Theisen, and J. Redemann

5:00 PM Joint 16.3

Closure of Aerosol Radiative Properties from Remote Sensing and In Situ – Pertinent to Assessing the Accuracy of AERONET Sky Scan Inversions Logan Thomas Mitchell, University of Oklahoma, Norman, OK; and C. J. Flynn, J. Redemann, K. Pistone, and S. LeBlanc

5:15 PM Joint 16.4

Near-Source Characteristics and Evolution of Wildfire Smoke Plumes Isabel L. McCoy, Colorado State University, Fort Collins, CO; and J. C. Chiu, S. M. Kreidenweis, T. Yuan, and T. Johnson

5:30 PM Joint 16.5

Secondary Ice Production Occurs Predominantly within -8 °C to -3 °C in Midlatitude Winter Mixed-Phase Marine Cold Air Outbreaks Genevieve Rose Lorenzo, Rosenstial School, Miami, FL; and P. Zuidema, S. Kirschler, E. Crosbie, G. S. Diskin, D. Menekay, R. H. Moore, L. W. Siu, A. Sorooshian, K. L. Thornhill, C. Voigt, and L. Ziemba

5:45 PM Joint 16.6

Testing the Consistency of Ice Crystal Scattering Properties Across Active Radar and Passive Radiometer Frequencies Using CCREST-M Observations Anthony Baran, Met Office, Exeter, United kingdom; UKMO, Exeter, United kingdom; UKMO, Exeter, Devon, United Kingdom; and S. Fox

Session 16 - Aerosol-Cloud Indirect Effects II

4:30 PM - 6:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Isabel L. McCoy / Florida State University / Tallahassee, FL / Siddhant Gupta / ANL / Lemont, IL / Saurabh Patil / School of Meteorology, University of Oklahoma

4:30 PM - 16A.1 Could the 2020 Australian Wildfires have triggered La Niña?
Christopher J Wright, University of Washington, Seattle, WA; and R. Wood, J. Thornton, and R. M. Eastman

4:45 PM - 16A.2 Synoptic Transport Drives Episodic Increase of Cloud Condensation Nuclei Over the High-Latitude Southern Ocean
Camille Fielding, Univ. of Utah, Saratoga Springs, UT; and G. G. Mace

5:00 PM - 16A.3 How Cloud Organization Shapes the Cloud Transition Zone and Aerosol Optical Depth on the Mesoscale.
Eshkol Eytan, CIRES, Boulder, CO; and M. Wirth, J. C. Chiu, S. Groß, T. Yamaguchi, J. J. Gristey, and G. Feingold

5:15 PM - 16A.4 Examining Aerosol Impacts on Southeastern Convective Clouds Using High-Resolution Numerical Modeling
Kyra Britton, University of Alabama in Huntsville, Huntsville, AL; and S. W. Freeman

5:30 PM - 16A.5 Revisiting MBL Aerosol–Cloud Interactions: The Role of Sub-Cloud CCN and Cloud-Base Microphysics from the DOE ARM MAGIC Campaign
Xiaomi Teng, University of Arizona, Tucson, AZ; Zhejiang University, Hangzhou, Zhejiang, China; and X. Dong, B. Xi, J. Brendecke, and W. Li

5:45 PM - 16A.6 Applying AI and Singular Value Decomposition Techniques in the Study of Aerosol Effect on Thin Cirrus Clouds over the Global Oceans
Xuepeng Zhao, NOAA/NESDIS/National Centers for Environmental Information, Silver Spring, MD; and M. J. Foster and A. Heidinger

12:00 AM WITHDRAWN
Withdrawn: A Mechanism for Aerosol Modification of the Shallow-Deep Convective Transition Frank Robinson, Sacred Heart University, Fairfield, CT; and T. Storelvmo, D. Kirshbaum, and S. Sherwood Madison Ballroom D (Monona Terrace Community and Convention Center) Cancelled

Session 16 - Other Topics on Weather Analysis and Forecasting

4:30 PM - 6:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Andrew C. Winters / Franklin, WI / Burkely Twiest Gallo / 557th Weather Wing / Offutt AFB, NE

4:30 PM - 16.1 Warning the Rarest Tornadoes: Examining Forecaster Decision-Making in Tornado Emergency Issuance
Edward C. Wolff IV, University of Illinois Urbana-Champaign, Urbana, IL; and J. E. Trujillo-Falcón, L. Blind-Doskocil, E. Conklin, and E. Gustafson

4:45 PM - 16.2 Seamless Nowcast/Forecast Blending of Geostationary Cloud Properties
James P Gibbons, MyRadar, Arlington, VA; and R. Lagerquist and S. Garimella

5:00 PM - 16.3 A Cloud-Native Python Framework for Scalable Access and Analysis of Environmental Data for Forecasting and Decision Systems
Ryan Paul Lafler, MS, Premier Analytics Consulting, LLC, San Diego, CA

5:15 PM - 16.4 Contrasting Boundary Layer Evolution during the Morning Transition and a Solar Eclipse using the High-Resolution Rapid Refresh (HRRR)
Julia M. Simonson, CIRES, Univ. of Colorado, Boulder, CO; Developmental Testbed Center, Boulder, CO; NOAA/ESRL/GSL, Boulder, CO; and D. D. Turner, T. Heus, G. N. Raghunathan, T. J. Wagner, S. Kimball, and J. Kenyon

5:30 PM - 16.5 The Role of Land Cover Type in the Evolution of Flash Drought: WRF-ARW Model and Sensitivity
Yibo Chen, University of North Dakota, Grand Forks, ND; University of North Dakota, Grand Forks, ND; and J. I. Christian

5:45 PM - 16.6 Evaluation of Vegetation Dryness Indices across Climatic Regimes in the United States
Aditya Kumar, CIMSS, Madison, WI; and R. B. Pierce and C. Schmidt

Session 16A - Idealized Modeling of Supercells and Tornadoes

4:30 PM - 6:00 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Robert A. Saba / Univ. of Oklahoma / Norman, OK / Morgan Nichole Siers / Lafayette, IN

4:30 PM - 16A.1 Analysis of 4D Vertical Vorticity Evolution and Resolution Sensitivity in Idealized Supercell Simulations

Nicholas William Camp, Texas Tech University, Lubbock, TX; and J. Dahl

4:45 PM - 16A.2 Effects of Land Cover on Multi-Vortex Tornado Structure and Subvortex Dynamics
YI ZHAO, Missouri Univ. of Science and Technology, Norman, OK; Missouri Univ. of Science and Technology, Rolla, MO; and G. Yan

5:00 PM - 16A.3 The Mechanism of Tornadogenesis from the Perspective of Vortex Tubes
Peng Yue, Missouri University of Science and Technology, Rolla, MO; and Y. C. Li, L. Orf, Y. ZHAO, and G. Yan

5:15 PM - 16A.4 The Genesis and Maintenance of Tornadoes in Supercells: The Critical Role and Origin of Lower Low-level Mesocyclone
Ming Xue, School of Meteorology, Norman, OK; and W. Huang and M. Oliveira

5:30 PM - 16A.5 Why Does Boundary Layer Turbulence Weaken Cold Pools in Simulated Supercells?
Brandon Antonio Garcia, Pennsylvania State University, University Park, PA; and K. A. Bowley and P. Markowski

5:45 PM - 16A.6 The Influence of Deep-Layer Shear on Severe Convective Storm Mode as a Function of Initial Updraft Morphology
Kristen L Axon, Purdue University, West Lafayette, IN; Purdue University, Bloomington, IL; and D. T. Dawson II, R. L. Thompson, A. R. Dean, and E. R. Mansell

Session 16B - Novel Observations of Hail

4:30 PM - 6:00 PM | Hall of Ideas - HI (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Francesco De Martin / U.S. National Science Foundation, National Center of Atmospheric Research / Boulder, CO / Zachary B. Wienhoff / HAAG ENGINEERING CO. / Champaign, IL

4:30 PM - 16B.1 An In-Depth Analysis of the Fall Behavior of Real Hailstones
Sean M. Waugh, NSSL, Norman, OK; NSSL, NORMAN, OK; and J. Segall

4:45 PM - 16B.2 An Experimental Investigation of Large Hailstone Aerodynamics
Matthew Mason, University of Queensland, St Lucia, QLD, Australia

5:00 PM - 16B.3 Hail Yeah! Design and Deployment of Portable Hail Cameras during ICECHIP
Talia Dawn Kurtz, University of North Dakota, Grand Forks, ND; and A. D. Kennedy, R. Adams-Selin, and J. Mascio

5:15 PM - 16B.4 In Situ Measurements of Wind-Driven Hail Impacts Using an Omni-Directional Disdrometer
Jacob Sorber, Insurance Institute for Business & Home Safety, Richburg, SC; and A. Prabhakaran, R. Marandi, and I. M. Giammanco

5:30 PM - 16B.5 Damage Investigation and Claims Analysis of the June 5th, 2025, Lubbock, Texas Hail and Wind Event
Brenna Meisenzahl, The Insurance Institute for Business and Home Safety, Fort Mill, SC; and I. M. Giammanco and J. Sorber

5:45 PM - 16B.6 Using Citizen Science Hail Observations to Validate Hail Melt and Terminal Velocity Parameterizations and their Sensitivity to Hail Shape and Environmental Conditions
Hannah C. Vagasky, JANUS Research Group, LLC, Atmospheric and Environmental Research (AER), Evans, GA; and R. Adams-Selin, C. Calkins, and J. Mascio

Event - Video Night

7:30 PM - 9:30 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Daniel T. Dawson / Purdue Univ. / West Lafayette, IN / Robin Tanamachi / Purdue University / West Lafayette, IN / View Related / 32nd Conference on Severe Local Storms

Friday, August 7, 2026

Joint 17 - Advances in Cloud Microphysics Towards the Improvement of Severe and High Impact Weather Modeling

8:30 AM - 10:00 AM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Matthew C. Brown / NSSL / Norman, OK

Cochairs: Andrew Michael Dzambo / Cooperative Institute of Severe and High-Impact Weather Research and Operations (CIWRO) / Norman, OK / Leah D. Grant / Colorado State University / Fort Collins, CO / Mateusz Taszarek / na / Poznań Poland

8:30 AM Joint 17.1
Sensitivity of Convective Storms to Initial Conditions in a Microphysical Piggybacking Framework William Y. Cheng, NSF NCAR, BOULDER, CO; and L. Xue, PhD, Y. Huang, D. J. Mise, G. M. McFarquhar, M. Xue, A. V. Ryzhkov, and J. C. Snyder

8:45 AM Joint 17.2
Quantifying Errors in 2-moment Sedimentation and a Simple Pseudo-3-Moment Correction Method Edward Mansell, NOAA, Norman, OK

9:00 AM Joint 17.3
Multi-Measurement Perspectives on Ice Microphysics In Convective Environments Julia A Shates, SIO, La Jolla, CA; and D. J. Posselt, PhD and M. D. Lebsock

9:15 AM Joint 17.4
The Impacts of Process-Based Initiation of Multiple Ice Categories in the Predicted Particle Properties (P3) Microphysics Scheme on Simulated Supercell Storm Structure and Behavior. Daniel T. Dawson II, Purdue Univ., West Lafayette, IN; and H. Morrison, J. A. Milbrandt, E. Mansell, M. Cholette, M. Oue, and R. Tanamachi

9:30 AM Joint 17.5
Effects of Multi-Moment Microphysics on Tornado Production and ZDR Column Behavior in Ensemble Simulations of Supercells Morgan Nichole Siers, Purdue University, West Lafayette, IN; Purdue University, Lafayette, IN; and R. Tanamachi, D. T. Dawson II, and E. R. Mansell

9:45 AM Joint 17.6
Regulation of Severe Hail Growth by Ice-Nucleating1Particle-Mediated Raindrop Freezing Tobias Innes David Ross, University of Illinois Urbana-Champaign, Urbana, IL; and S. Lasher-Trapp

Session 17 - Aerosol-Cloud-Radiation Interactions and Feedbacks on Weather and Climate I

8:30 AM - 10:00 AM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Maria Z Hakuba / Jet Propulsion Laboratory, California Institute of Technology / Pasadena, CA

Cochairs: Xiquan Dong / The Univ. of Arizona / Tucson, AZ / Ryan Kramer / GFDL / Alexandria, MD

8:30 AM - 17.1 Aerosol and Cloud Observation using the Space-borne Lidar ATLID and Imager MSI onboard EarthCARE
Tomoaki Nishizawa, National Institute for Environmental Studies, Tsukuba, Japan; and R. Kudo, E. Oikawa, Y. Jin, A. Higurashi, A. Shimizu, N. Sugimoto, K. Sato, and H. Okamoto

8:45 AM - 17.2 Revisiting Aerosol-Cloud Interactions with the CALIPSO-CloudSat-Aqua/MODIS A-Train Record
David Painemal, NASA Langley Research Center, Hampton, VA; and Z. Li, Y. Feng, and X. Zheng

9:00 AM - 17.3 Weak, Low-Level Dry Convection over Angola Determines Offshore Stratocumulus Cloud Droplet Number Concentrations
Tyler Tatro, Rosenstiel School of Marine, Atmospheric, and Earth Science, Miami, FL; and P. Zuidema

9:15 AM - 17.4 Using Cloud-Controlling Factors to Probe a Perturbed Parameter Ensemble of E3SMv3
Timothy Myers, University of Wyoming, Laramie, WY; and T. W. Aerenson, D. McCoy, M. D. Zelinka, J. M. Nugent, and H. Y. Brown

9:30 AM - 17.5 Sensitivities of CCCma-Calculated SW Fluxes at Surface and TOA to Different Profiles of Liquid Cloud Microphysical Properties
Xiquan Dong, The Univ. of Arizona, Tucson, AZ; and J. Brendecke, B. Xi, and H. Barker

9:45 AM - 17.6 Radiative-Microphysical Coupling in a One-Dimensional Turbulence Model
Suryadev Pratap Singh, Michigan Technological University, Houghton, MI; and S. K. Krueger, C. Bois, and R. A. Shaw

Session 17 - Deep Convective Clouds

8:30 AM - 10:00 AM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Julia A Shates / SIO / La Jolla, CA

Cochairs: Anita D. Rapp / College Station, TX / Siddhant Gupta / ANL / Lemont, IL

8:30 AM - 17.1 What Drives Negative KDP Above the Melting Layer? The Hidden Role of Secondary Ice Production
Yongjie Huang, University of Oklahoma, Norman, OK; and G. M. McFarquhar, A. V. Ryzhkov, M. Xue, D. J. Mise, J. C. Snyder, L. Xue, PhD, W. Y. Cheng, M. Oue, P. Kollias, M. P. Jensen, T. Logan, and E. C. Bruning

8:45 AM - 17.2 Tracking Nascent Convective Cells to Compare their Attributes and Fates for Differing Initiation Mechanisms
Christopher J. Nowotarski, Texas A&M University, College Station, TX; and M. Sharma, A. D. Rapp, and S. D. Brooks

9:00 AM - 17.3 Impact of Aerosols and Meteorology on Updrafts and Microphysics in Deep Convective Clouds during ESCAPE
Saurabh U Patil, University of Oklahoma, Norman, OK; and G. M. McFarquhar, Y. Huang, M. Wolde, L. Nichman, C. Nguyen, K. Ranjbar, N. Bliankinshtein, K. Bala, G. Roberts, and P. Kollias

9:15 AM - 17.4 The Influence of Aerosol Environment on the Microphysics of Deep Convective Thermal Chains
Charles Davis, Colorado State University, Fort Collins, CO; and S. C. van den Heever

9:30 AM - 17.5 Environmental Controls on the Sensitivity of Convective Mass Flux to Model Grid Spacing
Itinderjot Singh, Colorado State University, Longmont, CO; and J. Bukowski, P. J. Marinescu, L. D. Grant, R. Schulte, G. R. Leung, and S. C. Van Den Heever

9:45 AM - 17.6 Coupling Between Deep Convective Cores and Anvil Ice Cloud Properties
Anita D. Rapp, Texas A&M University, College Station, TX; and S. Elkins and J. Pilewskie

Session 17 - Model Development and Application: Global Model Development and Updates

8:30 AM - 10:00 AM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 30th Conference on Numerical Weather Prediction

Submitter: Austin A. Coleman

Cochairs: Jun Du / James A. Nelson

8:30 AM - 17.1 Overview of the Next Global Forecast System GFSv17
Catherine A. Thomas, NCEP, College Park, MD; and J. Meixner and R. Sun

8:45 AM - 17.2 Implementation of a Cold Pool Parameterization in the NCEP GFS
Jongil Han, NOAA/NWS/NCEP/EMC, College Park, MD; and C. H. Boyer, W. Li, K. Fung, and F. Yang

9:00 AM - 17.3 Novel Real-Time, Global, Kilometer-Scale Numerical Weather Prediction Model Forecasts with the Model for Prediction Across Scales (MPAS)
Craig Schwartz, NCAR, Boulder, CO; and F. Judt, B. Skamarock, M. Duda, R. Johnson, and R. A. Sobash

9:15 AM - 17.4 Representing Convective Organization and Memory in NOAA’s Unified Forecast System
Lisa Bengtsson, CIRES, Boulder, CO

9:30 AM - 17.5 Integration of Machine Learning In Weather Prediction: From Pure Data-Driven to Hybrid Modelling
Syed Zahid Husain, ECCC, Dorval, QC, Canada; and L. Separovic, J. F. Caron, C. Subich, J. Yang, R. Aider, M. Buehner, S. Chamberland, E. Lapalme, R. McTaggart-Cowan, A. Z. Moraes, and P. Vaillancourt

9:45 AM - 17.6 Overview of the Global Chemistry and Aerosol Forecast System (GCAFS) Version 1
Cory R Martin, PhD, NOAA/NWS, College Park, MD; and B. D. Baker II, L. Pan, B. Fu, P. S. Bhattacharjee, A. Tangborn, Y. Wang, Y. Tang, D. Holdaway, and F. Yang

Session 17 - Recent High-Impact Weather Events

8:30 AM - 10:00 AM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO

Cochairs: Austin Coleman / NOAA / Gina M. Eosco / NOAA Federal / Silver Spring, MD

8:30 AM - 17.1 AI-Enhanced Prediction of an Extreme Rainfall Event over the United Arab Emirates using the Pangu-Weather Model.
Abdulla AlRaisi, Abu Dhabi Polytechnic, Shakhbout City, AZ, United arab emirates; and A. S. alkhemeiri, N. A. ALRashdi, A. parottil, M. Elge, and S. Tilev

8:45 AM - 17.2 New York State of Mind: Assessing NY State Weather Risk Communication Center Forecasts of Two High-Impact Coastal Storm Events from 2025–2026.
Jordan Lee Tweedie, State Weather Risk Communication Center, University at Albany, SUNY, Albany, NY

9:00 AM - 17.3 Analysis of the Extreme April 2023 Ice Storm Over Eastern Canada
John Richard Gyakum, McGill University, Montreal, QC, Canada; and J. M. Theriault, Y. Low, M. Lachapelle, J. E. M. Wray, H. D. Thompson, S. Basnet, and M. Girouard

9:15 AM - 17.4 Numerical Simulations of Mesoscale Processes During the Historic Rio Grande Valley Extreme Precipitation Event of March 2025

Anastasia Joy Tomanek, Colorado State University, Fort Collins, CO; and R. S. Schumacher

9:30 AM - 17.5 Convection Allowing Model Solutions of Severe Convection on June 28, 2025
David A. Imy, Retired NOAA/SPC, Norman, OK; and V. A. Gensini

9:45 AM - 17.6 An Observational and Model Analysis on the Flash Flooding Event In the Texas Hill Country on Independence Day 2025: The Role of a Mid-Level Vortex and Antecedent Tropical Moisture
Kwan-yin Kong, Weather Prediction Center, College Park, MD

Coffee Break

10:00 AM - 10:45 AM | Grand Terrace (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Joint 18 - Community Modeling Efforts at Convection-Allowing Scales: Part II (Joint between the 30th Conference on Numerical Weather Prediction and the 32nd Conference on Severe Local Storms)

10:45 AM - 12:00 PM | Hall of Ideas - EFG (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitters: Austin A. Coleman / CIRES / WPC / Boulder, CO / Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Gina M. Eosco / NOAA Federal / Silver Spring, MD / Jeff Beck / NOAA/GSL and DTC / Boulder, CO

10:45 AM Joint 18.1
Impact of In-Situ Observations from Three TORUS Cases on Model Analyses and Forecasts of Severe Convection Robert Szot, Univ. of Nebraska-Lincoln, Lincoln, NE; and A. L. Houston and M. B. Wilson

11:00 AM Joint 18.2
Potential Benefits of Assimilating Phase Array Radar Data on Short-Term Convective Scale Data Assimilation and Forecasts - An Idealized Case Study Jidong Gao, OAR/NOAA, Norman, OK; and D. R. Stratman

11:15 AM Joint 18.3
Use of ZDRColumns with Additive Noise in Short-Term Convection-Allowing Model Forecasts of Developing Thunderstorms Christopher A. Kerr, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), Norman, OK; and J. T. Carlin, E. Mansell, and N. Yussouf

11:30 AM Joint 18.4
Convection-Allowing Model Forecasts of Pre-Convective Initiation Environments of Missed and False Alarm Mesoscale Convective Systems Sam Ritter, Iowa State Univ., Ames, IA; Iowa State University, Ames, IA; and N. D. Sonntag and W. A. Gallus Jr.

11:45 AM WITHDRAWN
Joint 18.5 Withdrawn: Sensitivity of Tornadic and Nontornadic Mesocyclone Intensity to Model Grid Spacing in Simulated Supercells Austin Dixon, CIWRO, Norman, OK; and P. Skinner, L. J. Wicker, L. Orf, and B. E. Coffer Hall of Ideas - EFG (Monona Terrace Community and Convention Center) Cancelled

Session 18 - Advances Facilitated By New Observational Platforms and Capabilities Part II

10:45 AM - 12:00 PM | Madison Ballroom B (Monona Terrace Community and Convention Center) | 34th Conference on Weather Analysis and Forecasting

Submitter: Austin A. Coleman / CIRES / NOAA WPC / Boulder, CO / CoChair / Matthew Wilson / NSF NCAR / Boulder, CO

10:45 AM - 18.1 Advances Facilitated by New GOES-R Observational Platforms: Impacts on Data Assimilation and Extreme Weather Prediction
Zhaoxia Pu, Univ. of Utah, Salt Lake City, UT; and C. Feng and M. C. Johncox

11:00 AM - 18.2 Observing System Simulation Experiments to Investigate Impacts of Deep-Tropospheric UAS Profiles on NWP Forecasts
Kenneth Alexander Swan, University of Nebraska - Lincoln, Lincoln, NE; and A. L. Houston, S. Murdzek, T. T. Ladwig, E. P. James, and M. Tsey

11:15 AM - 18.3 Using OSSEs to Study the Assimilation of Observations from a PBL Profiling Network for Improving Numerical Model Forecast of Severe Convective Weather
Matthew Bradley Ammon, Cooperative Institute for Severe and High-Impact Weather Research and Operations (CIWRO), University of Oklahoma, Norman, OK

11:45 AM - 18.5 Real-Time Three-Dimensional Wind Retrievals from a Multistatic Passive Radar Network
Benjamin Harwell Moose, Univ. of Oklahoma / CIWRO, Norman, OK; and P. S. Skinner, P. E. Kirstetter, S. Emmerson, R. D. Palmer, D. Schvartzman, D. J. Bodine, B. Cheong, C. Fulton, and T. Lindley

Session 18 - Advances in Tornado Damage Assessment and Survey Methodology

10:45 AM - 12:15 PM | Madison Ballroom A (Monona Terrace Community and Convention Center) | 32nd Conference on Severe Local Storms

Submitter: Victor A. Gensini / Northern Illinois Univ. / DeKalb, IL

Cochairs: Kayla Marie Wheeler / CIWRO, Univ. of Oklahoma / Norman, OK / Jacob T. Carlin / NSSL / Norman, OK

10:45 AM - 18.1 Relating WSR-88D Observations of Tornadoes with Fine-Scale Surface-Based Damage
Jana B. Houser, PhD, The Ohio State University, Columbus, OH; and G. Yan, J. Wang, and P. Yue

11:00 AM - 18.2 An Overview of the Enderlin, ND Tornado Event of 20 June 2025: A Review of the Environment, Storm Evolution and Damage Tracks
Melinda Beerends, National Weather Service, Davenport, IA; and C. S. Miller, D. M. L. Sills, E. Rasmussen, M. A. Wagner, J. G. LaDue, and C. J. Peterson

11:15 AM - 18.3 Damage to the Natural Environment in the 20 June 2025 Enderlin, ND Tornadoes
Erik N. Rasmussen, Disaster Imaging Inc., Ward, CO; and M. A. Wagner, C. J. Peterson, C. S. Miller, D. M. L. Sills, M. Beerends, and J. G. LaDue

11:30 AM - 18.4 An Overview of the Lofted Train Car Analysis Used to Determine Wind Speeds in the 20 June 2025 EF5 Enderlin, ND Event
Connell S. Miller, Canadian Severe Storms Laboratory, London, ON, Canada; and D. M. L. Sills, M. J. Beerends, E. Rasmussen, M. Wagner, and J. G. LaDue

11:45 AM Concluding Remarks

Session 18 - Aerosol-Cloud-Radiation Interactions and Feedbacks on Weather and Climate II

10:45 AM - 12:00 PM | Madison Ballroom C (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitter: Andrew John Buggee / Boulder, CO

Cochairs: Ryan Kramer / GFDL / Alexandria, MD / Kelly Balmes / CIRES, Univ. of Colorado / Boulder, CO

10:45 AM - 18.1 Tropospheric Brightening via Stratospheric Aerosols
Jake J. Gristey, CIRES, Boulder, CO; and G. Feingold and T. Yamaguchi

11:00 AM - 18.2 Atmospheric Aerosol Properties Derived from Airborne High Spectral Resolution Lidar Measurements using Machine Learning Regression
Richard A. Ferrare, NASA, Hampton, VA; and J. W. Hair, T. Shingler, A. R. Nehrir, C. Hostetler, M. Fenn, M. Clayton, A. J. Scarino, D. B. Harper, J. Collins, R. H. Moore, L. Ziemba, M. A. Shook, and J. Lee

11:15 AM - 18.3 Stronger Fires, Darker Particles? Linking Fire Intensity to Aerosol Properties
Abdulamid Abiola Fakoya, University of Oklahoma, Norman, OK; and J. Lee, L. Gao, I. Chang, C. J. Flynn, S. M. Loria-Salazar, and J. Redemann

11:30 AM - 18.4 Ocean Coupling Moderates Aerosol-Driven Changes In Hadley Circulation and Cloud Radiative Effects
Takanobu Yamaguchi, CIRES, Boulder, CO; NOAA, Boulder, CO; and R. Yoshida, Y. S. Chen, I. L. McCoy, and G. Feingold

11:45 AM Conference Closeout
Maria Z Hakuba, JPL, Pasadena, CA; and Y. L. Shea, A. J. Buggee, C. Dang, X. Dong, and K. S. Schmidt

Session 18 - Remote Sensing of Clouds I

10:45 AM - 12:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Siddhant Gupta / ANL / Lemont, IL / Julia A Shates / SIO / La Jolla, CA

10:45 AM - 18.1 Consistency of Cloud Amount and Vertical Structure Derived from Surface, Airborne and Spaceborne Cloud Radar Measurements
Xiquan Dong, The Univ. of Arizona, Tucson, AZ; and A. Das, B. Xi, and J. Brendecke

11:00 AM - 18.2 A Probabilistic Framework for Ice Crystal Habit Classification in Midlatitude Winter Storms using Synergistic Lidar-Radar Observations
Valeria Garcia, University of Washington, Seattle, WA; and L. A. McMurdie, R. M. Rauber, J. A. Finlon, and J. E. Yorks

11:15 AM - 18.3 Characterizing Thin Super-Cooled Liquid Water Clouds Using Airborne Microwave Radiometry during ARCSIX
Michael Angel Perez, University of Miami Rosenstiel School, Miami, FL; and P. Zuidema

11:30 AM - 18.4 Cloud Optical Depth Retrievals from Ground-based Radiometers: Comparison and Improvement
Damao Zhang, Pacific Northwest National Laboratory, Richland, WA; and E. Kassianov, S. Giangrande, and L. Ma

11:45 AM - 18.5 Drizzle Droplet Distributions Retrieved from Vertically Pointing Doppler Lidar and Ka-Band Radar Observations
Christopher R. Williams, University of Colorado Boulder, Denver, CO; and K. Loftus and M. van Lier-Walqui

Lunch Break

12:00 PM - 1:45 PM | View Related | The Waste Basket

Session 19 - Remote Sensing of Clouds II

1:45 PM - 3:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Cloud Physics

Submitter: Fan Yang / Brookhaven National Laboratory / Mastic, NY

Cochairs: Siddhant Gupta / ANL / Lemont, IL / Julia A Shates / SIO / La Jolla, CA

1:45 PM - 19.1 Retrieving the 3D Distribution of Atmospheric Temperature Inside and Outside of Clouds from Multi-Angle Imagery using Atmospheric Tomography with 3D Radiative Transfer (AT3D)
Michie Vianca De Vera, University of Illinois Urbana-Champaign, Urbana, IL; and L. Di Girolamo and J. R. Loveridge

2:00 PM - 19.2 Evaluating Snow Particle Habits and Microphysics Schemes in Microwave Brightness Temperature Simulations of Hurricane Dorian
Abhisek Das, Penn State, State College, PA; The Pennsylvania State Univ., University Park, PA; and E. E. Clothiaux, Y. Zhang, PhD, Z. Yao, A. C. Didlake Jr., and X. Chen

2:15 PM - 19.3 Retrieval of Polar Cloud Properties from IASI observations Using EarthCARE Geometrical Constraints
Tiziano Maestri, University of Bologna, Bologna, BO, Italy; and E. Fabbri, M. Martinazzo, G. Masiello, G. Liuzzi, K. Sato, and H. Okamoto

2:30 PM - 19.4 The NASA Continuity MODIS-VIIRS Imager-Sounder Cloud Products
Dongwei Fu, Space Science and Engineering Center, University of Wisconsin-Madison, Madison, WI; and R. Holz, S. Platnick, K. Meyer, P. Veglio, A. Heidinger, S. Dutcher, M. J. Foster, Y. Li, E. Borbas, E. Zhan, D. Botambekov, G. Quinn, and A. Gumber

2:45 PM - 19.5 A 22-Year Two-layered Cloud Climatology through the Fusion of MISR & MODIS on Terra
Larry DiGirolamo, University of Illinois at Urbana Champaign, Urbana, IL; and G. Zhao, A. Mitra, M. D. Zelinka, H. A. Mahmoud, and K. J. Mueller

Coffee Break

3:00 PM - 3:45 PM | Grand Terrace (Monona Terrace Community and Convention Center) | Eighth Conference on Weather Warnings and Communication

Joint 20 - Mixed-Phase Clouds Across Regimes: Observations, Modeling and Theory

3:45 PM - 5:00 PM | Madison Ballroom D (Monona Terrace Community and Convention Center) | 17th Conference on Atmospheric Radiation

Submitters: Yongjie Huang / CAPS / Norman, OK / Greg M. McFarquhar / CIWRO/SOM / Norman, OK / Alexei V. Korolev / Environment and Climate Change Canada / Dorval, QC / Canada

Cochairs: Yongjie Huang / CAPS / Norman, OK / Greg M. McFarquhar / CIWRO/SOM / Norman, OK

3:45 PM Joint 20.1
Investigating the roles of inversion strength and subsidence on the persistence of mixed-phase cloud cover using MOSAiC Observations and the CM1 LES Model Andrew Michael Dzambo, Cooperative Institute for Severe and High Impact Weather Research and Operations, Norman, OK; and H. Morrison, K. K. Chandrakar, PhD, and G. M. McFarquhar

4:00 PM Joint 20.2
What Drives the Morphological Variability of Ice Crystals in Mixed-Phase Clouds? Zachary Nice, Penn State University, University Park, PA; and J. Y. Harrington

4:15 PM Joint 20.3
Characteristics of the Mixed-Phase Cloud Morphology In the Arctic Identified with Volume Scanning Radar Observation Ryuto Hamada, Kochi University of Technology, Kami, 39, Japan; and T. Hashino

4:30 PM Joint 20.4
Untangling the Microphysics and Turbulent Factors Governing the Phase Structure of Mixed-Phase Stratiform Clouds Kamal Kant Chandrakar, PhD, NCAR, Boulder, CO; and H. Morrison, A. M. Dzambo, and G. M. McFarquhar

4:45 PM Joint 20.5
Observed (Bio)Aerosol Responses to Convective Cold Pools in the US High Plains and the ITCZ Leah D. Grant, Colorado State Univ., FORT COLLINS, CO; and K. Britton, R. Carrasco-Alvarez, N. M. Falk, T. K. Feldman, S. W. Freeman, T. Gobble, J. Horstmann, S. M. Kreidenweis, C. A. Neumaier, M. Nieto-Caballero, R. J. Perkins, E. A. Stone, O. Wurl, and S. C. van den Heever